Built to Sell: Strategic Entry in Pharmaceutical Markets *

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The majority of new cancer drugs are invented by small biotechnology firms but are brought to market by large ones, with control of the compound changing hands during clinical development. For a potential entrant, the prospect of being acquired therefore acts as a subsidy to entry, while antitrust authorities may view the same transaction as the removal of a future competitor. The policy remedy under debate bars incumbent firms from acquiring development-stage compounds in markets where they hold overlapping products. Using a quarterly panel that links oncology drug development to ownership of control rights, we show that acquirers are a small, identifiable set of large firms whose demand for new compounds rises predictably as their own products approach loss of exclusivity. We then model the sale of a development-stage compound as a first-price auction among the incumbents that engage, so that the set of permitted acquirers enters the price, and estimate how the acquisition payoff affects whether small firms enter development and which cancer they target. The ban cuts the expected acquisition payoff in half, redirects about 3 percent of small-firm entry away from the most contested cancers, and lowers the number of entering compounds by about 2 percent. When valuing deterred entrants and prevented killer acquisitions at the same approval odds, the ban raises expected approvals only if more than a fifth of the acquisitions it prevents would have been killer acquisitions, about three times the highest estimate in the literature.

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1 Introduction

Most new cancer drugs are invented by small biotechnology firms and brought to market by large ones, with control changing hands during clinical development. Of the 50 innovative, first-in-class cancer drugs approved in the United States between 2010 and 2020, 23 were originated by small biotechnology firms, and only four of those 23 reached the market under the firm that discovered them (Kennedy et al., 2023). By comparison, only seven of the first-in-class cancer drugs were originated by large, established firms. Despite this substantial disparity, large firms owned or licensed 38 of the 50 by the time those drugs launched.

The pattern of small firms disproportionately innovating and large firms commercializing via acquisition is present throughout pharmaceutical development and in other innovative markets, such as technology platforms (Kamepalli et al., 2020; Ederer et al., 2026). Across all therapeutic areas, companies with less than $500 million in sales originated 53 percent of the novel active substances launched in the United States from 2015 to 2019 (IQVIA Institute for Human Data Science, 2025). The trend is particularly pronounced in oncology, the dominant therapeutic area by research spending and sales. In the United States, spending on cancer drugs is expected to reach $180 billion by 2028 (Jazowski et al., 2024).

The ownership transfer happens during clinical development. The small originators are risky, typically owning a single asset with a 5% approval probability. They are also capital constrained. Bringing a compound from first human dose to approval takes about a decade and an estimated $100 million to $1 billion of capitalized cost (DiMasi and Grabowski, 2007), and most of that bill arrives at Phase III. Their venture capital funding usually cannot fund the Phase III trial that follows a successful Phase II, nor do they have the manufacturing capacity, regulatory function, payer contracts, and specialty sales force that a United States oncology launch requires (Teece, 1986). In 2024, 39% of public biotechnology firms held less than one year of cash (EY, 2025). For these firms, a sale to a large firm is the expected end of successful early development.

The timeline of Cougar Biotechnology illustrates this path. Founded in 2003 around a single licensed compound, abiraterone, Cougar ran its Phase II trial in prostate cancer. After promising results in 2008, its board hired an investment bank and not a sales force. Public filings show that 17 potential buyers signed confidentiality agreements, two submitted formal bids, and Johnson & Johnson ultimately acquired Cougar for about $1 billion. After acquiring the compound, Johnson & Johnson completed Phase III trials and launched the drug as Zytiga, which went on to earn over $2.5 billion in 2017 alone. Venture investors

describe choosing which biotechnology firms to fund by asking who the likely buyers would be. For a potential entrant, the prospect of acquisition is an entry subsidy. It raises the expected value of beginning development even for a firm that never intends to commercialize a drug.

The same transactions worry antitrust authorities. An incumbent that acquires a development-stage competitor in its own market may shelve it to preempt future competition, a practice referred to as a killer acquisition. Cunningham et al. (2021) estimate that 5.3 to 7.4% of pharmaceutical acquisitions are of this kind. Many of these deals fall below the Hart-Scott-Rodino notification threshold and are never reviewed (Wollmann, 2019). The policy response targets the acquirer's existing position, since a firm with an overlapping product on the market or in development is the one with a motive to shelve. The perceived welfare risk of killer acquisitions is large enough that the 2023 United States merger guidelines treat an acquirer's position in the target's market as a basis for scrutiny (U.S. Department of Justice and Federal Trade Commission, 2023), and the European Commission's 2024 ex-post evaluation recommends preventing incumbent acquisitions of overlapping pipeline products (European Commission, 2024). These remedies are forward-looking, blocking the horizontal acquisition before the competition it would eliminate exists. What they have not confronted is the effect on entry, and therefore innovation. If acquisition is an entry subsidy, a ban on a class of acquirers lowers the subsidy, and the small firms whose innovation the policy is meant to protect may respond by entering different cancers with a lower probability of approval or by not entering at all.

In this paper, we study how the prospect of acquisition shapes the entry decisions of small biotechnology firms in oncology on the extensive margin, whether to enter development, and on the intensive margin, which cancer to enter. We model the sale of a development-stage compound as a first-price auction among the incumbents that engage with the sale process, so that the set of incumbents permitted to bid enters the price a compound fetches and the probability that it sells. A policy that bars a class of acquirers therefore has a defined route to the entry decision. We estimate the model, quantify how a ban on overlapping acquisitions would move entry along both margins, and compare the entry it deters with the killer acquisitions it would prevent.

We construct a quarterly panel that follows 2,539 oncology compounds from 2006 through 2019, linking each compound's development history to the firm controlling its United States rights. Three descriptive facts from this panel discipline the model. First, the set of incumbent acquirers is small and identifiable. The 39 large firms we define as incumbents, between 22 and 26 of them holding an oncology asset in any quarter, made 220 of the 580 acquisitions in our sample, paid several times more per deal than other acquirers,

and are essentially the only repeat acquirers. A potential entrant forming expectations about acquisition is forming expectations about a known set of likely buyers. Second, demand for acquisition is predictable. An additional $1 billion of incumbent revenue within two years of losing patent exclusivity in a cancer raises the rate at which incumbents acquire entrants in that cancer by 48%, and entry into the same cancers rises alongside it, suggestive of entrants anticipating incumbent demand. Third, the sale of a development-stage compound is an auction among few participants. In the merger chronologies that selling companies are legally required to file, the median process puts two parties under a confidentiality agreement when a whole company is sold and five when the canvass is for a license, and in both the median number that go on to price the asset is two. The variation the estimation uses follows from these facts. The share of incumbents holding an overlapping position differs across cancers, from about one in ten in the smallest markets to seven in ten in lung and breast cancer, the exclusivity cliffs that drive incumbent demand arrive on a schedule known years in advance, and a compound's mechanism of action makes some cancers feasible for it and others not.

The model has two estimated stages. In the first, a compound is sold through a first-price auction among the incumbents that engage with the sale, a small set drawn from the pool by costly diligence, and the seller runs a process only when its option value covers the cost. Incumbents holding an overlapping position in the compound's cancer, a Phase II or later development asset or an approved product still on patent, draw their valuations from a stronger distribution. Following Guerre et al. (2000), we recover the valuation distributions by inverting the equilibrium bid function on observed acquisition prices. In the second stage, a compound that has completed Phase I, which need not name a cancer, chooses the cancer for its first Phase II trial or does not proceed. We estimate this choice as a conditional logit over 27 cancer indications and the outside option. The payoff in each cancer is the expected acquisition value plus the expected value of commercializing alone, each scaled by a match quality measure built from the compound's mechanism of action. This match quality term is what determines the possibility set of indications for a compound. For example, it dictates that a HER2 antagonist is a plausible breast cancer drug, a weaker candidate in gastric cancer, and no candidate at all in acute leukemia. Among the cancers a compound's mechanism makes feasible, entrants tilt toward the ones where acquisition pays more. A $10 million increase in the match-weighted acquisition payoff raises the odds that a compound enters a given cancer by about 38%.

The counterfactual bars incumbents from acquiring development-stage compounds in cancers where they hold an overlapping position. Within the auction, the ban removes the strongest bidders from each market. At the working calibration, the expected winning bid falls by 32% on average and the probability

that any engaged party prices the asset falls by 18%, so the expected acquisition payoff at entry falls from $36 million to $18 million. Entry responds on both margins. Aggregate entry falls by 2.2%, and 2.9% of entrants move to a different cancer than they would otherwise have entered. Non-small cell lung cancer loses 16% of its entrants and non-Hodgkin lymphoma 8%, while cancers with few overlapping incumbents gain 7 to 11% each.

These are entry effects and not welfare effects, but they support a useful accounting. Valuing deterred entrants and prevented killer acquisitions at the same Phase II approval odds, the ban raises expected approvals only if about 22% of the acquisitions it eliminates would have been killer acquisitions, roughly three times the highest estimate in the literature (Cunningham et al., 2021; European Commission, 2024). The step from approvals to welfare is left for future work. On the entry margin, the policy's main effect is to move entrants toward cancers with fewer overlapping incumbents, with potential benefits for patients there and costs for patients in the cancers that lose entrants.

Our question sits between the literatures on entry and on innovation. A theoretical literature recognizes that entry can occur in anticipation of buyout, so that the likelihood of acquisition raises the return to entry (Rasmusen, 1988; Gans and Stern, 2000; Gans et al., 2002; Norbäck and Persson, 2009). Recent work studies the settings in which acquisitions are made only to deter competition (Cabral, 2021; Fumagalli et al., 2023; Letina et al., 2024; Denicolò and Polo, 2026), and Teh and Wang (2025) survey this literature. Evidence from Cunningham et al. (2021) moved killer acquisitions to the center of merger policy, and a small set of structural papers now quantifies the entry subsidy from acquisition in other industries (Eisfeld, 2025; Wabiszewski, 2025). In these models, however, the acquisition payoff is either exogenous or the outcome of a bargaining game with one acquirer, so the number of potential acquirers never enters the price. In ours it does, which is what allows the counterfactual to act on the object the policy debate is about, the set of firms permitted to bid.

Our methods follow the empirical auctions literature. We build on Guerre et al. (2000) to invert the distribution of observed winning bids into a distribution of private valuations, and our use of merger chronologies as records of potential and actual bidders follows the takeover-auction work of Boone and Mulherin (2007) and Gorbenko and Malenko (2014). Because we observe only winning bids across heterogeneous assets, we estimate the ban's effect on prices as a ratio and do not recover valuations in levels, an approach that may transfer to other thin asset markets where only transaction prices are observed (Haile and Tamer, 2003; Krasnokutskaya, 2011). On the entry side, a substantial literature shows that market size, expected prof-

itability, and regulatory design steer pharmaceutical development (Blume-Kohout and Sood, 2013; Budish et al., 2015; Yin, 2008; Dranove et al., 2022; Gupta and Kao, 2025), and that incumbents buy development-stage assets as their own products approach the end of exclusivity (Higgins and Rodriguez, 2006; Krieger et al., 2022). We add the prospect of acquisition itself to the list of what steers development.

The paper proceeds as follows. Section 2 provides background on oncology drug development and fixes the definitions of markets, entrants, and acquisitions. Section 3 describes the data sources and the construction of the panel. Section 4 presents the model, and Section 5 provides reduced-form evidence for its assumptions. Section 6 takes the model to the data, Section 7 estimates it and reports the counterfactual, and Section 8 concludes with paths for future work.

2 Setting

Oncology drug development is an attractive setting to study how acquisitions affect entry and innovation. First, entry is observable. Phase II trials are a dated event that require regulatory filings and a named market choice. Second, development stage drugs change ownership through a sale process that may be conceptualized as an auction. Third, approved oncology drugs have a market exclusivity that is known to market participants years in advance. The end of these exclusivity periods is known to induce *revenue cliffs*, after which significant revenue loss is expected. This section describes each of these features in turn and fixes definitions of markets, entrants, acquisitions, and patent cliffs.

2.1 Drug Development and Clinical Trials

To become an approved oncology drug, a compound must move through a fixed sequence of stages. After preclinical work and an Investigational New Drug (IND) application, Phase I trials enroll a few dozen patients to test toxicity and set the dose. They say little about efficacy and need not name a cancer, so they often pool patients across several cancers as a trial in "solid tumors" or "hematological malignancies". Phase II trials enroll more patients, often around one hundred, and test for efficacy against a named cancer. Phase III trials are larger and more expensive still, evaluating efficacy across hundreds of patients against the standard of care. Finally, if the trial sponsor believes these results are compelling, it can file a New Drug Application (NDA) or Biologics License Application (BLA) for approval in a specific indication.

Development across these stages is slow, expensive, and usually ends in failure. Roughly one compound

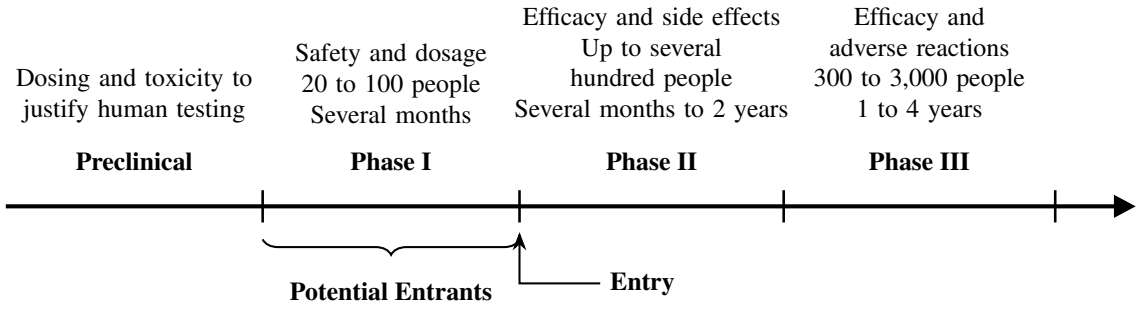
in twenty that enters Phase I reaches approval, the lowest rate of any therapeutic area (Biotechnology Innovation Organization et al., 2021; Wong et al., 2019), and the sequence from first human dose to approval takes roughly a decade, with trial lengths that vary widely by cancer (Budish et al., 2015). The full cost of bringing a compound through the regulatory process to commercialization has been estimated at $100 million to $1 billion (DiMasi and Grabowski, 2007).

Two definitions are required from this process to answer our research question. First, we define a market as a cancer indication. Biomedtracker (discussed in Section 3.1) records development against 64 oncology indications, several of which are clinically close enough that a drug approved in one is used in the others. We collapse these categories into 27 parent indications along the lines of the treatment regimens they share, following the market definitions used in the economics of cancer treatment (Dix and Lensman, 2026) and validated by a board certified oncologist. For example, marginal zone lymphoma and Waldenström macroglobulinemia are both treated with rituximab-based regimens and sit together under non-Hodgkin lymphoma in our categorization. The parent indications also align with the site groupings in which the National Cancer Institute reports incidence, which we use for market size. Biomedtracker also carries two placeholder categories, "Other Solid Tumors" and "Hematologic Cancer", which a record takes when it names no specific cancer. Neither is a market a compound can be approved in, and Section 3.1 describes how we use trial registry records to place these entries in the cancers they test. Markets are listed in Section 3. A compound can enter more than one market, 27% of first Phase II trials are basket trials naming more than one parent indication. We treat such a trial as entry into each market it names, with additional definitions for robustness.

Second, we define market entry as the start of a compound's first Phase II trial in the market that the trial names. Figure 1 places this definition into the regulatory sequence. An additional attractive feature of the market and this definition is that we are then presented with a defined set of potential entrants, or those that have finished Phase I trials but not yet initiated a Phase II trial.

Three features make Phase II the correct definition of entry. It is the first point at which a sponsor must commit to a cancer, since a Phase II protocol tests efficacy in a defined patient population. It is the first stage that is expensive enough to require a new round of financing, representing a commitment of capital as well as of direction. Importantly, the sponsor is not bound by anything from Phase I trials aside from dosing safety. A compound that was dosed in a basket of solid tumors can begin Phase II in any of them, and compounds with the same mechanism routinely begin in different cancers. Industry participants we spoke

Figure 1: The Drug Development Process and Definitions



Notes: Enrollment and duration by phase are from the FDA ("The Drug Development Process, Step 3: Clinical Research"). The set of potential entrants is the set of compounds that have completed Phase I. Entry is the start of a compound's first Phase II trial, which names the cancer it tests.

with described the Phase II start as the point at which a company decides what its drug is for.

The firms that make this entry decision are small. Of the 50 first-in-class cancer drugs approved in the United States between 2010 and 2020, seven originated solely at a large pharmaceutical firm, while 38 were owned or licensed by one at launch (Kennedy et al., 2023). This gap between where drugs originate and who brings them to market is the subject of our paper. The typical originator is a company built around a single compound and financed by venture capital, and it has neither the capital nor the sales force that are required to run a Phase III trial and a launch. In 2024, for instance, 39 percent of public biotechs held less than a year of cash (EY, 2025). Licensing revenue from alliances with large firms can also exceed all of the venture capital raised by biotechs in the same year, as it did by a wide margin in 2024 (McLaren, 2025). In our panel, firms that we define as Big Pharma initiate 25% of first Phase II trials, large biotech firms initiate 6%, and small firms initiate the remaining 69%. Additionally, the median small firm entrant has only a single compound in development at the time it initiates a Phase II trial.

2.2 Pharmaceutical Acquisitions

Most cancer drugs are started by one firm and launched by another. The transfer happens during development, most often between Phase I and Phase III. Our analysis sample holds 580 acquisitions of control rights to a development-stage oncology compound from 2006 through 2019 (Table 2).[1] Of these, 32 percent took place while the compound was in Phase I, 49 percent while it was in Phase II or III, and 5 percent during NDA review or after approval, with the remaining 14 percent preclinical. Large firms, defined in

[1]The full panel records 1,026 transactions over the same years in which a firm took exclusive United States or worldwide rights to a development-stage oncology compound, excluding mergers between two established large firms. The 580 transactions are those that enter at least one analysis in the paper. Section 3.9 describes the sample.

Section 3.4, were the acquirer in 220 of the 580, about 38 percent by count, and they paid substantially more per compound than other acquirers, which Section 5.1 documents.

We consider acquisitions take one of two general legal forms. The first is an exclusive license, otherwise known as an "alliance" in the industry. In an exclusive license, the biotech which owns the compound grants a partner the exclusive right to develop and sell the compound in a territory in exchange for an upfront payment, payments contingent on development and sales milestones, and a royalty on sales. In this case, the biotech survives as a company and may retain rights in other territories or to other compounds. Among exclusive licenses with a disclosed upfront payment, the mean upfront is $123 million when the acquirer is a Big Pharma firm and $28 million when it is not (Table 2). The second form of acquisition we consider is a company acquisition in which the buyer purchases the biotech's equity, usually through a tender offer or a merger vote, and acquires all the firm's assets.

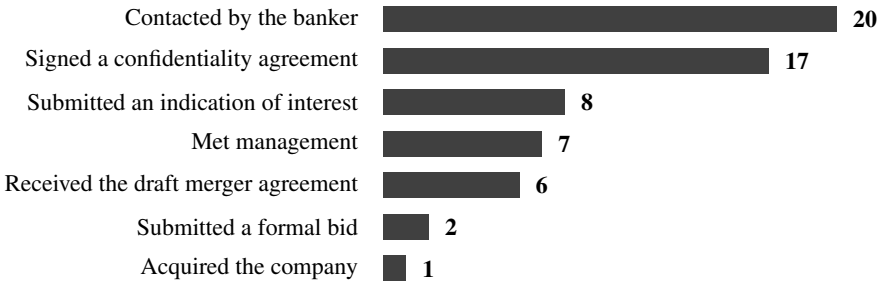
Seven in ten of the 580 transfers are exclusive licenses, and the remainder are company acquisitions. Both forms move control of the compound's United States commercialization from the biotech to the acquirer, and the alliance literature has found that it is the allocation of control rights that matters for how the asset is subsequently developed (Higgins, 2007). For this reason, we treat both forms as an acquisition of rights, and we describe how we construct the ownership chain in Section 3.2. When we require a price per compound, we use the licenses together with the whole-company acquisitions for which a price can be tied to a single compound, following the rule given in Section 3.3.

In practice, the sale of a development-stage compound is a process that the biotech itself initiates and runs. A board that has decided to sell hires an investment bank, which then contacts a list of potential buyers, signs confidentiality agreements with those that want to look at internal data, and opens a data room. Interested parties submit non-binding indications of interest that include a price range, a subset of these parties are invited to meet with management and receive a draft agreement, and a smaller subset submit final bids. The board then compares the best offer against the value of continuing alone, and it either signs the agreement or ends the process. A licensing canvass often runs alongside the sale, and boards treat the canvass as the market check for a sale, on the reasoning that any party with an interest in acquiring the company would already have surfaced. As a result, the process is close to a sealed-bid auction with a reserve price, which we formalize in Section 4. Bidders do not observe one another's offers, the seller accepts the highest bid that clears its own continuation value, and losing bidders are common.

Figure 2 shows one such process, detailed in SEC Filing SC 14D-9 (discussed in Section 3.7). Cougar

Biotechnology was founded in 2003 around one compound, abiraterone, licensed from a British research institute. When its Phase II trial in prostate cancer read out in 2008 the board hired a banker to sell the company. About twenty buyers were contacted, seventeen signed confidentiality agreements, eight submitted indications of interest, seven met management, six received a draft merger agreement, and two submitted formal bids. The highest early bidder withdrew as they wanted to first see Phase III results. Johnson & Johnson won at 43 dollars a share, about one billion dollars. In 2009, J&J finished Phase III and launched the drug as Zytiga in 2011.

Figure 2: Acquisition Funnel, Cougar Biotechnology



Source: Cougar Biotechnology Schedule 14D-9, June 2009, "Background of the Offer".

Both sides of the transaction have reasons to want a large-firm buyer. A small firm could in principle carry its own compound to market, and a few have, but the assets a United States oncology launch requires are large and indivisible: manufacturing capacity, a regulatory and pharmacovigilance function, contracting with payers, and a specialty sales force of several hundred representatives. A firm with one compound cannot spread those costs where a firm with forty can. Teece (1986) makes the general form of this argument, that an innovator captures the returns to an innovation only when it controls the complementary assets needed to commercialize it, and otherwise sells to the party that holds them. Oncology sits close to the limiting case, because the complementary assets are concentrated in the same small set of firms that Section 5.1 identifies as the buyers. The biotech's investors also need an exit, and venture investors describe choosing which biotechs to fund by asking who the likely buyers would be.[2]

The cost of financing what remains rises sharply at the end of Phase II. A biotech holding one unapproved compound cannot raise the hundreds of millions of dollars a Phase III oncology program requires at a price that reflects what the firm knows about the compound, while a large firm funds the same trial from internal cash flow, against a portfolio in which one failure is a single draw. The gap between the two costs

[2] "When he considers investing in a firm, he doesn't eye the public market, but rather looks at whom likely buyers would be." A managing director of a venture firm, quoted in BioWorld Today, 2006.

of capital is a gain from trade that requires no product market synergy, and it widens at the point where the remaining development bill jumps, which is one reason about half of the transfers we record occur while the compound sits in Phase II or Phase III. A large firm that already has a presence in the compound's market has still more to gain, since it holds the commercial organization in that cancer and, if it has an approved product there, faces the schedule of exclusivity losses that Section 2.3 describes. Acquisition has been the pharmaceutical industry's main channel for replacing internal research since at least the 1990s (Higgins and Rodriguez, 2006), and firms whose products erode respond by buying development-stage assets in the same area (Krieger et al., 2022).

At the same time, these same transactions have raised concerns among competition authorities. An incumbent that buys a development-stage competitor in its own market may shelve it, and the evidence that this occurs is strongest in pharmaceuticals (Cunningham et al., 2021). The remedies under discussion target exactly the transactions in which the acquirer already sells in the market. For instance, the 2023 United States merger guidelines and the European Commission's 2024 review both treat an acquirer's existing position in the target's market as the basis for scrutiny (U.S. Department of Justice and Federal Trade Commission, 2023; European Commission, 2024). The counterfactual that we study in Section 7 implements this policy. Under it, a firm with a development asset in Phase II or later, or an approved product still on patent, in a given cancer is not permitted to acquire a development-stage drug in that cancer.

2.3 Market Exclusivity and Patents

Market exclusivity is central to our setting, as an approved drug earns most of its revenue while it is protected from direct competition. Two layers of protection provide this exclusivity, patents and regulatory exclusivity, and both are set at or before approval. The end date is public knowledge for a decade or more before it arrives. We describe each layer in turn. The first is patent protection. The primary patent covers the active ingredient and is filed years before approval, so its remaining life at launch is typically eight to twelve years after adjustment. Secondary patents cover formulations, methods of use, and combinations, and expire later. The Hatch-Waxman Act restores up to five years of patent life lost to clinical testing and regulatory review, capped at fourteen years of protection after approval, and the sponsor chooses which patent receives the extension, usually the primary patent (Hemphill and Sampat, 2025). The second is regulatory exclusivity, which is granted at approval regardless of patent status and runs from three years for an approval supported by new clinical evidence to twelve years for a biologic. We give the terms we apply in Section 3.6. A firm's

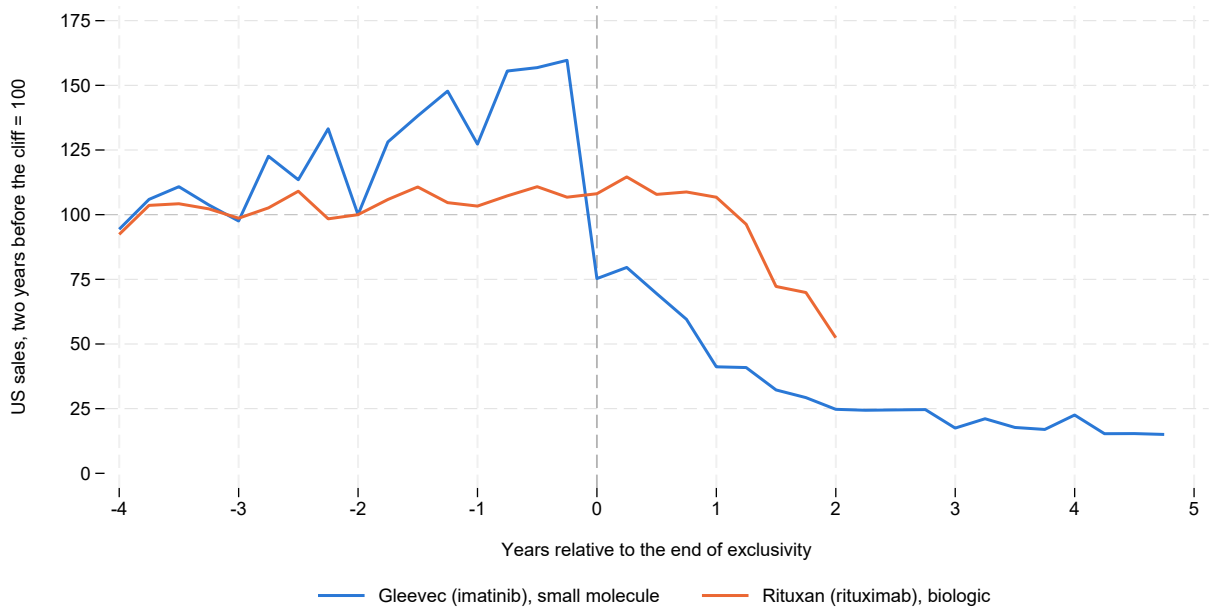
drug is protected until the later of the end of the exclusivity period and the end of patent protection.

The date of the last patent expiry is not always the end of a product's patent protection. Generic manufacturers are able to challenge patents before they expire, including the primary patents that cover the active ingredient. These challenges, however, fall disproportionately on the later-expiring secondary patents (Hemphill and Sampat, 2012), and they succeed there. Of challenges litigated to completion, generic manufacturers prevail against 68 percent of secondary patents but only 8 percent of primary patents (Hemphill and Sampat, 2013, 2025). The number of secondary patents has increased dramatically over the past twenty years due to a practice to delay generic entry, referred to as "evergreening" or "life-cycle management". Revlimid, for example, carried some thirty patents with different expiry dates. From outside the firm it is difficult to say which patent will be challenged and when, and as a result both analysts and the firms themselves work from a consensus expected date in place of the nominal one. In Section 3.6, we describe how we construct the exclusivity end for each drug and market from the Orange Book and from analyst consensus dates, as well as how we treat the challenge margin. Despite this complexity, the effective market life of pharmaceuticals has stayed close to 13 years since 2000 (Hemphill and Sampat, 2011, 2025).

When exclusivity ends, revenue falls, and it does so quickly. Figure 3 shows quarterly worldwide sales for two oncology drugs around their loss of exclusivity in the United States. Gleevec exemplifies the life-cycle of a small molecule, which typically loses most of its revenue within a year. For small molecules, the manufacturing process is simpler and produces chemically identical compounds. Biologics are more complicated in this regard. Biosimilars require specialty knowledge to produce and are not identical to the approved product, and as a result both their manufacture and their uptake by prescribers are slower, as the Rituxan series shows. In both cases, a firm's loss is concentrated in a single market, given that an oncology drug's sales come from the cancers in which it is approved.

Because exclusivity periods and patents are public knowledge, they provide us with predictable variation in incumbent demand. An incumbent firm with an approved, branded product in a given cancer indication has a commercial organization, expended fixed costs, and expertise in that cancer. An upcoming exclusivity cliff places their revenue at risk, and the market knows the date by which they will want to seek a replacement. Consequently, a development-stage compound in that cancer is worth more to that firm than the same compound is to a firm with no such product, and the value rises as the date approaches (Higgins and Rodriguez, 2006).

Figure 3: Revenue after the end of exclusivity



Notes: Quarterly worldwide sales from Evaluate Pharma. Each series is indexed to its own sales in the quarter two years before the end of exclusivity. End of exclusivity is designated by the vertical dashed line; 2016q1 for Gleevec and 2018q4 for Rituxan. Two years prior to the exclusivity end date, Gleevec sold $420 million a quarter and Rituxan $949 million. Two years after, they sold $104 million and $498 million, respectively. Series are deflated to 2015q1 dollars and end in 2020q4.

3 Data

In this section, we describe the construction of a quarterly panel that tracks all oncology compounds from their first clinical trial through acquisition, approval, and loss of exclusivity, and that links each compound in each quarter to the firm controlling its United States rights. The unit of observation is a drug by indication by quarter, covering 2,539 compounds across the 27 markets defined in Section 2 and the two placeholder categories, from 2006q1 through 2019q4. We view this link between ownership and development as the data contribution of this paper, as it gives us control right ownership at the quarter level. This allows us to count acquisitions by the acquirer's status at the deal date, price them, and measure the acquirer's portfolio in the drug's market at that moment.

We estimate the models in this paper using three aggregations of this panel. The first is the market-quarter panel of Section 5.2. This panel collapses the data to the 25 markets that ever hold an approved drug within eight years of losing exclusivity, for a total of 1,400 market-quarters (bladder cancer and myelofibrosis are the two markets that this excludes). The second is the deal-level sample of Sections 5.1, 5.3 and 7, with 580 acquisitions of control rights. The third is the entry sample of Section 7, with 1,195 compound-

level entry decisions. All dollar amounts in the paper are deflated to 2015q1 dollars with the CPI-U. In the remainder of this section, we describe the data sources, how we use them, and the cleaning decisions that matter for the analysis. Additional information is contained in Appendix A.

3.1 Drug Development, Trials and Classification

To track drugs through their development history, we use Biomedtracker, a Citeline product used extensively in prior work (Blume-Kohout and Sood, 2013; Branstetter et al., 2022; Cunningham et al., 2021). Biomedtracker assigns an identifier to every drug and indication pair and records clinical trials, regulatory events, and analyst assessments. From the drug records we take the mechanism of action and target, whether the compound is a biologic or a small molecule (excluding vaccines), and regulatory designations such as orphan status. From the commercially relevant trial and event records we build the life-cycle of each drug-indication pair. This includes the dates it enters each phase, its approval, and its discontinuation.

The life-cycle of a drug is not always cleanly defined in the data. For instance, Biomedtracker records some suspensions without a date, and a drug may run parallel trials in the same indication. In Appendix A, we describe how we date these suspensions and how we collapse the parallel trials.

Biomedtracker also supplies an analyst likelihood of approval, which is a phase-specific probability by indication. We use the Phase II value, as entry is defined as the first Phase II trial. We use this probability as a control in the market-level regressions of Section 5.2, and in the auction stage of Section 6 we include it in the seller's reserve and in the bid index. It is worth noting that the approval probability in the entrant's payoff is a different object, the realized rate at which Phase II programs in the indication reach approval (Section 6.4).

We supplement the development panel with two additional data sources. The first is the set of ClinicalTrials.gov records, which we access through the AACT database. We use these records to place the first Phase II trials whose Biomedtracker indication is designated with the placeholder "Other Solid Tumors". When the AACT registry record names specific cancers in the trial, we assign the entry decision to those cancers, which is the case for 27 of the 57 placeholder entries. The 30 entries that cannot be placed are tumor-agnostic programs, and these are excluded from the entry estimation (Section 6.5). The second is a compound-specific therapeutic class from the EphMRA ATC system at level 4, also taken from the AACT data, with 72 classes defined by modality and target family. This class is the coarsest of the three mechanism tiers in the match-quality measure of Section 6.5, and the finer two tiers come from Biomedtracker's

mechanism and target fields.

3.2 Ownership, Licenses and Acquisitions

The control right ownership of each compound through the panel is constructed as a chain from two tables. The first is a company identity chain which includes every company that appears anywhere in the data, together with who acquired it and when, so that the controller of any asset at any date resolves by walking the chain. The second is a drug-level ownership event table, which lists the originator of each compound and every subsequent transfer of control, assembled from Biomedtracker alliance records, Evaluate Pharma company M&A, Evaluate product-level deals, and four hand-collected files.

The object of interest that we track is United States control rights. The control-rights definition we use throughout covers company acquisitions, asset purchases, and exclusive worldwide or US licenses. Transactions excluded from this chain include non-exclusive co-development deals, co-commercialization deals, and reverse mergers. For the analysis sample, we also remove "mega-mergers", or whole-company M&A between large, established, diversified firms. These amount to 14 deals, identified by hand, detailed in Appendix A. These are excluded because we are unable to assign a value to any one development compound within them, and because the acquisitions are motivated by different market forces than those we consider here. Finally, licensing deals and whole-company M&A occasionally cover several assets. In these cases, we assign the deal to the compound the deal description identifies as the lead, or to the compound furthest in development. If we are unable to determine a clear lead, the deal is removed from the analysis sample. We report robustness to dropping multi-drug deals in Appendix A.

3.3 Deal Terms and Prices

We draw deal terms from both Biomedtracker and Evaluate Pharma, which allows us to validate one source against the other. For product level alliances, we primarily use Biomedtracker records, which report upfront payments, milestone schedules and royalty provisions for licenses along with text fields containing descriptions of the deals. Evaluate Pharma provides an additional check on these alliances with the product-deal file, and supplements for business line and direct asset purchases. Finally, Evaluate Pharma provides the terms for company M&A. From these terms, we carry through two main measures of deal value: total potential deal value and the upfront payment. The total potential deal value is the total amount committed if

all development and commercialization milestones are realized, which is often not the case. The upfront payment is paid at signing. For robustness, we also carry the upfront payment plus 50% of the milestone payments, as an approximation to the expected value of the deal at the point of sale.

To avoid concerns about correlation between deal structure and drug quality, our primary price metric in Section 6 is the total potential deal value. This measure captures the whole of what the acquirer commits to, and it is also the measure on which the acquisition route is comparable to self-commercialization. We do not risk-adjust the milestones at the deal level, because the data do not allow us to decompose the payments by trigger. Instead, we carry the upfront payment, which is the only component paid with certainty, as an alternative metric throughout for robustness. Given that most milestones never pay, the realistic transfer lies between the two measures, and in Section 7.2 we also report a risk-adjusted price (the upfront payment plus the milestone schedule at a fixed discount) for the deals that report both components.

A price is disclosed for 246 of the 410 exclusive licenses on total value and 211 on the upfront, and for most whole-company acquisitions. A whole-company price covers everything the target owns, so it enters the priced-deal analyses only when it can be tied to one compound. This is done either when target holds a single drug or when a clear lead drug is identifiable. This adds 91 whole-company deals on total value and 35 on the upfront, for a priced-deal sample of 337 and 246. One limitation of the data is that deal term disclosure is not legally compelled and subject to selection. Conditional on phase, scope, orphan status, drugs per deal and year, a large-firm acquirer is 37 percentage points more likely to report a total value (see Appendix A, Table 14). Because the missing prices are concentrated in deals between smaller firms, the large-firm premia in Section 5.1 are best read as lower bounds.

3.4 Firms and the Acquirer Pool

Because firms enter and leave the incumbent set through acquisitions and divestitures, we define incumbency at the quarter level and treat it as a state that can change over time. We define two types of incumbents. The first are Big Pharma firms, roughly twenty diversified firms meeting size and scope thresholds on market capitalization, revenue, research intensity, and geographic reach. The second are Large Biotech firms, smaller in size, scope, and diversification than Big Pharma firms, but still holding substantial commercialized assets. Our primary measure is the superset, which we refer to as "large firms". Appendix B lists both sets with their inclusion dates.

The acquirer pool in the structural model, K_t, consists of all incumbents that hold an oncology asset

in the quarter, between 22 and 26 firms over the sample. We intentionally do not define the pool at the indication level, given that incumbents routinely buy into cancers in which they hold no assets, and we instead measure the composition of the pool by market in Section 6.1. In Table 2, we additionally report a residual category of other large firms. We define this category by an oncology presence in the panel together with either $500 million of revenue and three control-rights acquisitions or $250 million of revenue and five acquisitions.

3.5 Sales, Market Revenue and Indication Attribution

Quarterly drug sales combine Evaluate Pharma and Biomedtracker revenue series, at the worldwide and United States level. About 22 percent of on-patent approved drug-quarters have no sales row, all of which are products that fall below disclosure thresholds. These are set to zero and the count of covered drugs is carried alongside.

A drug approved in several cancers reports one sales figure, and the paper needs revenue by cancer. Attributing the full amount to each cancer double counts, and an equal split is arbitrary. We attribute sales with indication shares built from MarketScan claims. For molecule d, cancer c and quarter t, the share is patient-months on d in c over patient-months on d in all cancers, with boundary quarters carried outward beyond the 2009 to 2019 claims window. Where the claims data cannot see the market, shares are assigned by hand from labels and analyst sources. This is done where the claims data do not identify specific site groups, such as CML, GIST, neuroendocrine tumors, sarcoma, and thyroid cancer. Claims shares carry about 71 percent of attributed sales, hand shares about 11 percent, and single-indication drugs, which need no attribution, the remaining 18 percent. Revenue at risk, the paper's demand shifter, is built from this attributed series and defined in Section 5.2.

3.6 Exclusivity and Patents

We define the end of an approved drug's protection as the later of its patent expiry and the end of its regulatory exclusivity. The regulatory side is relatively straightforward to construct from the approval record, with three years of protection from an approval in the market, seven from an orphan approval, and a floor of five years from first approval for a new chemical entity. The patent side, however, is harder to observe ex ante. As Section 2.3 describes, secondary patents are challenged often and successfully, so the nominal

last-to-expire date overstates protection and market participants work from an expected date instead.

We follow the same practice in our data construction. We take the expiry date to be Evaluate Pharma's consensus analyst forecast of the first commercially significant patent expiry in the drug's key market, which incorporates term extensions as well as the thicket of secondary patents. Where Evaluate has no record, we fall back to the Orange Book active-ingredient patent with its term extension, taken from the NBER-maintained database (Durvasula et al., 2023). A biologic with neither record is assigned fifteen years from first approval, which is the shorter end of realized biosimilar entry in oncology, and a small number of drugs whose litigation outcome or settlement date is public take that date instead. We then convert the expiry to a patent term at the drug level and extend it with market-level regulatory exclusivity, such that a combination approval earns its additional protection in the market in which it is approved. We also construct a realized exclusivity end, defined as the later of the expected date and the Orange Book's first generic approval, in order to mark the cliffs that the expected date understated. The expected date is the relevant date for an entrant that is planning years ahead, and it is the date that enters the quarters-to-expiry measures behind revenue at risk and the deal-price regressions. The realized date, on the other hand, defines whether a drug is on patent in a given quarter.

3.7 Merger Disclosures

A public US target that sells itself must explain the process to its shareholders. The Schedule 14D-9 filed for a tender offer and the DEFM14A filed for a merger vote both contain a "Background of the Merger" chronology, written by deal counsel from the banker's log. Boone and Mulherin (2007) establish these chronologies as the standard source on how corporate sales actually run, and Gorbenko and Malenko (2014) use the bidding records they contain to estimate what acquirers were willing to pay. The filings are legally compelled, contemporaneous, and name or consistently alias ("Company A") the parties that lost, which no transaction database does. Of 171 Evaluate M&A targets that could plausibly have filed, 135 have a filing on EDGAR, 95 yield a usable chronology, and 88 support the full coding, covering 2000 to 2024.

With these data, we are able to identify the distributions of the parties that are contacted in the event of a sale, the parties that signed a confidentiality agreement in order to receive information on the compound, and the parties that eventually submit bids. We use these counts in Section 6 to calibrate the rate at which pool members engage with a sale process. We define the engaged-bidder count and report the results in Section 5.4, and the full codebook and audit protocol are provided in Appendix C.

3.8 Claims Data

The final data source is the MarketScan Commercial Claims and Encounters and Medicare Supplemental databases, which cover the commercially insured under 65 and retirees with employer supplemental coverage from 2009 to 2019. We use these data to construct the indication shares of Section 3.5 and the quantity outcome for the competitor-approval event study of Section 5.5. The Medicare supplement matters because cancer incidence is concentrated over 65. A patient is assigned to a cancer by any inpatient or outpatient claim carrying a cancer diagnosis. Drug exposure comes from outpatient J-codes and pharmacy NDCs in the antineoplastic classes, 266 molecules after crosswalking to our drug identifiers. The working panel is an enrollee by cancer by month file with 10.4 million patient-months across 21 primary-site groups.

The claims data have two well known limitations that affect our analysis. First, inpatient drug administration is under-captured, and for this reason we exclude the acute leukemias from the claims-based shares. Second, the base of contributing employers shrinks over the window, which means that patient counts in levels require year effects while shares do not. Additionally, every table that we export suppresses cells with fewer than 11 patients.

3.9 Samples and Descriptive Statistics

In Table 1, we describe the 25 markets that make up the market-quarter panel. Market size is measured as annual cancer incidence from the SEER program of the National Cancer Institute, whose site groupings align closely with our 27 parent indications. Across the 25 markets, 1,716 distinct compounds ran 3,966 drug-market development programs from 2006 through 2019, a compound developed for several cancers counting once in each, and only 20% of programs reached Phase III and 7% reached approval. Revenue is concentrated, with the top seven markets carrying $15.0 billion of the $19.1 billion in mean quarterly sales, but entry and deal activity track development rather than revenue. The acute leukemias, for example, have 70 entries and 105 deals on $0.1 billion of quarterly revenue, while breast cancer, the largest revenue market, has 79 entries. This divergence between revenue and development activity is one reason that we include the match-quality and crowding controls described in Section 6.

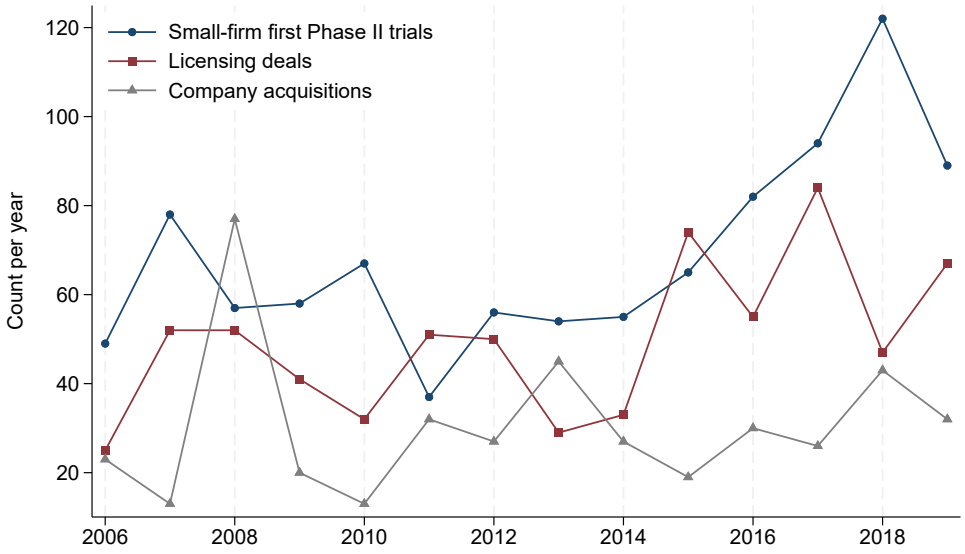
In Figure 4, we plot small-firm entries and deals by year. Entry rises throughout the sample period, roughly doubling from the early 2010s to the late 2010s. Licensing deals are the more common form of acquisition in all years, and they move together with entry.

Table 1: The 25 markets: development, entry, deals and revenue, 2006–2019

Market	Drugs in development				Entries	Deals		
	All	Ph. II	Ph. III	Appr.	Small firm	Lic.	M&A	Sales (USD bn)
Breast Cancer	353	286	72	30	79	47	32	3.78
Multiple myeloma	208	131	27	11	44	32	26	2.29
Non-small cell lung cancer	410	362	94	28	103	65	38	1.89
Colorectal cancer	244	194	34	14	52	37	31	1.87
Non-Hodgkin's lymphoma	252	188	43	23	67	52	36	1.84
Chronic leukemias	151	121	36	18	34	34	19	1.73
Prostate Cancer	243	193	47	13	53	43	26	1.57
Neuroendocrine tumors	56	39	15	8	13	15	6	0.83
Melanoma	215	166	41	12	57	30	20	0.63
Renal cell cancer	125	111	27	14	33	21	10	0.59
Brain cancer	170	126	25	2	44	38	11	0.34
Female reproductive cancers	258	207	43	9	60	46	37	0.25
Sarcoma	93	79	17	6	25	19	10	0.24
Gastrointestinal stromal tumor	24	21	9	3	2	3	3	0.23
Head and neck cancer	145	123	28	6	49	18	11	0.20
Hepatobiliary cancers	147	124	35	8	31	19	12	0.17
Acute leukemias	279	180	52	20	70	63	42	0.13
Pancreatic cancer	198	160	38	3	48	33	12	0.11
Small cell lung cancer	93	80	15	4	19	15	10	0.09
Gastro-esophageal cancers	120	108	34	6	23	21	9	0.09
Hodgkin's lymphoma	47	40	5	3	17	7	7	0.07
Non-melanomatous skin cancers	32	25	7	6	7	3	7	0.05
Cutaneous lymphomas	41	37	13	8	14	18	5	0.05
Thyroid cancer	27	26	11	8	11	6	4	0.03
Mesothelioma	35	30	11	1	8	7	3	0.02
Total	3,966	3,157	779	264	963	692	427	19.11

Notes: The 25 markets of the market-quarter panel, 2006q1 to 2019q4, sorted by revenue. Drugs in development are distinct compounds in the analysis sample holding a Phase I or later record in the market during the window; the phase columns count those reaching at least Phase II, at least Phase III, and approval in that market by 2019q4. Entries are first Phase II trials by small firms, with a trial naming several markets counted in each. Lic. is control-transferring licensing deals and M&A company acquisitions of drugs with a record in the market; a deal for a drug developed in several markets appears in each. Sales are mean quarterly US revenue of approved drugs in the market, indication-weighted, USD billions, 2015q1 dollars. A drug and a deal can appear in more than one row, so the total row exceeds the count of distinct drugs and deals.

Figure 4: Small-firm entry and acquisitions of control rights by year



Notes: Annual counts over the 25 markets of the market-quarter panel. Entries are first Phase II trials by small firms, each named market counted. Licensing deals are control-transferring exclusive licenses, and company acquisitions are whole-company M&A of firms with a drug in the market. A deal for a drug developed in several markets is counted in each.

We define the two remaining samples, the deal sample and the entry sample, as follows. The deal sample consists of the 580 control-rights acquisitions that are reachable by at least one analysis, of which 410 are exclusive licenses and 170 are whole-company acquisitions, covering 659 distinct drugs and 330 acquiring firms. Of these 580 deals, 385 enter an estimation sample. The entry sample consists of 1,195 cases, made up of 633 compound by named-market entries and 562 compounds that completed Phase I but never started a Phase II trial. The choice set for these cases is the 27 indications plus the outside option of not proceeding.

4 Model

In this section, we develop a model of entry in which the prospect of acquisition is priced by an auction among incumbents. For every market it could enter, a potential entrant compares two payoffs in expectation. The first is the value of developing and commercializing the compound alone. The second is the price an acquirer would pay for it, weighted by the probability that a sale occurs. The purpose of the model is to express the acquisition payoff as a function of the set of incumbents that are permitted to bid, which provides a defined route through which a policy that restricts that set reaches the entry decision.

Entrants make three decisions in the model. First, at $t = 0$ the entrant chooses the market in which

to begin Phase II, or it chooses not to proceed. Second, at each later date the entrant chooses whether to initiate a sale process, which is costly. Third, conditional on a process, the incumbents that have engaged with the entrant submit sealed bids, and the entrant accepts the highest bid if it clears the value of continuing alone. We solve the model backward. In Section 4.1, we define the primitives. In Section 4.2, we take a sale process as given and derive the price, and in Section 4.3 we derive the conditions under which an entrant runs a process. We then return to the entry decision in Section 4.4, and in Section 4.5 we state the three results that the counterfactual relies on.

4.1 Definitions and Setting

4.1.1 Agents, Markets, and Timing

We partition firms into two mutually exclusive sets. The first set consists of potential entrants, which are small biotechnology firms holding a compound that has completed Phase I trials. These firms are indexed by $i \in \mathcal{I}$.[3] The second set consists of incumbents, the large firms of Section 3.4, indexed by $k \in \mathcal{K}$. Incumbents are capable of acquiring development-stage assets and hold the means for commercialization.

We define markets as cancer indications, indexed by $j \in \mathcal{J} = \{1, \ldots, J\}$, and we denote the outside option of not entering any market by $j = 0$. Time is discrete and indexed by $t \in \{0, 1, \ldots, T\}$. Here, $t = 0$ designates the entry decision period, in which the entrant commits to begin Phase II trials in a chosen indication, and T is the expected date of the regulatory decision. We let d_{it} denote the development phase of compound i at date t. Finally, all firms share a common discount factor $\delta \in (0, 1)$.

Market j at date t is characterized by a vector ψ_{jt} that is known to all firms. This vector collects the market's revenue M_{jt}, the stock of compounds in development by phase and by owner, and the indication's scientific promise. Entry into market j requires a fixed cost F_j, which we assume is common across potential entrants, and development from t to $t + 1$ requires a flow cost c_{it}.

4.1.2 Asset Match Quality

Each compound has quality θ_{ij} that is specific to the drug-indication pair. The quality parameter is a match term that reflects how well the compound's target and mechanism of action satisfy the biology of indication j. For example, the same compound may have potential in pancreatic cancer but not be indicated in

[3] Potential entrants are indexed at the drug level, since the overwhelming majority of these firms hold a single product. A multi-product firm is treated as several single-compound firms.

leukemia. The originating biotechnology firm and, upon disclosure, any counterparty assess a noisy signal of θ_{ij} from the compound's mechanism and the clinical evidence accumulated through date t, and expectations $\mathbb{E}_t[\cdot]$ are taken with respect to that information. The probability of eventual regulatory approval is $\pi_j^{\text{app}}(\theta_{ij})$, and conditional on approval, the firm that retains the asset earns V_{ij}^{own}, the discounted value of self-commercialization. We construct the empirical counterpart of θ_{ij} in Section 6.5.

4.1.3 The Acquirer Pool and Its Types

We restrict attention to acquisitions by incumbents.

Assumption 1 *Only incumbents may acquire a biotechnology compound, and the identity of the incumbent set is common knowledge.*

Incumbency is time-specific, and all incumbents are able to bid in all markets. Let

$$\mathcal{K}_t = \{k : k \text{ is an incumbent active in oncology at date } t\}, \qquad K_t = |\mathcal{K}_t|, \tag{1}$$

denote the set of potential acquirers. We intentionally do not restrict the pool to firms that are already present in market j, given that incumbents acquire compounds in indications where they hold no assets. The composition of the pool, however, does vary by market. Let

$$\mathcal{O}_{jt} \subseteq \mathcal{K}_t \tag{2}$$

denote the incumbents holding an overlapping position in indication j, meaning a Phase II, Phase III or NDA-stage development asset, or an approved product still on patent. Let $H_{jt} = |\mathcal{O}_{jt}|$, and let $L_{jt} = K_t - H_{jt}$ be the number of incumbents without such an overlap. This distinction is central to the counterfactual policy. The firms in $\mathcal{O}_{jt}$ are the ones that could conduct a killer acquisition, and they are the firms prohibited from bidding in indication j under the policy we study.

Throughout the paper, we refer to the incumbents in $\mathcal{O}_{jt}$ as high types and to the remaining incumbents as low types. An incumbent's valuation of compound i is private information, and we assume that the two types draw their valuations from different distributions.

Assumption 2 *Conditional on $(\theta_{ij}, \psi_{jt}, d_{it})$, incumbent valuations are independent across incumbents. A high type draws its valuation from $F_H(\cdot \mid \psi_{jt})$ and a low type from $F_L(\cdot \mid \psi_{jt})$, where F_H first-order stochastically dominates F_L.*

There are two reasons to expect the overlapping incumbents to value a compound more highly than the other incumbents. The first is that a firm with an established commercial and clinical presence in the indication is able to spread the fixed costs of a launch over an existing organization. The second is the incentive to replace the revenue of a product that is approaching its loss of exclusivity, a mechanism documented by Higgins and Rodriguez (2006). Both motives are tied to the position that the counterfactual restricts, and for this reason we do not separate them in the model and instead treat them together as the type. We provide evidence for the exclusivity motive in Section 5.2 and for the overlap premium in prices in Section 5.3.

4.1.4 The Sale Process

A sale process is costly, as it requires engaging an advisor, preparing diligence materials, and disclosing clinical information to potential rivals (Section 2.2). We model this as a market-invariant fixed cost $\kappa > 0$, which is incurred whether or not a transaction eventually results. The entrant decides whether to bear this cost, and when it does so. Conditional on a process being initiated at date t, the sale proceeds in four steps, which we describe in turn.

First, incumbents decide whether to engage with the process. Evaluating a clinical-stage asset requires costly diligence, and we assume that an incumbent undertakes this diligence only when the expected return exceeds the cost. We let $\mathcal{E}_{it} \subseteq \mathcal{K}_t$ denote the set of engaged incumbents, meaning those that open diligence on compound i at date t, with $m_{it} = |\mathcal{E}_{it}|$.

Assumption 3 *Each pool member engages independently with probability $p \in (0, 1]$, common across types and invariant to the pool.*

The engaged count and its composition are therefore

$$m_{it} = m_{it}^H + m_{it}^L, \qquad m_{it}^H \sim \text{Bin}(H_{jt},\, p)\,, \qquad m_{it}^L \sim \text{Bin}(L_{jt},\, p)\,. \tag{3}$$

A pool member engages when the option value of learning its valuation exceeds the cost of diligence, and we treat the engagement rate p as a primitive. Equation (3) is important for what follows, as it is the channel

through which the counterfactual affects the price. A restriction that removes $\mathcal{O}_{jt}$ from the set of acquirers has two effects. First, it reduces the expected number of engaged parties by pH_{jt}. Second, it removes the high-type draws from the upper tail of the value distribution.

Second, an engaged party may exit the process before it forms a valuation. Diligence proceeds in rounds, and a party that drops out early is observed to do so by the other parties through the banker. We refer to this exit as the diligence screen, and we assume that engaged parties pass the screen independently, with probability $\sigma \in (0, 1]$. We let $n_{it} \leq m_{it}$ denote the number of parties that remain past the screen.

Third, the parties that remain past the screen learn their valuations v, drawn as in Assumption 2, and decide whether to bid. The entrant, in turn, compares any offer against the value of holding the asset for one more period. We let W_{it} denote the value of holding compound i at date t with the option to initiate a process, and we write

$$R_{it} = -c_{it} + \delta \, \mathbb{E}_t\big[W_{i,t+1}\big], \tag{4}$$

the net cost of one more period of development plus the discounted value of holding the asset into the next period. An offer must clear R_{it} in order to be accepted, and as a result a party bids only when $v \geq R_{it}$. A party that draws a value below R_{it} walks away, and this exit, unlike the diligence screen, is not observed by its rivals.

Fourth, the parties that choose to bid submit sealed offers b_{kit}, and the entrant observes the highest of these offers, $b_{it}^{(1)}$. The entrant sells at $b_{it}^{(1)}$ if that offer exceeds R_{it}, and it retains the asset otherwise. We assume that both R_{it} and the number of parties past the screen, n_{it}, are common knowledge among those parties, while the bids and the bid-stage exits are not.

The probability that an engaged party prices the asset is $\omega_{it} = \sigma\,(1 - F(R_{it}))$, the product of the two pass rates. We measure this product with the merger filings of Section 5.4.

4.2 Bidding

Throughout this subsection we take as given that a process has been initiated at date t, that m_{it} parties engaged, and that n_{it} of them passed the screen.

We first consider the case of a single acquirer type, in which all of the parties past the screen draw from a common distribution F. A bidder holding valuation $v \geq R_{it}$ wins when all of its rivals either abstain or bid

less than it does, and both of these events are contained in $F(v)$. The bidder's expected surplus is therefore $(v-b)\,F(v)^{n_{it}-1}$, and the symmetric equilibrium bid solves

$$b(v) = v - \frac{1}{F(v)^{n_{it}-1}} \int_{R_{it}}^{v} F(x)^{n_{it}-1}\,dx. \tag{5}$$

Importantly, bid shading depends on n_{it}, the number of parties past the screen, and not on the number that ultimately submit an offer. A bidder does not observe which of its rivals drew valuations below R_{it}, and it therefore shades against all of the parties that remained past the screen. A party that is in fact alone past the screen is not told so. Exits at the screen, by contrast, are observed by the rivals and reduce the field directly. Because shading falls as n_{it} rises, a smaller number of parties past the screen increases the gap between valuations and the prices that we observe.

With two types of potential bidders, the bid function is no longer symmetric, as high-type and low-type incumbents draw from F_H and F_L respectively and best respond to the composition of their rivals. We let n_{it}^H and n_{it}^L denote the numbers of high and low types past the screen, with $n_{it} = n_{it}^H + n_{it}^L$, and we let f_H and f_L be the densities of F_H and F_L. Both types bid according to a strictly increasing function, and the equilibrium is characterized by the inverse bid functions $\phi_H(b)$ and $\phi_L(b)$, which give the valuation of a bidder of the given type that bids b. A bidder's first-order condition equates the marginal cost of a higher bid to the marginal gain from beating the rivals it faces, and the pair of inverse bid functions therefore solves

$$\begin{aligned}
\frac{1}{\phi_H(b)-b} &= \left(n_{it}^H - 1\right) \frac{f_H\big(\phi_H(b)\big)\,\phi_H'(b)}{F_H\big(\phi_H(b)\big)} + n_{it}^L \frac{f_L\big(\phi_L(b)\big)\,\phi_L'(b)}{F_L\big(\phi_L(b)\big)}, \\
\frac{1}{\phi_L(b)-b} &= n_{it}^H \frac{f_H\big(\phi_H(b)\big)\,\phi_H'(b)}{F_H\big(\phi_H(b)\big)} + \left(n_{it}^L - 1\right) \frac{f_L\big(\phi_L(b)\big)\,\phi_L'(b)}{F_L\big(\phi_L(b)\big)},
\end{aligned} \tag{6}$$

with boundary conditions $\phi_H(R_{it}) = \phi_L(R_{it}) = R_{it}$ at the reserve and a common maximum bid at the top of the two supports. Setting $F_H = F_L$ collapses both lines to the symmetric first-order condition whose solution is equation (5). The system has no closed form, but the solution exists, is unique, and is solved numerically (Lebrun, 1999; Maskin and Riley, 2000, 2003; Hubbard and Paarsch, 2014). Because each type shades against the realized composition (n_{it}^H, n_{it}^L), we assume that composition, and not only the count, is common knowledge among the parties past the screen.

Two properties of the equilibrium are important for the counterfactual. First, shading falls as the number of parties past the screen rises. Second, the expected winning bid rises when a high type replaces a low

type among them. In the symmetric treatment that we use in the estimation, both properties follow from the results in Appendix D. In the asymmetric equilibrium of (6), the first property holds generally, while the second holds in the typical case but is not a theorem, as a low type facing stronger rivals bids more aggressively than a high type with the same valuation (Maskin and Riley, 2000).

An entrant sells the asset when at least one engaged party prices it at or above R_{it}, and it retains the asset otherwise. Conditional on a process with m_{it} engaged parties, the probability of a sale is given by

$$\Pr\big(\text{sale} \mid \text{process}, m_{it}\big) = 1 - \big(1 - \omega_{it}\big)^{m_{it}}, \tag{7}$$

which reduces to $1 - F(R_{it})^{m_{it}}$ when there is no screen. Equation (7) also describes the selection that the estimation sample is subject to, given that we observe a transaction price only for a process that produced a sale. We derive the correction for this selection in Appendix F.

4.3 Initiating a Sale Process

We next turn to the biotech's decision of whether and when to bear the cost κ of running a sale process. Running a process at date t yields the best offer when that offer clears R_{it}, and it yields the asset otherwise, which means that the value of holding the compound satisfies

$$W_{it} = \max\Big\{ \underbrace{\mathbb{E}_t\big[\max\{b_{it}^{(1)},\, R_{it}\}\big] - \kappa}_{\text{initiate at } t},\;\; \underbrace{R_{it}}_{\text{wait}} \Big\}, \tag{8}$$

with R_{it} given by equation (4). At $t = T$ the compound has reached a regulatory decision and no option remains, which gives $W_{iT} = \pi_j^{\text{app}}(\theta_{ij})\, V_{ij}^{\text{own}}$.

We define the option value of running a process,

$$S_{it} \;\equiv\; \mathbb{E}_t\big[\max\{b_{it}^{(1)},\, R_{it}\}\big] - R_{it} \;=\; \mathbb{E}_t\big[(b_{it}^{(1)} - R_{it})^{+}\big]. \tag{9}$$

Because the biotech keeps the asset whenever the best offer falls short of R_{it}, we always have $S_{it} \geq 0$. Comparing the two branches of (8) then gives us the condition under which a process occurs.

Proposition 4.1 *The biotech initiates a sale process at date t if and only if $S_{it} \geq \kappa$.*

Proposition 4.1 states that a process runs when the expected upside over holding the asset covers the

cost of soliciting one. We use three implications of this proposition in our analysis.

First, the static auction is nested in the model as a special case. When $\kappa = 0$, the condition holds in all states because $S_{it} \geq 0$, which means that all assets are marketed at all dates and the only reason a sale fails is a low realization of $b_{it}^{(1)}$. A first-price auction with a reserve price and no initiation decision is therefore the $\kappa = 0$ case of this model, and the initiation channel matters economically only to the extent that κ is large.

Second, the acquirer pool enters the initiation decision. S_{it} is an integral over the upper tail of the winning-bid distribution,

$$S_{it} = \int_{R_{it}}^{\infty} \left(1 - G_{it}^{(1)}(x)\right) dx, \tag{10}$$

where $G_{it}^{(1)}$ is the distribution of the maximum bid. Removing the high-type incumbents compresses the upper tail of this distribution, which lowers the option value of running a process. By Proposition 4.1, it therefore also shrinks the set of states in which the entrant chooses to run a process at all.

Third, the probability of acquisition factors into two pieces:

$$\pi_{it}^{A} = \underbrace{\Pr\left(S_{it} \geq \kappa\right)}_{\text{a process runs}} \times \underbrace{\Pr\left(\text{sale} \mid \text{process}\right)}_{\text{the best offer clears } R_{it}}. \tag{11}$$

4.4 Entry and Market Choice

We now return to the entry decision at $t = 0$. At this date, an entrant holds a compound that has completed Phase I and assesses its quality from the Phase I evidence. The entrant then chooses an indication in which to begin Phase II, or it chooses not to proceed and exits. Entering market j requires the fixed cost F_j and provides the entrant with the value of holding a compound in that market, so that its expected profit is

$$\Pi_{ij} = W_{i0}\left(\theta_{ij},\, \psi_{j0},\, \mathcal{O}_{j0};\, \mathcal{K}_0\right) - F_j. \tag{12}$$

The entrant enters market $j^* = \arg\max_{j \in \mathcal{J}} \Pi_{ij}$ when $\Pi_{ij^*} > 0$, and selects $j = 0$ otherwise. The match θ_{ij} varies across indications for the same compound.

Solving (8) forward and substituting into (12) separates the two routes by which a compound generates

value,

$$\Pi_{ij} = \mathbb{E}_0\Big[\underbrace{\pi^A_{ij}\, A_{ij}}_{\text{acquisition}} + \underbrace{\big(1-\pi^A_{ij}\big)\,\pi^{\text{app}}_j(\theta_{ij})\,\delta^T V^{\text{own}}_{ij}}_{\text{self-commercialization}}\Big] - F_j, \tag{13}$$

where π^A_{ij} is the cumulative probability that the compound is acquired over its development horizon and A_{ij} is the expected transaction price conditional on a sale. Both terms depend on the composition of the pool in market j, and this dependence is the channel through which a restriction on the acquirer pool affects the entry decision. A policy that removes the high-type incumbents from market j lowers A_{ij} through the bidding stage of Section 4.2, and it lowers π^A_{ij} through both factors of (11). Importantly, neither channel changes π^{app}_j or V^{own}_{ij}, which means that a restriction on the acquirer pool changes the acquisition payoff without changing the value of developing the compound independently.

Because a restriction binds in some markets and not in others, the model predicts reallocation as well as deterrence. A restriction that binds in market j lowers Π_{ij} without lowering it elsewhere, and compounds at the margin therefore move toward indications where the acquirer pool is unaffected. Whether aggregate entry falls depends on how many compounds are near $\Pi_{ij^*} = 0$, as opposed to near the boundary between two markets.

4.5 How a Restriction on the Pool Reaches Entry

In this subsection, we state three results that are relevant to the counterfactual policy of restricting the incumbents in $\mathcal{O}_{jt}$ from bidding, with proofs in Appendix D. All three results treat the parties past the screen as symmetric, with all of them drawing from the mixture of F_H and F_L implied by the pool, which is the case that we estimate and simulate in Section 6.

Proposition 4.2 *Entry value is nondecreasing in the acquirer pool. Adding an incumbent to $\mathcal{K}_t$ weakly raises Π_{ij}, and the increase is weakly larger when the added incumbent is a high type.*

The first part of the proposition is a standard result. The second part is the one that matters for the counterfactual, given that the policy we study does not remove a randomly chosen incumbent. Instead, it removes the incumbents that hold an overlapping position in the indication, which Assumption 2 identifies as the firms with the highest valuations. Removing these firms therefore truncates the value distribution from above, as opposed to lowering the number of draws uniformly.

Proposition 4.3 *Consider a restriction that removes $\mathcal{O}_{jt}$ from the pool in market j. Under Assumption 3, the expected number of engaged parties falls by pH_{jt}, and Π_{ij} weakly falls in every market where $H_{jt} > 0$ and is unchanged elsewhere.*

In Proposition 4.3, we hold the engagement rate at its baseline value under the restriction. If the rate were instead to adjust until the remaining low types earned zero surplus from engaging, then a banned high type would be replaced by more than one low type, and the engaged count would fall by less than pH_{jt}. We derive this replacement ratio in Appendix E. Holding the rate fixed therefore gives us an upper bound on the participation channel.

Proposition 4.4 *Let $\Pi_{ij}^{auction}$ denote entry value when $\kappa = 0$, in which case all assets are marketed at all dates. Then $\Pi_{ij}^{auction} \geq \Pi_{ij}$. Holding R_{it} fixed, a restriction that removes $\mathcal{O}_{jt}$ weakly lowers S_{it} at all dates, and the set of states in which a process runs therefore weakly shrinks. Under $\kappa = 0$ a process runs in all states, which means that $\Pi_{ij}^{auction}$ responds to the restriction only through the sale stage, and a counterfactual that holds the initiation factor of (11) at its baseline understates the fall in the acquisition payoff.*

We rely on Proposition 4.4 when we implement the counterfactual. The hazard that we measure is the product in (11), and the counterfactual moves only its second factor and the price, holding the initiation factor fixed. The proposition tells us the direction of the error from this omission. As a result, the entry effects that we report are conservative with respect to the initiation channel.

5 Descriptive and Reduced Form Evidence

In this section, we provide evidence for the assumptions underpinning the model in Section 4. These assumptions are required for the estimation in Section 6. The first, Assumption 1, is that only incumbents acquire development-stage compounds and that the identity of the incumbent set is known to potential entrants. Section 5.1 measures that set and shows that it is small. The second, Assumption 2, is that incumbents holding an overlapping position in an indication draw their valuations from a distribution that first-order stochastically dominates the one other incumbents draw from. We address this in two parts, with Section 5.2 taking up the exclusivity motive behind the ordering and Section 5.3 taking up the ordering itself. The third, Assumption 3, is that each member of the pool engages with a given sale independently and with a probability common across types. Section 5.4 measures that engagement rate from merger chronologies.

We then provide reduced form evidence in Section 5.5 which estimates the share a newly approved product takes from the incumbents already selling in a cancer. The share is the quantity from which the self-commercialization payoff V^{own} is built in Section 6.4.

5.1 Who Acquires Development-Stage Oncology Drugs

The first assumption the model makes is Assumption 1: all potential entrants know the set of incumbent firms K_t capable of acquisition. These firms must have the resources to make acquisitions and the expertise to gain from acquisitions. Table 2 summarizes the deal sample of Section 3.2 by acquirer type: 580 control-rights acquisitions between 2006 and 2019, comprised of 410 product-level acquisitions and 170 company acquisitions, covering 659 unique compounds, with some compounds acquired multiple times. There are 19 Big Pharma firms and 20 Large Biotech firms in our sample. In contrast, the remaining set of firms that make at least one acquisition in our sample is comprised of 294 distinct entities, as shown in Figure 5.

Table 2 establishes that the set of Big Pharma and Large Biotech firms, which we refer to as *large firms*, feasibly constitute the set of potential acquirers. While the pure count of the number of deals presents a less compelling metric with large firms comprising 220 of 580 (37.9%) of all deals, the dollar value share presents a different picture. Large firms constitute 72% and 79% of the upfront value and total value, respectively, of exclusive licensing deals, 89% of company acquisition value, and 87% of deal value across both categories. Looking at mean deal values, this gap is also evident, with mean whole-company deal values $3.3 billion and $1.7 billion higher for Big Pharma and Large Biotech firms, respectively, compared to other deals.

Big Pharma firms are less likely to acquire Phase II/III assets than the residual firms, consistent with a story that the residual firms are not investing as speculation on promising technology, but due to financial pressures to continue with existing research and development. This is then driven by the high level of deals at these stages from the residual firms, given that the plurality of deals for Big Pharma firms still occurs at this stage. Given the differences in prevalence, volume, and frequency, the typical entrant will have reasonable expectations that if an acquisition were to occur, one of the large firms is by far the most likely to do so.

Table 2: Acquisitions of rights to development-stage oncology drugs, by acquirer type

		Large firms			Difference from others	
	All	Big Pharma	Large Biotech	Others	Big Pharma	Large Biotech
Acquisitions of control rights, 2006q1 to 2019q4						
Distinct drugs	659	220	140	361		
Distinct deals	580	149	71	362		
Exclusive licenses	410	92	33	287		
Whole-company acquisitions	170	57	38	75		
Distinct acquiring firms	330	19	20	294		
Deals entering an estimation sample	385	149	60	178		
Share of Phase II+ events	1.00	0.25	0.19	0.57		
Sample drugs per deal (mean)						
Exclusive licenses	1.2	1.3	1.3	1.2	0.1	0.1
					(0.1)	(0.1)
Whole-company acquisitions	2.6	3.2	3.7	1.7	1.6***	2.1***
					(0.5)	(0.5)
Phase at the deal (shares)						
Preclinical or IND	0.14	0.18	0.14	0.11	0.07***	0.03
					(0.03)	(0.03)
Phase I	0.32	0.35	0.31	0.30	0.05	0.01
					(0.04)	(0.04)
Phase II or III	0.49	0.40	0.46	0.56	-0.15***	-0.09*
					(0.04)	(0.05)
NDA review or approved	0.05	0.06	0.09	0.04	0.02	0.05**
					(0.02)	(0.02)
Prices, exclusive licenses (2015q1 USD m)						
Upfront payment, mean	66.7	123.0	78.8	27.9	95.1***	50.9**
					(21.4)	(19.8)
Licences with an upfront reported	211	70	30	111		
Total potential value, mean	425.3	929.8	445.8	139.1	790.6***	306.7***
					(86.6)	(52.4)
Licences with a total value reported	246	77	31	138		
Prices, whole-company acquisitions (2015q1 USD m)						
Total value, mean	1,896.0	3,404.0	1,804.5	124.1	3,279.9***	1,680.4***
					(861.4)	(350.0)
Acquisitions with a value reported	135	55	35	45		

[1] Acquisitions that transfer control rights to a development-stage oncology drug from 2006q1 to 2019q4, in the form of either a whole-company acquisition or an exclusive license or asset purchase conveying US or worldwide rights. Co-development, co-commercialization, reverse mergers and shell transactions are excluded, as are mega-mergers of two large, established, multi-product firms. A deal enters if the drug is between its first Phase I trial and NDA review, or if it is an exclusive license reporting a price.

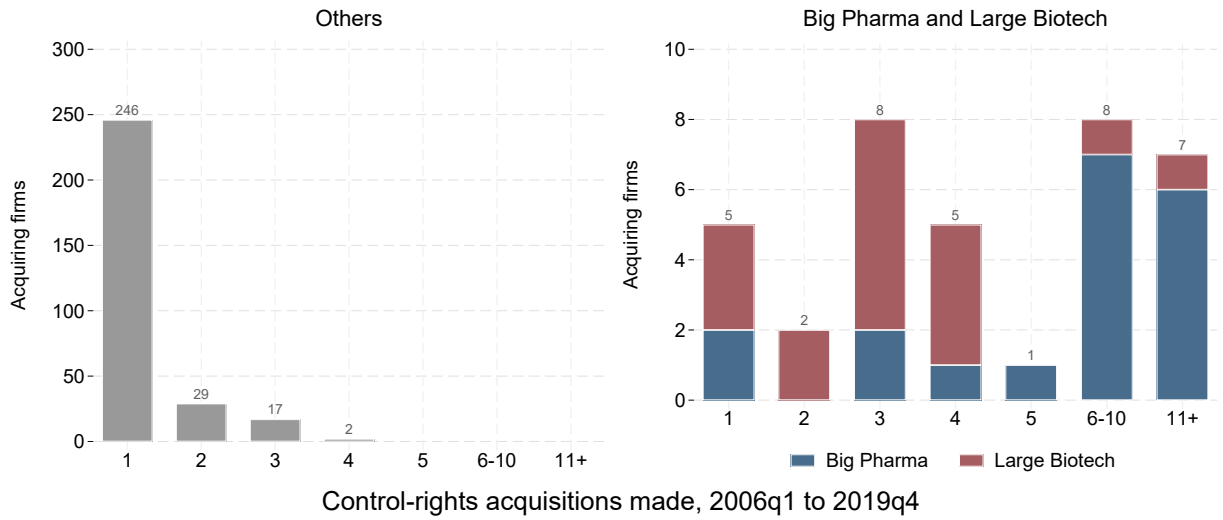
[2] Acquirer type is attributed to the ultimate parent company in the quarter of the deal. Big Pharma firms have about $40 billion of market capitalization, $20 billion of annual revenue, research spending near 15 percent of revenue, and approvals and commercial operations across the United States, Europe, Japan and China. Large Biotech corresponds to at least $15 billion of market capitalization and $2 billion of revenue, a pipeline-driven business, independence from any Big Pharma parent, and commercial presence in more than one major market. The Large Biotech column also holds large firms outside both segments that have an oncology presence and either $500 million of revenue with at least 3 acquisitions here, or $250 million with at least 5.

[3] Prices are deflated to 2015q1 prices and reported in millions, and only where terms were disclosed, so each price row carries its own count. Prices for whole-company acquisitions represent the full acquisition of the target firm. Disclosure is more common among large-firm acquirers, by 37 percentage points (SE 5) conditional on phase, scope, orphan status, drugs per deal and year, so the price gaps below are lower bounds.

[4] The last two columns are the difference between each large-firm group and the residual group, with the standard error of the difference in parentheses. Counts of drugs, deals and firms are taken within each column, so a drug or deal involving two types of acquirer appears in more than one column.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

Figure 5: Only large firms are likely to be repeat buyers



Notes: Each firm that acquired control rights to a development-stage oncology drug between 2006q1 and 2019q4 is counted once, in the bin given by how many it acquired. Deals are those in Table 2. A firm that crosses an incumbency threshold during the window is assigned the highest type it attains, so it appears in one panel only. The left panel is the 290 firms outside both large-firm segments, of which 243 acquired once and none more than four times. The right panel is the 36 Big Pharma and Large Biotech firms. The vertical scales differ between the panels, so heights are not comparable across them. *Source:* Biomedtracker and Evaluate Pharma.

5.2 Revenue at Risk, Acquisitions, and Entry

We next provide evidence suggestive of the exclusivity motive behind Assumption 2. We show that incumbents value a compound in a given indication more when there is revenue at risk in that indication due to an exclusivity cliff. This effect is pooled across all incumbents in the indication, indicative of market-level incentives. Expiry opens the therapeutic space that the expiring product defended to whichever incumbent refreshes it (Caves et al., 1991; Magazzini et al., 2004; Ellison and Ellison, 2011).

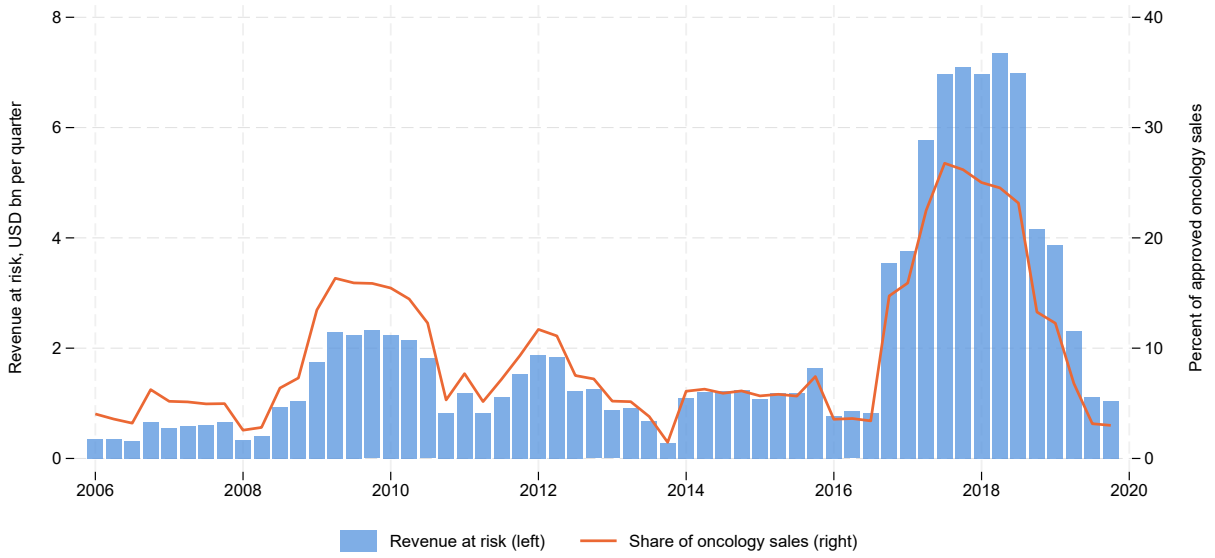
Section 3.6 defines revenue at risk (RAR) in market j and quarter t, RAR^8_{jt}, as the sales of approved drugs in j whose exclusivity ends within eight quarters, in billions of 2015q1 dollars per quarter.[4] Figure 6 shows how RAR moves in aggregate and Figure 7 shows where RAR falls. The revenue cliffs provide rich variation across our sample. The cliffs are large, lumpy, and staggered across cancers. RAR averages \$1.95 billion a quarter across all markets and reaches \$7.4 billion at the 2017 peak. On the other end of the spectrum, there are three quarters which have zero RAR across all cancers and there are four markets that never carry any revenue risk at all.

We estimate

$$\mathbb{E}[N_{jt} \mid \cdot] \;=\; \exp\!\left(\beta\, RAR^8_{jt} + Z'_{jt}\gamma + \alpha_j + \delta_t\right), \tag{14}$$

[4]Table 23 in Appendix G reports robustness checks using RAR measures with different horizons.

Figure 6: Revenue at risk across all 25 markets, by quarter



Notes: Revenue at risk in a market is the sales of approved drugs in that market whose exclusivity ends within eight quarters, with a drug approved in several cancers having its sales split across them as described in Section 3.5. Bars sum revenue at risk over the 25 markets in each quarter, on the left axis. The line is that total as a percentage of the sales of all approved drugs in the 25 markets, on the right axis.

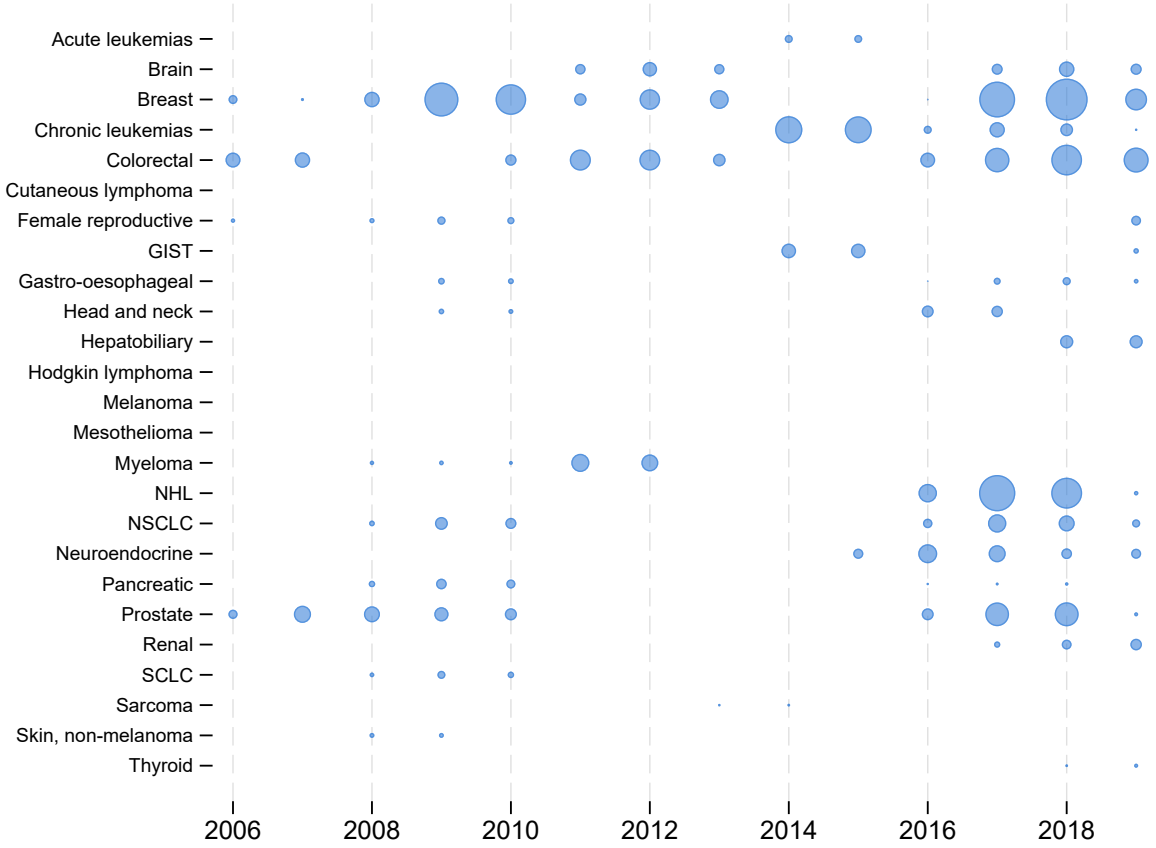
where N_{jt} is either N_{jt}^{acq}, the number of large firm acquisitions in market j and quarter t, or N_{jt}^{ent}, the number of entries into market j in quarter t. The controls Z_{jt} are market revenue M_{jt} and the market's Phase II likelihood of approval π_j^{app}, α_j and δ_t are market and year effects, and standard errors are clustered by market. Because the mean is in log terms, β is a proportional change in the count per billion dollars at risk rather than a number of events.[5]

Figure 8 visualizes the cross-sectional evidence that is estimated in Table 3, Column (1). Without the inclusion of market fixed effects, however, this specification runs the risk of simply measuring market density. That is, markets that are more profitable will persistently have more entry and revenue at risk will drive the relationship. If entry is only responding to market profitability, we would be unable to separately identify the entry incentives of acquisition and market profitability. To address this, Column (2) includes market fixed effects to isolate within-market variation.

Table 3, Column (2) reports regression output with acquisition as the outcome of interest. With market and year effects, $1 billion in revenue at risk increases the incumbent acquisition rate by 0.394 log points,

[5]We estimate by Poisson in place of ordinary least squares in levels for two reasons. First, acquisitions average 0.17 per market-quarter and their base rate varies by an order of magnitude across cancers, which requires a response that scales with the base rate. Second, the counterfactual in Section 7 lowers RAR, and a linear model then predicts negative counts. Least squares places 199 of the 1,400 fitted values below zero for acquisitions.

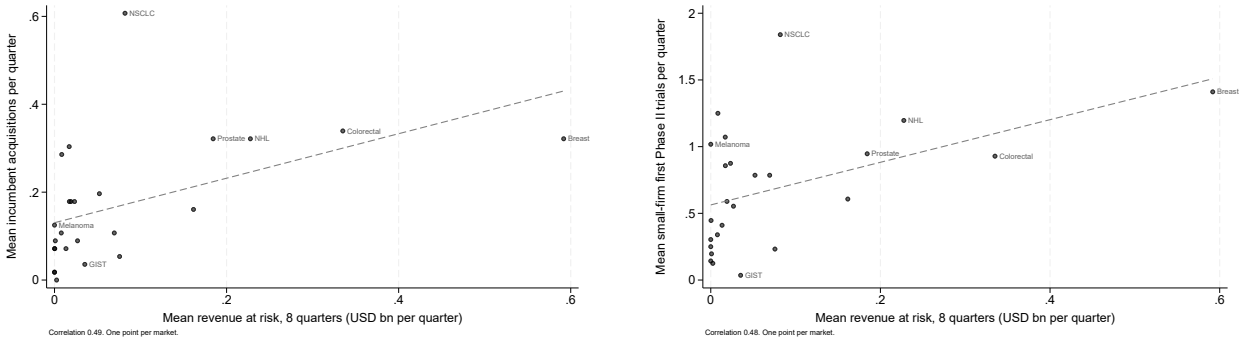
Figure 7: Revenue at risk by cancer and year



Notes: The same quantity as Figure 6, plotted by cancer and year with circle area proportional to revenue at risk. Cancers are listed alphabetically. Blank space is a cancer-year with no approved drug within eight quarters of losing exclusivity, which is most of the panel: 76.5 percent of the 1,400 market-quarters carry no revenue at risk, 21 of the 25 cancers ever carry any, and 104 of the 350 cancer-years do. Conditional on being positive the mean is $0.33 billion a quarter against an unconditional mean of $0.08 billion.

or by 48% over baseline. Column (3) includes a linear trend term to rule out the effect being driven by secular trend of increased entry or acquisition. Inclusion of this trend term lowers the point estimate and precision slightly, but does not meaningfully alter the interpretation. In levels, the marginal effect of a $1 billion increase in revenue at risk is an increase of 0.067 acquisitions per market-quarter against a mean count of 0.17. This measure is difficult to interpret, and most compounds never face $1 billion at risk. A more meaningful measure is that a market moving from no revenue at risk to the average positive amount ($332 million/quarter) sees its acquisition rate rise by 14%.

Figure 8: Markets with more revenue at risk have more acquisitions and more entry



(a) Acquisitions (b) Entrants

Notes: One point per market, 25 markets, averaging over 2006q1 to 2019q4. Revenue at risk is the sales of approved drugs in the market whose exclusivity ends within eight quarters, in billions of 2015q1 dollars per quarter, with a multi-market drug's sales split across its markets as described in Section 3.5. The dashed line is an unweighted least-squares fit. This is the cross-sectional variation that the market fixed effects in the regressions absorb. It motivates the design and does not identify it.

Table 4 reports analogous regression output with entry as the outcome of interest. Entry by non-incumbents increases by 0.232 log points per $1 billion at risk, which only counts trials dedicated to at most three cancers, and by 0.179 when we count every named cancer in a basket trial. Importantly, incumbents' own entry into Phase II trials does not move, with a point estimate of -0.147 statistically indistinguishable from zero. This heterogeneity is informative of several narratives consistent with our model. The first is that incumbent firms are not driven by acquisition incentives in the same way that entrants are motivated. We expect this is because they are not resource-constrained and do not require additional support to profitably commercialize their products. Second, the compounds that appear where a cliff is coming are the ones held by incumbent firms that would have to sell to realize their value.

There are several limitations of this reduced-form design. First, we are unable to separate the acquisition channel from other mechanisms associated with a cliff. For instance, a cliff arrives fifteen to twenty years

Table 3: Revenue at risk and incumbent acquisitions of development-stage drugs

	(1)	(2)	(3)
Revenue at risk, 8 quarters (USD bn)	0.272	0.394***	0.303**
	(0.218)	(0.136)	(0.127)
Controls			
Market sales (USD bn)	0.293***	-0.073	0.091
	(0.101)	(0.182)	(0.235)
Likelihood of approval	-0.371**	0.011	-0.174
	(0.173)	(0.176)	(0.225)
Mean of dependent variable	0.170	0.170	0.170
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	No	Yes	Yes
Market linear trend	No	No	Yes

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales (the sales of approved drugs in the market, USD billions per quarter) and the likelihood of approval. Column (3) adds a market-specific linear time trend.

[2] The dependent variable counts acquisitions by a Big Pharma or Large Biotech incumbent of drugs in Phase II, Phase III or NDA review in the market that quarter: company acquisitions, asset purchases and exclusive worldwide or US licenses, including co-development agreements that carry exclusive rights; one event per acquirer, drug and quarter. The definition is the control-rights definition of the acquirer-types table: non-exclusive co-development and co-commercialization agreements transfer no control and are not counted, and neither are reverse takeovers, in which a private company merges into a listed shell to obtain the listing. Whole-company mergers between diversified firms are also excluded (14 deals, Table 13).

[3] Revenue at risk is the sales of approved drugs in the market whose exclusivity ends within 8 quarters, in USD billions per quarter. A drug approved in several markets has its sales split across them by MarketScan patient-month shares, or by label and analyst shares where the claims data do not identify the market (Appendix A).

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

Table 4: Revenue at risk and Entry

	Small-firm first Phase II trials		Incumbent first Phase II trials
	(1) Dedicated	(2) All	(3) Incumbent
Revenue at risk, 8 quarters (USD bn)	0.232***	0.179**	-0.147
	(0.059)	(0.074)	(0.142)
Controls			
Market sales (USD bn)	-0.043	-0.045	0.175
	(0.080)	(0.116)	(0.130)
Likelihood of approval	-0.116	-0.116	0.354**
	(0.101)	(0.104)	(0.145)
Mean of dependent variable	0.541	0.688	0.296
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	Yes	Yes	Yes

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales (USD bn) and the likelihood of approval.
[2] Entry is a drug's first Phase II trial in any indication, assigned to the market(s) of that trial. Column (1) excludes trials that name four or more markets at once; column (2) counts each named market in full. Small firms are neither Big Pharma nor Large Biotech.
[3] Revenue at risk as in the acquisitions table, 8-quarter horizon.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

after a therapeutic class's first approval, which is also when the follow-on science in that class matures, and generic availability lowers the cost of developing a combination therapy around it. Market and year effects absorb the level and the common trend, but not the within-market correlation. In our structural model, Section 6 separates these channels. Second, this regression approach cannot estimate aggregate entry effects. Because we condition on year effects, we are measuring where entry goes across cancers within a year and not how much entry there is in total. Our structural model is able to recover the aggregate margin, since it places the outside option of developing nothing, which a count regression never observes, in the entrant's choice set. This is shown in Section 7.3.

We provide further robustness checks in Appendix G, all of which are consistent. First, we show that the response is concentrated in the two years before a cliff rather than after it. Revenue that has already lost exclusivity does not move acquisitions. If anything, there is suggestive evidence of a "kill-zone" effect (Kamepalli et al., 2020), where small firms are less likely to enter after a cliff (and presumably an acquisition) occurs, since they now have decreased expectations of acquisition. We also provide consistent results using alternative estimators (ordinary least squares and zero-inflated negative binomials).

5.3 Acquirer Portfolios and Deal Prices

Assumption 2 argues that incumbents with a drug in development for a given cancer indication value compounds in that market more than other incumbents. This assumption separates high type and low type incumbents.

To test this, we study the association between portfolio overlap and acquisition price (i.e., winning bid). Index a deal by d, its drug by i, its acquirer by a and its lead market by j, in the quarter t of the deal, and let P_d be the price in 2015q1 dollars. Let o_{ajt} be the number of drugs the acquirer already holds in the drug's market from Phase II through approval, counting an approved drug only while it remains on patent, and measured the quarter before the deal takes place. Because a drug can be developed for several indications at once, we aggregate o_{ajt} in 2 ways: the sum over the drug's markets, and the largest position the incumbent holds in any one market. We estimate:

$$P_d \;=\; \beta' o_{ajt} + \theta\,\text{LARGE}_{at} + Z_d'\gamma + \alpha_{j(d)} + \delta_t, \tag{15}$$

where LARGE_{at} is an indicator variable for whether the acquirer is a large firm, α_j and δ_t are lead-market and year fixed effects, and Z_d is a vector of deal controls. The controls that we include in this vector are the phase of the compound at the time of the deal, whether the deal is for US or worldwide rights, an indicator variable for the presence of a royalty to the licensee, an indicator for orphan designation, the number of sample drugs in the deal, the acquirer's oncology sales, total sales in the market, the total count of the market's Phase II to NDA pipeline, and the number of markets that the drug is currently in.

In these regressions, we use the sample of deals with observable prices as discussed in Section 3.3. Omitted deals (i.e., those that do not report prices) are likely between smaller non-incumbent firms, so the premia below may be interpreted as lower bounds.

Table 5 reports both prices. On total potential value, each overlapping drug that a high-type incumbent has adds \$74.1 million when the holdings are summed over the drug's markets and \$79.5 million when the largest position in any one market is used instead. On the upfront payment, the summed count gives \$11.1 million and the largest-position count \$7.7 million. The whole-company indicator is \$122 million on total value and \$187 million on the upfront; the upfront premium is mechanical, since a company price has no milestone schedule to strip out, and the total-value premium is modest once the drug and market are controlled.

Table 5: The acquirer's portfolio in the drug's market and the deal price

	Upfront payment		Total potential value	
	(1) Summed Across Markets	(2) Max Across Markets	(3) Summed Across Markets	(4) Max Across Markets
Portfolio overlap, count	11.057*	7.704	74.080**	79.501**
	(5.958)	(7.720)	(31.046)	(35.673)
Whole-company deal	186.516***	187.814***	122.340	131.770
	(45.586)	(47.188)	(132.191)	(136.007)
Acquirer is a large firm	96.695***	115.317***	497.859***	546.194***
	(31.604)	(29.418)	(127.739)	(117.100)
Market sales (USD bn/qtr)	5.580	7.229	-3.286	8.908
	(6.739)	(8.233)	(18.677)	(26.405)
Mean of dependent variable	95.8	95.8	486.4	486.4
Observations	270	270	366	366
R^2	0.50	0.49	0.40	0.38
Year FE	Yes	Yes	Yes	Yes
Lead-market FE	Yes	Yes	Yes	Yes

Dependent variable: the deal price in 2015q1 USD m, the upfront payment in columns 1 and 2 and the total potential value including milestones in columns 3 and 4. One observation per acquirer parent, drug and deal quarter. The sample is the exclusive licenses with a reported price, plus the whole-company acquisitions whose price can be tied to one compound: a single sample drug, or two to three sample drugs with a unique most-advanced drug still in development and a company value under USD 5bn; the whole-company indicator absorbs the deal-form level. The regressor throughout is the number of drugs the acquirer already holds in the drug's market from Phase II through approval, counting an approved drug only while it is still on patent, and measured the quarter before the deal. An expired product is not a franchise the acquirer can defend, which is the position the counterfactual restricts. A drug can be developed for several parent indications, so the acquirer's holdings have to be aggregated across them. The first column of each pair sums the holdings over the drug's markets; the second takes the largest position the acquirer holds in any one of them. Counting approved drugs whether or not they remain on patent gives 10.3 and 67.3 in place of the first and third columns. On the 223 deals for a drug in a single indication (164 on the upfront), where the aggregation choice does not arise, the coefficient is not distinguishable from zero on either price. Estimated with no holdings measure at all the large-firm indicator is 132.6 with a standard error of 26.0 on the upfront and 713.4 with a standard error of 101.2 on total value. Controls: phase at the deal, worldwide rights, royalty in the deal, orphan designation, sample drugs in the deal, acquirer oncology sales (USD bn), market sales (USD bn), Phase II-NDA drugs in the market, number of markets, year and lead-market fixed effects. Standard errors clustered by acquirer parent.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

A high-type incumbent has a sales force, formulary access and prescriber relationships there, and it has a franchise with an expiry date, which Section 5.2 shows is when incumbents buy. Counting only its on-patent approved drugs and nothing in development gives $53.4 million on the upfront, a standard error of 27.1, and $165.7 million on total value, a standard error of 94.1: a larger premium per drug than the broader count, on far fewer deals. Whether approved products are counted only while on patent makes little difference to the broader measure, $11.1 million against $10.3 million on the upfront, because an acquirer with an approved drug in a cancer usually has one still on patent.

5.4 Engaged Bidders in Merger Disclosures

In this subsection, we provide evidence for Assumption 3, which requires that members of the acquirer pool engage with a given sale independently of one another and with a probability that is common across types. In other words, the parties that bid in a given auction are a subset of the full acquirer pool, as opposed to the pool itself. We draw this evidence from the merger disclosures described in Section 3.7, which allow us to observe how many potential acquirers were contacted, how many signed confidentiality agreements, and how many ultimately submitted a bid.

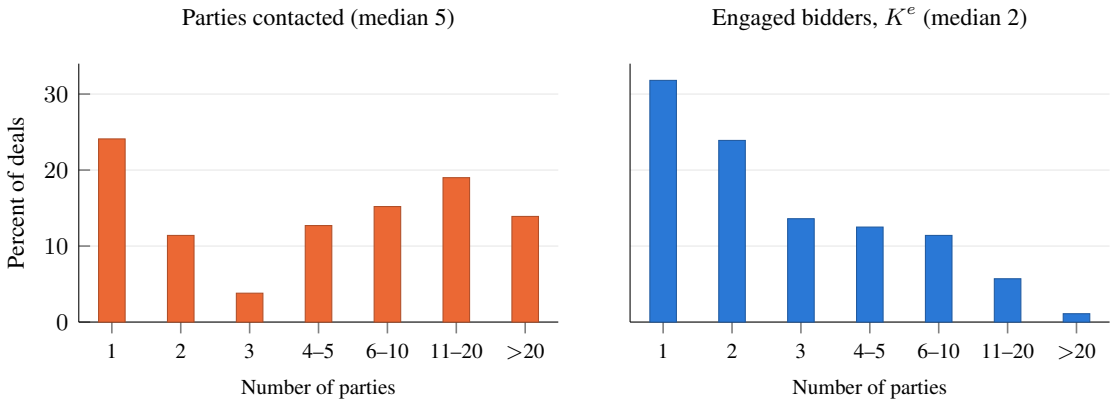
Throughout the model, the sale price that an entrant expects for its own product is a function of the number of incumbents that bid in a first-price auction and the distribution of incumbent private valuations. Indeed, we are explicitly modeling the price formation process with these dynamics. While we have thus far defined the set of large firms, allowing each to be a bidder would result in auctions in which firms bid their private value, and any counterfactual would be hard pressed to increase the bid shading in an economically meaningful capacity. Thus, the model defines a process in Section 4.1.4 in which the full set of potential acquirers, defined as large incumbent firms, is transformed to a set of potential bidders (i.e., engaged) drawing from their valuation and submitting bids.

The merger chronologies of Section 3.7 record all of the parties that the seller's banker approached, as well as those that went on to receive confidential information or to price the asset. Because the losing parties are tracked through the process, we are able to measure participation after engagement and to identify the firms that undertook costly diligence but ultimately chose not to bid. We define the engaged bidders K^e in a sale as the parties that received confidential information under a confidentiality agreement executed for the transaction, or that made a proposal carrying a price, with the acquirer always counted. The number of parties that the banker approached, in turn, provides us with an empirical floor for the potential acquirer

pool.

Figure 9 and Table 6 report the distribution of the sales processes. The number of banker-contacted parties is large, with a median of five parties and a mean of 11.9 contacted. The funnel below this to engaged bidders is thinner. The median sale has two engaged bidders with a mean of 3.8, 75th percentile of five, a 90th percentile of eight, and a maximum of 22. The modal value is one, in 28 of the 88 sales, and 82 percent have five or fewer. From this, a median of two parties bid at all and a median of one submits a final bid.[6]

Figure 9: How many parties engage in a sale



Notes: 88 sales of US-listed oncology targets, 2000 to 2024, coded from the "Background of the Merger" chronology in the target's Schedule 14D-9 or DEFM14A. Left: parties the seller's banker contacted, where the filing states it (79 deals, median 5). Right: engaged bidders, parties that received confidential information under a confidentiality agreement for the transaction or made a price-bearing proposal (median 2, modal value 1, 82 percent at five or fewer).

Table 6: Parties at each stage of the sale and of the parallel licensing canvass

	N	Mean	p25	Median	p75	p90	Max
Sale of the company							
Parties contacted by the banker	79	11.9	2	5	14	30	100
Engaged bidders, K^e	88	3.8	1	2	5	8	22
Parallel licensing canvass, same filings							
Parties contacted	17	18.9		12			100
Informed (confidentiality agreement and information)	19	7.5		5			33
Priced (submitted a term sheet)	18	2.6		2			6

[1] Sales of US-listed oncology targets in the Evaluate M&A sample, 2000 to 2024, with a usable "Background of the Merger" chronology in the Schedule 14D-9 or DEFM14A (88 of 171 candidate targets). An engaged bidder received confidential information under a confidentiality agreement executed for the transaction or made a price-bearing proposal; the acquirer always counts. Fifteen of the 88 are lower bounds where the filing describes informed parties without counting them. Parties contacted is reported where the filing states it.
[2] The licensing rows code the partnering canvass that ran alongside the sale in the same chronologies, where the filing states the stage counts (22 of 95 sections). Six canvasses solicited a sale and a license together; excluding them, the informed maximum is 13.

[6] Fifteen of the 88 counts are floors, where the filing describes informed parties without enumerating them. These counts are flagged and retained, which biases the measured count downward.

These filings are reported only by public companies undergoing a whole-company merger or acquisition. The auction sample of Section 6 contains deals of exactly this form, the 91 whole-company acquisitions whose price can be tied to one compound, so the funnel evidence bears directly on part of the estimation sample. The larger part of that sample remains exclusive licenses, and licensing deals are less costly on average for the acquiring firm, so they may have a lower threshold for engagement and more bidders per process. Two measurements inside the same filings address this possibility. First, twenty-two chronologies in the data describe a partnering canvass running alongside the sale, with a median of 12 parties contacted, 5 informed and 2 pricing the asset. Second, eighteen processes describe a parallel exploration of licensing or asset purchase with a median of 62 contacted, 8 informed, and 2 priced. From these data, we see that licensing draws a wider field at the top of the funnel but the same field at the bottom, and no single-asset process in either measurement has more than eight parties price the asset. Boards themselves treat the partnering canvass as the market check for a sale, which is the micro-foundation for using partnering counts to bound the set.

In Section 6.1, we keep the potential acquirer set and participation as separate objects, and we calibrate the engagement rate to these counts. About a third of the informed parties go on to price the asset (47 of 134 on the sale side), and this is the total walk-away that the model takes as given (Section 6.2).

5.5 Competition Among Approved Drugs

The last input that we require from the reduced-form evidence is the share of a market that a newly approved product takes from the incumbents already selling in that cancer, which we use to build V^{own} in Section 6.4. Unlike the previous subsections, this estimate is an input to the model as opposed to evidence for one of its assumptions.

The expected value of self-commercialization is the market's revenue times this share, and we measure the share in reduced form, as opposed to estimating a structural demand system.

The choice to estimate expected revenue from self-commercialization using a reduced-form approach is informed by two observations. First, the relevant object that is needed in the structural model is entrant revenue expectations over market entry, not contemporaneous realized entry. In speaking with market participants, the reduced-form approach we describe is how entrants describe their own internal estimates. Second, the oncology market has idiosyncrasies which result in difficult-to-interpret parameters from traditional demand estimation. In the oncology market, physicians mediate patient demand, with price-inelastic

preferences. Additionally, oncology treatments are structured as lines of therapy, in which there are formal regimens defined for a given indication that physicians follow. Newly-approved drugs are almost always approved to be the final treatment in the line of therapy, and drugs added to the end of the line may be complements to prior treatment rather than direct substitutes, which is difficult to reconcile in traditional models of demand. Finally, prices are generally determined as a percent over the Medicare Average Sales Price (ASP), meaning they do not directly negotiate drug prices, and the decision-relevant price, net of rebates, is unobserved.

Empirically, price-mediated substitution appears to be absent. In distributed-lag regressions of log price on the number of competitor approvals, the average sales price of infused drugs moves by -0.007 over the twelve quarters after an approval, statistically indistinguishable from zero, and oral list prices keep rising throughout, about one point a year more slowly than they otherwise would (Appendix I). No incumbent cuts price after a competitor product is approved and enters the market.

There is, however, a modest but measurable quantity displacement from entry. Figure 10 and Table 7 report a stacked event study in the construction of Cengiz et al. (2019), one stack per competitor approval in a cancer over a window of eight quarters before to twelve after, so that each event carries its own clean controls and its own clock (Goodman-Bacon, 2021; Sun and Abraham, 2021). Treated incumbents are drugs approved and selling in the cancer at least eight quarters before the event and not sold by the entrant's seller, and controls are incumbents in other cancers over the same quarters. We measure outcomes in logs because the parameter that the entry model needs is the proportional effect: the entrant's share s of Section 6.4 is the incumbents' percentage loss, and the post-by-treated coefficient on a log outcome measures that loss directly. For drug d in the stack built around event e, observed in quarter t, we estimate:

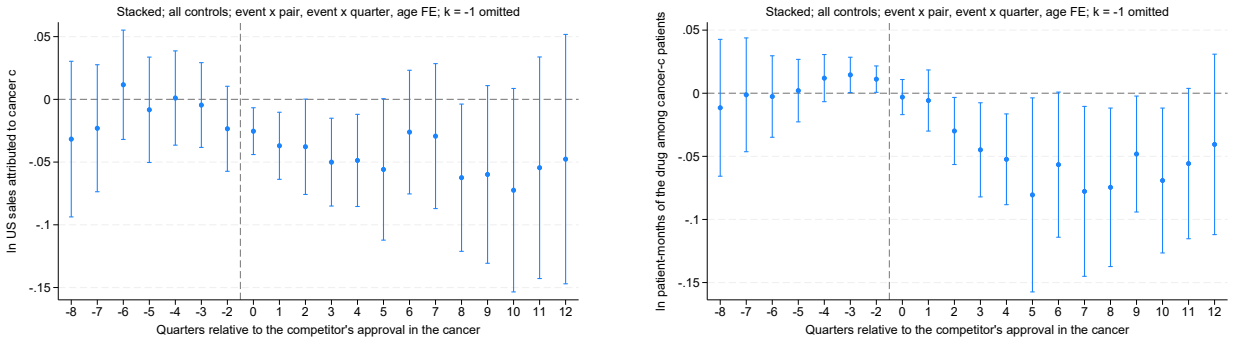
$$y_{det} \;=\; \beta \left(\text{Post}_{et} \times \text{Treated}_{de}\right) + \alpha_{de} + \lambda_{et} + \gamma_{a(d,t)} + \varepsilon_{det}, \tag{16}$$

where y_{det} is the log of attributed sales, of patient-months, or of the drug's share of its cancer's treated patients; Post_{et} marks quarters at or after the event; Treated_{de} marks the incumbents in the event's cancer; α_{de} are event-by-drug effects, λ_{et} event-by-quarter effects, and $\gamma_{a(d,t)}$ age-in-market effects that absorb the launch curve. Standard errors are clustered by drug.

The event-time paths in Figure 10 replace the single post term with indicators for each quarter relative to the event, -8 to 12 with -1 omitted. Important for identification, there is no statistically significant trend

in the quarters before the event. With the full control group, patient-months in the cancer fall by 0.048 log points, the incumbent's share of treated patients falls by 0.031, the total number of treated patients in the cancer does not move at all, and there is no statistically significant decrease in sales. The same regressions in levels agree in sign and implied magnitude, with attributed patient-months falling significantly and the dollar and point-share estimates noisier, because levels weight the largest incumbents (Appendix H).

Figure 10: Incumbent revenue and volume in a cancer around a competitor's approval



(a) Log US sales allocated to the cancer

(b) Log patient-months in the cancer

Notes: Stacked event study, one stack per competitor approval in a cancer, window -8 to $+12$ quarters, 118 events. Treated: incumbents approved and selling in the cancer at least eight quarters before the event and not sold by the entrant's seller. Controls: incumbents in all other cancers over the same quarters. Event by pair, event by quarter, and age-in-cancer fixed effects; standard errors clustered by drug. Sales in the cancer are US sales times the drug's MarketScan share of patient-months in that cancer; patient-months are MarketScan counts. Later approvals in the same cancer are not netted out, so the post-period average is the effect of about two net approvals.

Table 7: Incumbent outcomes after a competitor's approval in the cancer

	ln sales in c		ln patient-months in c		ln share of c		ln total treated in c	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post x treated	-0.118	-0.035	-0.122**	-0.048**	-0.106*	-0.031	0.012	-0.003
	(0.088)	(0.026)	(0.060)	(0.022)	(0.057)	(0.020)	(0.049)	(0.021)
Control cancers	clean	all	clean	all	clean	all	clean	all
Observations	31,394	151,243	31,927	154,556	31,927	154,556	12,043	38,137
Clusters	81	84	80	83	80	83	18	18

Notes: One stack per competitor approval in a cancer (event date: approval), window -8 to +12 quarters. Treated: incumbents approved and selling in the cancer at least 8 quarters before the event, not sold by the entrant's seller. Controls: incumbents in other cancers over the same quarters; clean = no approval in the control's cancer within 4 quarters of the event. Event x pair, event x quarter, and age fixed effects; SEs clustered by drug (columns 1-6) or cancer (7-8; 20 clusters). Sales in c = US sales x Marketscan share of the drug's patient-months in cancer c. * p<0.10, ** p<0.05, *** p<0.01.

Taken together, a competitor's approval reduces an incumbent's use in a cancer by two to three percent, with the reduction coming from the incumbent's share rather than from the number of patients treated.

Given that revenue falls by less than volume, and this change in revenue is indistinguishable from zero, a competitor count is an adequate summary of the competitive effects of an additional entrant.

6 Estimating the Structural Model

This section describes how we measure each object in the model of Section 4, and Section 7 then estimates the two stages and reports the counterfactual. Our estimation imposes one restriction on the model, the symmetric-within-type inversion, and we note where this restriction is implemented. We measure the acquisition hazard as the unseparated product in equation (11), which does not require us to take a stand on the process cost κ, and the counterfactual holds the initiation factor fixed in the conservative direction (Proposition 4.4).

6.1 The Pool and Engagement

We measure the pool of Section 4.1.3 as $\mathcal{K}_t$, the set of large incumbents that hold an oncology asset in quarter t, combining the Big Pharma and Large Biotech firms as defined in Section 5.1. This gives us $K_t = |\mathcal{K}_t|$ between 22 and 26 firms over the sample. We measure the overlap set $\mathcal{O}_{jt}$ using an overlap definition of development assets in Phase II or later or an approved product still on patent, consistent with Assumption 2. $H_{jt} = |\mathcal{O}_{jt}|$ averages nine, and the high-type share H_{jt}/K_t averages 0.36 with wide variation, from around 0.7 in non-small cell lung and breast cancer to near 0.1 in the smallest markets. $L_{jt} = K_t - H_{jt}$ is the permitted pool under the policy ban, with a mean of fifteen and a minimum of three in the sample.

The engaged count follows equation (3). The filings require at least one engaged party, providing us with

$$m_{it} \sim \text{Bin}(K_t, p) \text{ truncated at } m \geq 1, \qquad m_{it}^{\text{cf}} \sim \text{Bin}(L_{jt}, p) \text{ truncated at } m \geq 1, \tag{17}$$

in the baseline and under the ban. We calibrate p to the merger filings of Section 5.4 such that the mean engaged count reproduces the filings' mean of 3.8, which gives $p = 0.16$ at $K_t = 24$. For robustness, we report all of our results over a grid of p that includes the value that reproduces the filings' median of 2 ($p = 0.055$). Equation (17) holds the rate at its baseline value under the ban, consistent with Assumption 3 and Proposition 4.3. Allowing for p to be determined in equilibrium is an area of open work. Such a change would increase the engagement probability for low-type firms, increasing the number of bidders on average

in the counterfactual (Appendix E). Thus, the fixed probability is the case in which the counterfactual has the largest effect.

6.2 Valuations, the Reserve, and Bidding

In this subsection, we describe how we measure three objects from the auction stage of Section 4.2. The first is the seller's reserve, which we measure and which is deliberately slack. The second is the walk-away, whose total the filings fix and whose split between its two margins we vary over a grid. The third is the pair of type-specific valuation distributions, which we recover by inverting the observed bids.

The reserve value of a given small biotech is the continuation value of developing and commercializing the product itself, given by equation (4). We measure it deal by deal as

$$R_i = \phi \, \pi^{\text{app}}_{ij(i)} \, V^{\text{own}}_{j(i)t(i)}. \tag{18}$$

where $\pi^{\text{app}}_{ij(i)}$ is the compound's analyst likelihood of approval in the deal's indication at the deal date, a drug-indication object that proxies for the quality term, θ_{ij}. This is measured as the drug-indication pair's own value where the analysts cover it and the indication mean otherwise. V^{own}_{jt} is the discounted revenue of one more approved product in its market and year (detailed in Section 6.4), and $\phi \in (0, 1]$ is the discount that a small firm applies to commercializing alone, and this discount is the reason that the firm sells at all. We are unable to separately identify ϕ from the entry coefficient on V^{own}, and as a result we do not estimate it. Instead, we use the requirement that an observed sale means that the winning bid beat the seller's accept threshold, which means that the estimated reserve should rarely, if ever, exceed the price actually paid. We set ϕ to the largest value at which the winning bid clears the reserve in 95 percent of the deals by incumbent pool members. This requirement constrains ϕ from one side only. In Appendix K.6, we report where the requirement binds and show that the counterfactual results are not sensitive to this choice.

The measured reserve is small relative to the prices that we observe, and this is an intentional feature of the model. Among deals by the incumbent pool members, the median price is about thirty times $\pi^{\text{app}} V^{\text{own}}$ on total deal value, nineteen times at the risk-adjusted price, and six times on the upfront payment alone (Appendix K.6). In other words, even the only component that is paid with certainty is well above the seller's continuation value. The reserve is intentionally slack because its only role in the model is to serve as the seller's accept threshold. We do not ask it to explain why engaged parties fail to bid, as this is instead

determined by the walk-away margins.

We measure the walk-away in two pieces. Recall from Section 4.1.4 that an engaged party that never prices the target exits either at the diligence screen, before forming a valuation and observed by its rivals, or at the bid stage, after forming a value. There might be multiple, unobservable, reasons that drive the choice to disengage after forming a value: internal operating frictions, the opportunity cost in the acquisition budget, or shocks to firm expectations. We let F denote this bid-stage share, or the probability that a party which passes the screen nevertheless never bids. The filings determine the product of the two margins, since 65 percent of informed parties never price, but the deal prices alone are not able to separate the split between the two channels. For this reason, we carry F as a grid. A bid-stage share of F leaves a screen exit rate of $1 - 0.35/(1 - F)$, and the grid runs from all of the walk-away occurring at the screen ($F = 0$) to all of it occurring at the bid stage ($F = 0.65$, the one-margin model). We invert and simulate at the same split in each cell, with $F = 0.35$ as the working value. The bids themselves constrain the split through a bid-fit check, in which the recovered valuations are pushed back through the bid function and should return the observed bids. A large F implies a pile-up of winning bids just above the no-bid threshold, which the smooth lower tail of the observed bids does not show. A simulation confirms that the check has power, as the fit gap stays small when bids are generated under a true F of any size on the grid, and we report the gap by cell in Section 7.2.

We invert surplus bids, as opposed to the prices themselves. Given that a bidder's valuation is the reserve plus a private surplus, we define the surplus bid as $s_i = b_i - R_i$ and apply the bid function of equation (5) to it. In this bid function, the bid-stage share F enters as a point mass at the bottom of the distribution against which all bidders shade:

$$b(v) = v - \frac{\int_R^v F(x)^{m-1}\,dx}{F(v)^{m-1}}, \qquad F(x) = F(R) + \big(1 - F(R)\big)\,\tilde{F}(x), \tag{19}$$

where $\tilde{F}$ is the distribution of surplus above the reserve and $F \equiv F(R)$ is that point mass. Here and in the rest of this subsection, m is the number of parties past the screen, which is the count n_{it} of Section 4.2, since those are the parties that a bidder shades against. The exits at the screen enter through that count and do not enter through F, because rivals are able to observe them. Given that we observe only winning bids, and only for completed sales, we recover the distribution of a representative engaged party's bid from the winning-bid distribution using the order-statistic identity $\tilde{G}(s)^m = F(R)^m + (1 - F(R)^m)\,G_{\text{win}}(s)$, and

valuations then follow from the first-order condition (Guerre et al., 2000),

$$v(s) = s + \frac{\tilde{G}(s)}{(m-1)\,\tilde{g}(s)}. \tag{20}$$

Please see Appendix F for the derivation of this correction for the reserve.

In order to make deals for different drugs in different markets comparable, we do not invert the raw surplus bids. Instead, we construct a bid index and invert the residuals from it. The index is a regression of the surplus bid on the drug's phase and promise, the deal's form and terms, the market's sales and pipeline, and year and indication effects (the full set of covariates is listed in Appendix J). We intentionally exclude the acquirer's own position in the market from the index, given that the counterfactual acts on this margin. Additionally, because the index is additive, we are able to add a markup recovered on the residual scale back to the fitted bid, which allows us to express simulated prices in dollars.

This inversion has two limitations. First, equation (20) is valid only if the residual dispersion is the private-value dispersion. The bid index explains 39 percent of the variance in surplus bids, and a symmetric independent-private-values model reads the rest as dispersion within an auction, leading us to estimate the scale factor instead of the level of valuations. This is described in more depth in Section 7.2. Instead, the inversion provides us with the shape of the valuation distribution above the reserve, by type, which is all that the two counterfactual ratios require. Second, we recover the types by applying equation (20) to the winning bids of the two types separately, split by the acquirer's type, which is a symmetric shortcut that we take due to data limitations. We only have 28 acquisitions in the inversion sample where the winning bids are by low types. The asymmetric inversion of Campo et al. (2003), in which each type best responds to the other's distribution, is the estimator that a larger sample would support, and it is a place that we highlight for future work.

6.3 The Acquisition Payoff

The expected payoff from the acquisition route to a compound entering indication j in year t is the price an acquirer pays, times the discounted probability that the compound is acquired at some point over the years it stays in development there:

$$\Pi^{A}_{jt} = A_{jt} \sum_{\tau=0}^{T_j-1} \delta^{\tau} (1 - \pi^{A}_{jt})^{\tau}\, \pi^{A}_{jt}, \tag{21}$$

This definition ensures that the payoff is on a comparable scale to self-commercialization. π^A_{jt} is the annual acquisition hazard, defined as the incumbent acquisitions of small-firm compounds in Phase I through NDA review in j during t, over the stock of relevant compounds. This measure is first smoothed via a Poisson regression on the pool mix H_{jt}/K_t, market revenue, the incumbents' revenue at risk from Section 5.2, and indication and year fixed effects. T_j is the median number of years a small-firm program stays in the indication after its first Phase II, 4.3 on average, and δ is the annual discount factor. A_{jt} is a hedonic price, predicted from the deal's and the market's characteristics.[7] We use the total potential deal value as the price measure in order to capture the total commitment, with the upfront payment and the risk-adjusted price of Section 7.2 carried as alternatives.

We summarize the auction's counterfactual adjustments in two ratios. The first, η^A_{jt}, is the ratio of the expected price under the ban to the expected price in the baseline, conditional on a sale. The second, η^π_{jt}, is the corresponding ratio for the probability that any engaged party prices the compound. Both ratios compare the restricted market to the baseline in the same indication and year, and as a result both are numbers close to one that scale a payoff we measure in the data, as opposed to replacing it.

$$\eta^A_{jt} = \frac{\mathbb{E}\big[\, b(v^{(1)}) \mid m^{\text{cf}},\, F_L \,\big]}{\mathbb{E}\big[\, b(v^{(1)}) \mid m,\, F_{\text{mix}}(H_{jt}/K_t) \,\big]}, \qquad \eta^\pi_{jt} = \frac{\Pr\left(m^{\text{cf}} \geq 1 \text{ party bids}\right)}{\Pr\left(m \geq 1 \text{ party bids}\right)}, \tag{22}$$

We simulate both ratios from the recovered type distributions, drawing the engaged count from equation (17). In the baseline, a party is a high type with probability H_{jt}/K_t, while under the ban all parties are low types drawn from the permitted pool (the simulation is described in Appendix K.4). These two ratios are the point at which the counterfactual policy interacts with the auction model. The price ratio η^A_{jt} captures the composition effect, as the remaining bidders draw weaker valuations and shade differently, together with the effect of a smaller field on shading. The probability ratio η^π_{jt} captures the effect of a smaller field on whether anyone prices the compound at all. We then apply these ratios to the payoffs that we measure in the data, which gives us the counterfactual price $\eta^A_{jt}A_{jt}$ and the counterfactual hazard $\eta^\pi_{jt}\pi^A_{jt}$.

[7] The hedonic regression uses all priced deals in the Section 3.3 sample, with the pool mix, market sales, and the deal's own terms as covariates. Predictions are formed at each indication-year's pool mix and market sales, with the deal characteristics held at their sample means. Appendix J reports the specification.

6.4 Self-Commercialization

If the compound is not acquired and is approved, the biotech earns

$$V_{jt}^{\text{own}} = \sum_{\tau=T}^{T+40} \delta^{\tau}\, s\, M_{jt}, \tag{23}$$

where M_{jt} is the US revenue of approved drugs in indication j in year t, s is the share of the market that a newly approved product takes, $T = 24$ quarters is the lag from a first Phase II trial to approval, forty quarters of exclusivity follow approval, and δ is the quarterly discount factor.

We take the entrant's share s from the stacked event study of Section 5.5. In that event study, a competitor's approval lowers an incumbent's share of treated patients by 3.1 percent, does not change the number of patients treated, and does not change incumbent prices, which means that the entrant's revenue is the incumbents' loss, about 3 percent of the market. We also carry two alternative measures throughout the analysis, the clean-control estimate of 10.6 percent and the equal-share benchmark $1/(n_{jt}+1)$ (Appendix J). The approval probability that multiplies V^{own} in the payoff is the realized rate at which Phase II programs in the indication reach approval, which we measure in our data and report in Table 8.

6.5 Entry

We model the entry decision as a conditional logit over the 27 indications and the outside option. A compound i that has completed Phase I chooses an indication for its first Phase II trial, or it does not proceed, and its payoff from indication j in decision year t is given by

$$U_{ijt} = \lambda_1\, q_{ij}\, \Pi_{jt}^{A} + \lambda_2\, q_{ij}\, \pi_j^{\text{app}} V_{jt}^{\text{own}} + \gamma \ln q_{ij} + Z_{jt}'\beta + \alpha_j + \varepsilon_{ijt}, \tag{24}$$

with ε_{ijt} type-I extreme value. Equation (24) is the econometric counterpart of the entry value in equation (13), where the entrant has two possible payoffs: acquisition or self-commercialization. The entry cost F_j of equation (12) is absorbed into the alternative constants α_j. Z_{jt} includes the indication's scientific promise, defined as the number of trials in the prior year run by non-industry sponsors, and a measure of crowding which captures the number of small-firm programs already in Phase II or later in the indication the prior year. A higher crowding measure results in lower expected payoffs from both acquisition and self-commercialization, and we lag it by one year in order to exclude the year's own entrants. α_j are

Table 8: Realized approval rates by indication

Indication	Phase II pairs	Approved (%)
Non-Small Cell Lung Cancer (NSCLC)	233	12.4
Breast Cancer	216	15.7
Prostate Cancer	169	11.8
Female reproductive system cancers	148	8.1
Colorectal Cancer (CRC)	142	9.2
Acute leukemias	134	22.4
Non-Hodgkin's lymphoma	130	20.0
Melanoma	117	12.0
Pancreatic cancer	111	4.5
Multiple Myeloma (MM)	96	18.8
Chronic leukemias	94	23.4
Brain Cancer (Malignant Glioma; AA and gliob	91	4.4
Hepatobiliary cancers	87	9.2
Renal Cell Cancer (RCC)	80	16.2
Gastro-esophageal cancers	70	5.7
Head and neck cancer	56	8.9
Sarcoma	55	23.6
Small Cell Lung Cancer (SCLC)	52	7.7
Bladder Cancer	42	14.3
Hodgkin's lymphoma	33	18.2
Neuroendocrine tumors	31	32.3
Myelofibrosis (MF)	29	6.9
Cutaneous lymphomas	28	39.3
Mesothelioma	20	5.0
Gastrointestinal Stromal Tumor (GIST)	16	18.8
Thyroid cancer	15	33.3
Non-melanomatous skin cancers	8	50.0

Notes: The sample is all drug-indication pairs in the development panel that reached Phase II in the indication and whose first record is 2012 or earlier, which gives all pairs a decade of runway before the panel ends. All firms and all of a drug's indications are included, not only the first. Approved means that the pair ever reaches approval in the indication. The approval share is the probability that the entry model's self-commercialization payoff uses (Section 6.4). No unconditional rate is reported, because the sample conditions on a clinical record. The registry placeholders ("other solid tumors" and "hematologic cancer") name no approvable indication and are excluded here, as in the estimation. Across all indications the rate is 13.3 percent (329 of 2,478 pairs), above the 6.7 percent that Wong et al. (2019) estimate from Phase II for oncology and the 10.8 percent in Biotechnology Innovation Organization et al. (2021). The cohort restriction accounts for the difference, since conditioning on a first record in 2012 or earlier both resolves the censoring and selects a period before oncology approval rates fell. The ranking matches the published one, with the hematologic cancers above the large solid tumors and pancreatic and brain cancer at the bottom. Six indications rest on fewer than 30 pairs, from myelofibrosis and cutaneous lymphomas at 29 and 28 down to non-melanomatous skin cancers at 8, and their rates are correspondingly imprecise.

alternative-specific constants, which absorb everything that is permanent about an indication. As a result, λ_1 and λ_2 are identified by variation across the alternatives that a single compound faces, as opposed to differences in levels between markets.

In measuring the possible payoffs, we scale them by a match quality measure, q_{ij}, which supplies most of that variation, since the indication-year terms move little once the constants are in (indication and year effects explain 88 percent of the fitted acquisition hazard) while a compound's mechanism makes some markets feasible for it and others not. This is the empirical analog to the quality measure, θ_{ij} in Section 4. The measure also serves to discipline the entrant's choice set. For example, the quality measure we use captures the fact that a HER2 antagonist has the highest match in treating breast cancer, a lower probability of treating gastric cancers, and no probability of treating an acute leukemia. The metric is constructed as the probability that a compound with i's mechanism of action is developed in indication j. It is the leave-one-out share of same-mechanism compounds in the development panel with a Phase II or later record in j, shrunk through three nested classifications of the compound's mechanism, from its specific action (584 values) to its target family (404) to its EphMRA ATC level-4 class (72), which allows a compound whose exact mechanism is rare to borrow from mechanisms aimed at the same target. For a compound i in class c at any tier, with n_{cj}^{-i} the number of other compounds in the class with a Phase II or later record in j and n_{c}^{-i} the number of other compounds in the class,

$$q_{ij}^{c} = \frac{n_{cj}^{-i} + \alpha\, q_{ij}^{\text{parent}}}{n_{c}^{-i} + \alpha}, \tag{25}$$

with prior weight $\alpha = 5$, where the parent of the specific action is the target family, the parent of the target family is the ATC class, and the parent of the ATC class is $\bar{\pi}_j$, the share of all compounds developed in j. The headline measure q_{ij} that we use is the specific-action tier. A tier falls back to its parent when the class is empty or holds a single compound, and as a result a first-in-class compound inherits the profile of its target family, then of its ATC class, and reaches $\bar{\pi}_j$ only when all three are uninformative. Returning to the example, this metric becomes very small for an acute leukemia, and this shrinks the possible payouts from entering that market toward zero. It does, however, stay strictly positive by construction, in order to ensure that a compound whose mechanism has no precedent anywhere is not recorded as infeasible everywhere. In Appendix J.1, we report how many compounds resolve at each tier.

The data also require two final adjustments. First, we must right-censor the outside option, as the

majority of the potential entrants (Phase I compounds) are still feasibly running their Phase I trials by the end of the analysis window. Second, we require the outside option to be in the choice set, and as a result the model is estimated at the compound level and the counterfactual's aggregate-entry effect is given by the change in the predicted no-entry probability. We also report aggregate-share versions of the model in Appendix K.2 as a reference, namely the multinomial logit over indications estimated as a Poisson regression on indication-year counts and the Berry log-odds form (Berry, 1994; Guimarães et al., 2003). These versions have little variation to work with, however, given that all of the payoff terms other than q_{ij} vary at the indication-year level and are largely absorbed by the constants.

7 Counterfactual and Estimation

The counterfactual policy bars incumbents holding an overlapping position in an indication (Phase II through approved, on-patent product) from acquiring development-stage compounds in the same indication, with the portfolio fixed at the entry date. We apply the two ratios in equation (22) to the measured payoff, hold V^{own}, π^{app}, Z, q and the constants fixed, and re-solve the choice probabilities, including the outside option of no entry.

7.1 Measured Inputs

In Table 9, we report the measured components of the acquisition payoff by indication, averaged over 2006 to 2019. The pool consists of the same 24 to 25 incumbents for all indications. There is substantial variation, however, in the share of high-type firms in the pool, which ranges from 10% to 70%. The annual acquisition hazard averages 3 percent, a program stays in its indication for a median of 4.3 years after its first Phase II, and the predicted price averages \$366 million on total deal value, so the discounted probability of being acquired over the horizon is 10 percent and the expected acquisition payoff of equation (21) averages \$36 million with variation across markets. Appendix J.3 reports the hazard and price regressions behind those averages and how each loads on the pool mix. The own payoff, the Phase II approval probability times V^{own}, averages \$41 million, which puts the two routes close to parity for the average entrant, with \$36 million from acquisition against \$41 million from commercializing alone. Both of these are expectations formed at the time of the entry decision, and both are small relative to the prices that change hands when a sale does occur, given that a compound reaches either outcome with a probability of around one in ten.

Table 9: The measured payoff components by indication, 2006 to 2019

Indication	Entries	K	H	H/K	π^A (%)	T_j	A (USD m)	Π^A (USD m)	V^{own} (USD m)
Non-Small Cell Lung Cancer	39	24.6	17.4	0.71	3.9	4.0	451.0	52.8	677
Breast Cancer	44	24.6	17.3	0.71	2.6	4.0	511.8	41.9	1353
Non-Hodgkin's Lymphoma	39	24.6	16.3	0.67	4.7	3.9	444.8	61.2	657
Female Reproductive Cancer	28	24.6	16.2	0.66	3.0	4.8	393.4	41.1	91
Prostate Cancer	29	24.6	15.5	0.63	3.0	4.8	432.8	44.9	561
Colorectal Cancer	28	24.6	15.4	0.63	4.2	3.5	442.2	51.4	670
Acute Leukemias	44	24.6	14.6	0.60	2.6	4.8	382.7	35.6	48
Pancreatic Cancer	26	24.6	13.2	0.54	1.7	4.2	376.1	20.6	40
Chronic Leukemias	19	24.6	12.8	0.52	3.6	4.0	426.3	45.2	619
Renal Cell Cancer	16	24.6	12.2	0.50	3.0	4.0	387.2	34.7	210
Melanoma	26	24.6	12.0	0.49	2.0	4.4	387.8	25.0	226
Multiple Myeloma	24	24.6	11.7	0.48	3.6	4.8	439.8	54.1	818
Brain Cancer	21	24.6	11.3	0.46	1.7	4.8	375.3	22.7	123
Gastroesophageal Cancers	12	24.6	10.5	0.43	2.4	3.4	364.0	24.0	32
Head and Neck Cancer	17	24.6	10.5	0.43	3.4	3.4	367.5	33.9	72
Hepatobiliary Cancer	17	24.6	10.3	0.42	1.3	3.8	365.9	13.8	68
Small Cell Lung Cancer	9	24.6	10.0	0.41	3.9	4.0	361.6	42.8	34
Sarcoma	11	24.6	9.3	0.38	3.0	4.1	363.7	33.3	85
Bladder Cancer	10	24.6	6.9	0.28	2.6	3.0	347.5	23.0	17
Myelofibrosis	8	24.6	6.1	0.25	5.9	4.8	348.5	66.0	107
Neuroendocrine Tumors	6	24.6	5.8	0.24	2.2	6.2	367.6	32.4	295
Hodgkin's Lymphoma	13	24.6	5.2	0.21	2.9	3.1	341.0	24.6	25
Gastrointestinal Stromal Tumor	2	24.6	4.9	0.20	3.7	3.8	344.8	36.1	83
Thyroid Cancer	4	24.6	4.6	0.19	4.9	4.5	336.9	52.3	11
Mesothelioma	7	24.6	4.3	0.17	1.6	7.5	335.1	24.8	5
Cutaneous Lymphomas	10	24.6	2.9	0.12	1.4	5.8	330.4	18.8	18
Non-Melanomatous Skin Cancers	2	24.6	2.8	0.11	2.2	2.6	330.0	16.0	18

Notes: Means over 2006 to 2019 by parent indication, sorted by the high-type share of the pool. Entries are first Phase II trials by small firms directed at the indication, v indications counted. K is the number of Big Pharma and Large Biotech incumbents holding an oncology asset in the quarter; H the number holding an overlapping p indication, a Phase II or later development asset or an on-patent approved product. π^A is the fitted annual acquisition hazard of a small-firm Phase I to NDA compou column 2); T_j the median years a small-firm program stays in the indication after its first Phase II; A the predicted total deal value (Table 27, column 1), 2015q1 $m; Π^A acquisition payoff over the horizon, equation (21). V^{own} is the discounted revenue of one more approved product in the indication, equation (23), at the event-study shar

We build V_{jt}^{own} on the entrant's share s, which is the 3.1 percent loss of treated patients that a competitor's approval imposes on an incumbent in the stacked event study of Section 5.5. At this share, V_{jt}^{own} averages \$242 million conditional on approval, with 83 percent of its variance occurring across indications, and the \$41 million own payoff reported above is this value times the indication's approval probability. The reserve of equation (18) applies the same risked value deal by deal, and it averages \$34 million across the deals by pool members. The calibration puts the commercialization discount at its bound of one on total deal value and at 0.40 on the upfront payment alone (Appendix K.6).

7.2 The Auction Stage

The bid index is fitted on all 337 priced deals, the 246 exclusive licenses with a reported total deal value and the 91 eligible whole-company acquisitions with clearly assignable lead assets, from 2006 to 2019, with the deal type included in the index covariates. For the inversion, we restrict the sample to the deals made by incumbents, which total 140 of the 337 priced deals. Additionally, we drop the top and bottom 5% of deals, leaving us with a total of 126 in the inversion. Valuations are recovered at $m = 3$ and the counterfactual is simulated with the engaged count drawn from equation (17). The low-type distribution is identified on 28 deals, which is thin, and prohibits us from using the preferred asymmetric estimator. We discuss this limitation further at the end of the section.

The estimand of this stage is the proportional effect of the ban, defined as the ratio of the expected winning bid with and without the barred bidders, and the ratio of the probability that anyone bids. Our current proportional approach is required as we only observe the winning bid of a heterogeneous asset, and we cannot separate dispersion of values within an auction from differences in asset quality across auctions. Bidder shading increases with this dispersion, leaving us with large implied markups. At the working cell the mean recovered valuation is about four times the mean winning bid, where Gorbenko and Malenko (2014), estimating the same object on takeover auctions, find winning slack of 7 to 15 percent of willingness to pay. Thus, as a validation of our estimation strategy, we run a Monte Carlo simulation that generates the predicted winning bids from a known distribution at our sample size and inverts them. This process recovers the truth to within 10 to 25 percent of the prediction, and the same Monte Carlo shows that unobserved auction-level heterogeneity of a plausible size roughly doubles recovered valuations (Krasnokutskaya, 2011). Instead, we estimate the ratios which are far less exposed. A common inflation of all valuations cancels in a ratio of expected bids. We also run this stage using the alternative price measures as a robustness check in

Appendix K.5, which supports the use of the proportional estimand and demonstrates that the choice of price measure does not meaningfully change the estimates.

In the merger filings, two thirds of the parties that sign a confidentiality agreement and receive information never go on to price the asset. In Section 4.1.4, we attribute this walk-away to two margins that affect the price differently: a diligence screen, where exits are observed by rivals, and an exit at the bid stage, where they are not. Given that the deal prices alone are not able to separate the two margins, we instead treat the bid-stage share F as a grid in Section 6.2 and report results across it.

In Table 10, we report what the values of F on the grid imply for the two margins and how well the bids support them. The check first inverts the observed bids at a given F, then pushes the recovered valuations back through the bid function, and finally compares the result to the bids that we started from. A large bid-stage share puts a large mass of valuations below the reserve, and as a result explaining the bids that we observe requires valuations far above them. The gap is \$18 million at $F = 0$ against a mean bid of \$761 million, \$125 million at 0.35, and \$981 million at 0.65, where it exceeds the mean bid itself. The bids therefore reject the story in which most of the walk-away happens after a valuation is formed, and instead place most of it at the diligence screen.

The best-fitting cell is the corner at $F = 0$, but we do not use it as our working value. This cell implies that no firm ever forms a valuation of an oncology asset and then declines to bid, which the data do not support. Instead, we take $F = 0.35$, which divides the walk-away about evenly, with 46 percent of engaged parties exiting at the screen and 35 percent of those that pass declining to bid. This choice is conservative for the results that follow, given that the price effect of the restriction is larger at the cells that fit better. We also report all of the cells, in addition to the working one.

We also run a separate exercise to isolate the composition effect, in which we hold the engaged count fixed and draw all parties from the low-type distribution in place of the sample's mix (Appendix K.4, Table 30). In this exercise, the expected winning bid falls by 14 percent at $m = 2$, 18 percent at $m = 3$ and 21 percent at $m = 5$, which indicates that the composition of the field influences the price even when its size is held fixed. The expected maximum valuation falls by about 15 percent at all values of m. Interestingly, the price falls by less than the valuations in the smallest field and by more in the larger ones, as equilibrium shading depends on the composition of rivals as well as on their number.

Table 10: Where the walk-away sits, and whether the bids support it

	Where the walk-away sits (%)		Bid-fit check	
F (bid stage)	Screen exit	Bid stage	Gap (USD m)	Rank corr.
0.00	65.0	0.0	17.9	1.000
0.25	53.3	25.0	46.2	1.000
0.35*	46.2	35.0	125.2	1.000
0.50	30.0	50.0	422.6	0.996
0.65	0.0	65.0	981.3	0.910

Notes: F is the bid-stage share, the probability that a party which has formed a valuation stays out of the bidding. The filings fix the total walk-away at 65 percent of informed parties, which means that the probability that an engaged party prices is 0.35 at all values of F and the screen-pass rate is the residual $0.35/(1-F)$. The first two columns report the resulting exit rates at the two margins, as a percent of engaged parties for the screen and of screen-passers for the bid stage. For the bid-fit check, the surplus bids are inverted at the stated F, the recovered valuations are drawn as the per-party primitive, the equilibrium winning bid is simulated at $m = 3$, and the gap is its mean less the observed mean surplus-bid residual, in 2015q1 $m, against a mean bid of $761m. Rank corr. is the Spearman correlation between the bid and its recovered valuation. * Working value.

7.3 Entry Estimates

In this section, we report the estimates from the conditional logit of equation (24). A compound that has completed Phase I trials chooses among the 27 indications, or chooses to develop nothing. In each market, the compound weighs two expected payoffs, acquisition and self-commercialization, both of which are the indication-year payoff scaled by the compound's match quality q_{ij}. The scaling factor serves to alter the probability of both approval and acquisition by taking the ex-ante indication-specific quality match into the expectation. For example, a predicted price of $366 million is worth little to a compound whose mechanism cannot reasonably be expected to serve the given market. Match quality also enters on its own, in logs, to govern the indication feasibility separately from how it scales expected payoffs. For instance, a drug with few plausible markets is more likely to exit in the counterfactual, independent of the scaled price. This term is what allows a poor match to push a compound out of development entirely. The alternative constants absorb everything permanent about an indication, including the entry cost. Thus, λ_1 governs whether, among the markets a compound's mechanism fits, it tilts toward the ones where acquisition pays more. The sample holds 633 compounds with a named indication and 562 that never start a Phase II trial, giving 1,195 cases in all.

In Table 11, we report the estimates. In Column 1, the model makes no attempt to determine the ex-ante quality of indication match for a given drug. In other words, each compound is equally likely to be approved in any indication - an unrealistic assumption. As a result, this specification finds nothing. Columns 2 and 3 report the same specification but using different samples. Column 2 utilizes the full data,

and the coefficient on the match-weighted acquisition payoff is significant with a point estimate of 0.032 per $1 million. Column 3 is our preferred estimate, as it corrects for a right censoring issue in the data. This specification uses the sample of potential entrants that we are certain have made a choice to not enter any market to identify the extensive margin. This correction prevents us from erroneously assigning the "no indication" choice to a drug that is actually still in Phase I development. Reassuringly, the coefficient of interest is not substantially altered from Column 2 to Column 3, indicating a stable parameter. The own payoff parameter does change, however, and becomes marginally significant under the right censoring correction, at 0.009 per $1 million. The feasibility term $\ln q_{ij}$ is the strongest single predictor of where a compound goes through all specifications. We report additional robustness checks in Appendix K.1 varying how we measure the feasibility term as well as using different deal value metrics. Across all price measures and all match-quality tiers, the response has the same sign and the same economic size.

Table 11: Entry at the compound level, with match quality

	(1) Without q	(2) With q	(3) With q, right censor correction
Acquisition payoff Π^A	-0.0004		
	(0.0074)		
Own payoff $\pi^{app}V^{own}$	-0.0023		
	(0.0023)		
Crowding (log)	-0.0143	-0.1055	-0.2086
	(0.4324)	(0.4368)	(0.1702)
Promise (log)	0.1032	0.0950	-0.3425***
	(0.2054)	(0.2246)	(0.0769)
$q\times$ acquisition payoff		0.0316***	0.0322***
		(0.0091)	(0.0096)
$q\times$ own payoff		0.0035	0.0086*
		(0.0059)	(0.0046)
$\ln q$		0.3791***	0.3355***
		(0.0615)	(0.0624)
Cases	1,195	1,195	835
Choice rows	33,460	33,460	23,380
Clusters (years)	25	25	24

Notes: Conditional logit with alternative constants (not shown) over 27 indications and the outside option, so one coefficient governs both the choice of market and the choice to enter at all. A case is a compound and a named market (a first Phase II naming k markets is k cases, with compounds naming more than four markets excluded), or a compound that never starts Phase II. Payoffs in USD m. Match quality q is the compound's specific mechanism of action's leave-one-out share of compounds developed in the indication, shrunk toward its target family, its EphMRA ATC level-4 class, and the indication's overall share; it multiplies both payoff terms and enters in logs. Crowding is log(1 + small-firm Phase II to NDA programs in the indication the prior year), and promise is log(1 + non-industry trials started in the prior year). Columns 1 and 2 are estimated on the same full sample and differ only in whether q is in the model. Column 3 is the headline and applies the right-censoring correction described in the text, dropping the non-entrants whose imputed decision date falls within seven years of the panel's end. Standard errors clustered by decision year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7.4 Counterfactual Results

We now turn to the results of the counterfactual. At the auction stage, the ban lowers the expected winning bid by 32 percent and the probability that any engaged party prices the asset by 18 percent at the working cell, which amounts to a combined fall of 42 percent in the price times the pricing probability. In Table 12 and Figure 11, we report the ban's effect on entry along both margins, using the headline model of Column 3. Under the ban the acquisition payoff falls from $36 million to $18 million on average, falling more in markets with larger high-type shares. Entry moves out of the indications where incumbents hold overlapping positions and into the indications where they do not: non-small cell lung cancer loses 16 percent of its entrants, non-Hodgkin lymphoma 8 percent, breast cancer 5 percent and prostate cancer 4 percent. The indications with few overlapping incumbents gain 7 to 11 percent each, and 2.9 percent of entrants change indication in total. On the extensive margin the predicted no-entry share rises from 24.2 to 25.8 percent of decisions, a fall in aggregate entry of 2.2 percent of its baseline. Priced at the upfront payment instead, the model reallocates 3.2 percent of entrants and the same indications move, and at the risk-adjusted prices it reallocates 3.4 percent. At the coarse ATC tier, the model reallocates 0.9 percent of entrants, which is the same pattern at a smaller size and reflects the attenuation that we document in Appendix K.1.

The reallocation is not neutral in terms of expected approvals. In Table 8, we report the empirical approval rate estimates by indication conditional on a Phase II trial, which vary substantially by market. For instance, averaged across indications among Phase II pairs with a decade of runway, compounds in the hematologic cancers reach approval at 18 to 23 percent, the large solid tumors are near 9 to 16 percent, and pancreatic and brain cancer are under 5 percent. A compound that the ban moves out of non-small cell lung cancer therefore enters a market with a different conversion rate from entry to approved drug.

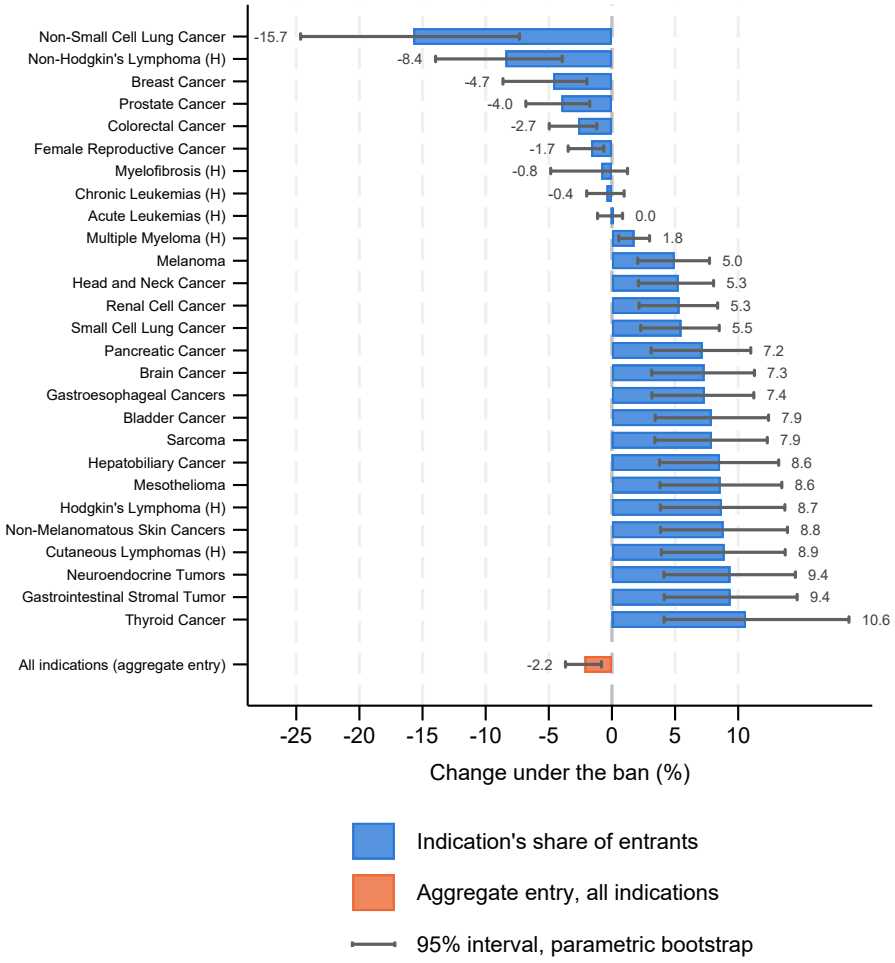
The size of the auction-stage effect depends on the walk-away split, and we report it across the grid of Section 7.2 in Appendix K.4. Over the cells of the grid, the combined ratio runs from 0.49 to 0.65, with a value of 0.58 at the working cell that we use for the entry counterfactual.

Table 12: The counterfactual: entry shares by indication under the overlap ban

Indication	Baseline share (%)	Restricted share (%)	Change (%)
Acute Leukemias	7.76	7.77	0.0
Bladder Cancer	1.88	2.03	7.9
Brain Cancer	4.13	4.44	7.3
Breast Cancer	8.44	8.05	-4.7
Chronic Leukemias	4.65	4.63	-0.4
Colorectal Cancer	5.28	5.14	-2.7
Cutaneous Lymphomas	1.96	2.14	8.9
Female Reproductive Cancer	4.77	4.69	-1.7
Gastroesophageal Cancers	2.47	2.65	7.4
Gastrointestinal Stromal Tumor	0.32	0.35	9.4
Head and Neck Cancer	3.27	3.44	5.3
Hepatobiliary Cancer	3.15	3.42	8.6
Hodgkin's Lymphoma	2.17	2.36	8.7
Melanoma	5.46	5.73	5.0
Mesothelioma	1.10	1.20	8.6
Multiple Myeloma	5.28	5.38	1.8
Myelofibrosis	1.57	1.56	-0.8
Neuroendocrine Tumors	1.07	1.17	9.4
Non-Hodgkin's Lymphoma	7.32	6.70	-8.4
Non-Small Cell Lung Cancer	9.06	7.63	-15.7
Non-Melanomatous Skin Cancers	0.32	0.35	8.8
Pancreatic Cancer	4.60	4.93	7.2
Prostate Cancer	5.87	5.64	-4.0
Renal Cell Cancer	3.14	3.31	5.3
Sarcoma	1.98	2.14	7.9
Small Cell Lung Cancer	2.15	2.27	5.5
Thyroid Cancer	0.81	0.90	10.6
All indications: entrants changing indication (%)			2.9
All indications: aggregate entry (%)			-2.2

Notes: Predicted share of small-firm first Phase II entries directed at the indication, mean over entrant cases 2006 to 2019, in the baseline and under the ban, from the compound-level conditional logit with match quality (Table 11, column 3), which includes both margins. Shares sum to 100 by construction, and the two aggregate rows therefore report different objects. The first is the share of entrants that change indication under the ban (half the sum of absolute share changes), and the second is the change in the number of entrants. The ban lowers the acquisition payoff in an indication-year by the product of the two ratios in Appendix Table 31 at the working cell. V^{own}, the approval probabilities, promise and the constants are held fixed.

Figure 11: Change in each indication's share of entrants under the ban



Notes: The blue bars show the percent change in the indication's share of small-firm first Phase II entries, mean over 2006 to 2019, from the compound-level conditional logit with match quality (Table 11, column 3) at the working cell, with bid-stage share $F = 0.35$ and $p = 0.16$. (H) marks hematologic indications. The last bar reports a different object, the percent change in the number of entrants across all indications, from the same model and the same coefficient. Whiskers are 95 percent intervals from a parametric bootstrap of the entry coefficients, with 200 draws from the estimated coefficient distribution of each column and the counterfactual re-solved for each draw. The auction-stage ratios are held fixed in all draws, and their variation is reported across the grid of Appendix Table 31 as opposed to a standard error. The interval for the share of entrants changing indication is 1.3 to 4.6 percent around the 2.9 point estimate, and for aggregate entry it is -3.7 to -0.8 around -2.2.

8 Conclusion

Small biotech firms do not choose where to develop cancer drugs in isolation from the market for their assets. We model the sale of a development-stage compound as a first-price auction among the incumbent firms that engage, measure every payoff the entry decision weighs, and estimate where entry goes when the acquisition payoff moves. Barring overlapping incumbents from acquiring, the policy debated as the remedy for killer acquisitions, cuts the expected acquisition payoff by 50 percent, redirects about 3 percent of small-firm Phase II starts away from the most contested cancers, and lowers the number of entering compounds by about 2 percent.

These headline figures that we report are entry effects, not welfare effects. They do, however, provide insight into the potential welfare effects, and particularly for expected approvals. Per 100 entering compounds, the ban deters 2.2 entrants. Over the years a compound stays in development, about 12 of the 100 are acquired, roughly 10 of them by the overlapping incumbents the ban bars. If some of those banned deals are killer acquisitions, each killer acquisition prevented restores the compound's chance of approval. Valuing both sides of the ledger at the Phase II likelihood of approval, which assumes a killed compound and a deterred marginal entrant carry the same approval odds, the ban pays for itself only if about 22 percent of the acquisitions it eliminates are killers. This would require that 18 percent of all acquisitions are killer acquisitions, far above measured rates. For example, Cunningham et al. (2021) put the killer share at 5.3 to 7.4 percent of pharmaceutical acquisitions, and the European Commission's ex-post evaluation finds a discontinuation of an overlapping project without a technical or safety explanation after 89 of 3,193 transactions, under 3 percent (European Commission, 2024). The Commission's conditional rate is higher, 37 percent of the 240 transactions whose projects overlapped narrowly, but narrow molecular overlap describes a small subset of the broad indication-level class this ban bars. On this margin the policy forgoes three to seven times the approvals it saves.

We stress that approvals are not the same as welfare, and we leave that gap for future work. The value of an approval depends on its market, and the ban moves entry out of crowded cancers and toward markets with fewer approved options, where variety plausibly matters most. At the same time, the reallocation has an approval cost of its own. The match-quality gradient in the entry model implies that a compound pushed from its best-fit indication toward its second choice is a worse scientific fit there, and as a result some of the 3 percent of compounds that reallocate are likely to arrive with lower approval odds.

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A Data Appendix

In this appendix, we provide the construction details behind Section 3, in the order in which the sources appear there.

A.1 Development Histories

There are two cleaning decisions that shape the drug-indication life-cycles. First, some drugs stop progressing without a disclosed termination, and Biomedtracker lists these drugs as suspended. We assign these a suspension date equal to the last recorded event plus the sample median duration of the final recorded phase (six quarters from preclinical, two from an IND filing, eight from Phase I, twelve from Phase II or Phase III, and two from NDA review). This imputation creates 126 artificial suspension events, 63 where the recorded history ends with a trial start and 63 where development stops with no further event of any kind, and a further 186 drug-indication pairs are marked as suspended at their last recorded event with no added time. Officially disclosed suspensions, of which there are 2,376 events, carry their reported dates. Second, a drug can run several trials in the same indication at once, most often a combination trial that begins after an approval. We collapse these parallel tracks by taking the running maximum of the development phase within the drug-indication pair, such that a drug's phase never falls while any program in the indication is alive. The additional exclusivity that a combination approval earns enters through the patent data instead of through the development panel. In practice, the maximum rarely binds, as recorded phases move backward in only 10 of 5,670 drug-indication event histories.

A.2 Ownership and Control-Rights Deals

We build ownership from two tables. The first is the company identity chain, which lists all of the companies that appear anywhere in the data together with their acquirers and the transfer dates. This chain allows us to resolve the ultimate parent of any firm at any date. The second is the drug-level ownership event table, which records the originator of a compound and all subsequent transfers of control. We assemble this table from Biomedtracker alliances, Evaluate company M&A, Evaluate product deals and four hand-collected files. We validate the chain against Biomedtracker's own lead-company field on its pull date, and the chain matches for 2,459 of 2,539 drugs (96.8 percent), and for 99.1 percent of drugs at Phase II or later.

We exclude four types of transactions from the deal analysis. The first is BMT Marker rows, which

are non-exclusive co-development or co-commercialization deals that transfer no control. The second is reverse takeovers, which are listing transactions in which the operating company merges into a listed shell (22 drug-rows). The third is the 14 whole-company mergers between diversified firms that we identified by hand, which are going concerns with marketed portfolios across therapeutic areas. The fourth is option deals, which involve no immediate change of control. We reproduce the hand-identified list in Table 13.

Table 13: The 14 excluded mergers between diversified firms

Acquirer	Target	Quarter	USD bn	Reason
Pfizer	Wyeth	2009q4	68.0	Diversified pharma target (305 products, 137 marketed)
Bristol Myers Squibb	Celgene	2019q4	74.0	Going concern with a marketed portfolio (167 products; Revlimid)
Roche	Genentech	2009q1	46.8	Buy-out of the minority in a majority-owned subsidiary
Merck & Co	Schering-Plough	2009q4	41.1	Diversified pharma target (243 products, 144 marketed)
Shire	Baxalta	2016q2	32.0	Diversified target (hematology and immunology)
Bayer	Schering AG	2006q3	21.3	Diversified pharma target (148 products, 95 marketed)
Sanofi	Genzyme	2011q2	20.1	Diversified target (rare disease)
Pfizer	Hospira	2015q1	17.0	Diversified generics and injectables target
Novartis	GSK oncology business	2015q1	16.0	Portfolio swap between two incumbents
Merck KGaA	Serono	2007q1	13.8	Diversified target (fertility, multiple sclerosis)
Teva	IVAX	2006q1	7.4	Diversified generics target
Teva	Cephalon	2011q4	6.8	Diversified pharma target (127 products, 87 marketed)
Novartis	Chiron	2006q2	5.7	Diversified target (vaccines and diagnostics)
Servier	Shire oncology business	2018q3	2.4	Business-unit purchase of a marketed franchise (Oncaspar)

Notes: Announced deal values in billions of dollars. A deal is on this list when the target is a going concern with a marketed portfolio across therapeutic areas, or when the transaction is between two incumbents, as opposed to the purchase of a development-stage compound.

We assign a licensing deal that covers several assets to the compound that the deal description identifies as the lead, or to the compound that is furthest in development. Of the 246 licenses priced on total value, 42 cover more than one sample drug. When we restrict the Table 5 regressions to deals covering exactly one sample drug, which also removes the clear-lead whole-company tier, we find an overlap premium of $8.8 million (SE 3.4) on the upfront and $80.5 million (SE 25.7) on total value, on 203 and 267 deals respectively. The result therefore does not depend on the assignment rule.

Whole-company acquisitions enter the priced-deal analyses only when the price can be tied to a single compound. The rule that we apply has two tiers. In the first, the target holds a single sample drug (95 deals, 61 with a reported value). In the second, the target holds two to three sample drugs with a unique most-advanced drug that is still in development (Phase I to NDA review) and a company value under $5 billion (45 clear-lead deals, 39 priced, 30 surviving the screen). This screen is important in practice, given that the

unscreened tier is dominated by multi-billion dollar takeovers of firms with marketed products. Including these deals roughly halves the fit of the bid index and eliminates the overlap premium of Section 5.3, because those prices reflect the acquired firm's full commercial assets as opposed to the lead compound alone. A whole-company indicator absorbs the deal-form level in all of the pooled specifications.

A.3 Prices and Deflation

All dollar amounts are deflated to 2015q1 dollars with the CPI-U (series CPIAUCSL). License terms come from Biomedtracker alliance records (upfront payment, milestone schedule, royalty provisions), and company deal values from Evaluate Pharma. The Evaluate product-deal file is the spine linking deals to drugs. Milestones enter total potential deal value at their nominal amounts because the data do not break them down by trigger. Selective disclosure is measured by a linear probability model of whether a deal reports a price, on acquirer type, phase, scope, orphan status, drugs per deal and year effects, one observation per exclusive-license deal. We report this model in Table 14. A large-firm acquirer is 37 percentage points (SE 5) more likely to report a total value and 40 points (SE 5) more likely to report an upfront payment.

Table 14: Selective disclosure of deal terms

	(1) Total value reported	(2) Upfront reported
Large-firm acquirer	0.370^{***}	0.398^{***}
	(0.046)	(0.048)
Mean of dependent variable	0.53	0.46
Observations	462	462
R^2	0.18	0.17

Linear probability that an exclusive-license deal reports a price, one observation per deal, 2006q1 to 2019q4. Controls: phase at the deal, worldwide rights, orphan designation, sample drugs in the deal, and year fixed effects. Standard errors clustered by acquirer parent. Missing prices are concentrated among smaller acquirers, so the large-firm price premia in the paper are lower bounds.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

A.4 Sales and Indication Attribution

We combine the Evaluate Pharma and Biomedtracker revenue series to construct quarterly drug sales. About 22 percent of on-patent approved drug-quarters have no sales row, and nearly all of these are products below disclosure thresholds. We set these to zero and carry the count of covered drugs alongside each market aggregate. For a drug approved in several cancers, we split its sales using MarketScan indication shares.

For molecule d, cancer c and quarter t, the share is patient-months on d in c over patient-months on d in all cancers, with boundary quarters carried outward beyond the 2009 to 2019 claims window. Where the claims data are unable to identify the cancer (CML, GIST, neuroendocrine tumors, sarcoma, thyroid), we assign shares by hand from labels and analyst sources. Claims shares carry about 71 percent of attributed sales, hand shares about 11 percent, and single-indication drugs the remaining 18 percent.

A.5 Exclusivity End Dates

We define the exclusivity end as the later of the expected patent expiry and the end of regulatory exclusivity. On the patent side, we use Evaluate Pharma's consensus analyst forecast of the first commercially significant expiry in the drug's key market, converted to a term at the drug level. Where Evaluate has no record, we fall back to the Orange Book active-ingredient patent with its term extension, from the NBER-maintained cleaned version, and a biologic with neither record is assigned fifteen years from first approval and flagged. We never set expiry earlier than five years after first approval, which is the new chemical entity exclusivity, and drugs with a public litigation outcome or settlement date take that date. On the regulatory side, we apply exclusivity at the market level, with three years for an approval in the market and seven for an orphan approval. The realized exclusivity end is the later of the expected date and the Orange Book first generic approval through 2024, and we hold the expected date where an override or a pre-1991 legacy product applies. The expected date enters the quarters-to-expiry measure and revenue at risk, while the realized date defines the on-patent indicator.

B Potential Acquirer Definitions

In this appendix, we list the firms behind the incumbent definitions of Section 3.4, together with the quarters in which they qualify inside the analysis window of 2006q1 to 2019q4. Table 15 lists the Big Pharma incumbents and Table 16 lists the Large Biotech incumbents. A firm that is acquired by an incumbent rolls up to its parent from the acquisition quarter onward, and a firm that crosses the Big Pharma thresholds moves from one table to the other in the quarter in which it crosses.

Table 15: Big Pharma Incumbents

Incumbent	Member Firm	First Quarter	Last Quarter
Abbott Laboratories	Abbott Laboratories	2006Q1	2012Q4
AbbVie	AbbVie	2006Q1	2019Q4
	Abbott Laboratories†	2013Q1	2019Q4
Amgen‡	Amgen	2011Q1	2019Q4
AstraZeneca	AstraZeneca	2006Q1	2019Q4
	MedImmune†	2007Q2	2019Q4
Bayer	Bayer	2006Q1	2019Q4
Biogen‡	Biogen	2012Q1	2019Q4
Boehringer Ingelheim	Boehringer Ingelheim	2006Q1	2019Q4
Bristol Myers Squibb	Bristol Myers Squibb	2006Q1	2019Q4
	Celgene†	2019Q4	2019Q4
Eli Lilly	Eli Lilly	2006Q1	2019Q4
Gilead Sciences‡	Gilead Sciences	2013Q1	2019Q4
	Kite Pharma†	2017Q4	2019Q4
GSK	GSK	2006Q1	2019Q4
Johnson & Johnson	Johnson & Johnson	2006Q1	2019Q4
	Janssen Pharmaceutica	2006Q1	2019Q4
Merck & Co	Merck & Co	2006Q1	2019Q4
	Schering-Plough†	2009Q4	2019Q4
Novartis	Novartis	2006Q1	2019Q4
Pfizer	Pfizer	2006Q1	2019Q4
	Wyeth†	2009Q4	2019Q4
Regeneron Pharmaceuticals‡	Regeneron Pharmaceuticals	2018Q3	2019Q4
Roche	Roche	2006Q1	2019Q4
	Genentech†	2006Q1	2019Q4
	Chugai Pharmaceutical†	2006Q1	2019Q4
Sanofi	Sanofi	2006Q1	2019Q4
	Aventis†	2006Q1	2019Q4
Schering-Plough	Schering-Plough	2006Q1	2009Q3
Takeda‡	Takeda	2011Q3	2019Q4
	Shire†	2019Q1	2019Q4
	Baxalta†	2019Q1	2019Q4
Wyeth	Wyeth	2006Q1	2009Q3

Notes: To qualify as Big Pharma, in a given year the firm must have over $40 billion in market capitalization, $20 billion in annual revenue, 15% of revenue invested in research and development, and significant commercial operations in the US, Europe, China, and Japan.

†Member firm mapped to the incumbent through the company identity chain from the quarter shown. Wyeth and Schering-Plough qualify independently until their 2009 acquisitions, and Abbott Laboratories qualifies independently until the AbbVie separation in 2013. Genentech and Chugai map to Roche through majority ownership, with Genentech also qualifying independently as a Large Biotech until 2009Q2 (Table 16).

‡Qualified as a Large Biotech incumbent before the first quarter shown (Table 16).

Table 16: Large Biotech Incumbents

Company Name	**First Quarter**	**Last Quarter**
Amgen[‡]	2006Q1	2010Q4
Baxalta[†]	2015Q3	2016Q1
Biogen[‡]	2006Q1	2011Q4
Celgene[†]	2006Q1	2019Q3
Daiichi Sankyo	2006Q1	2019Q4
Genentech[†]	2006Q1	2009Q2
Genmab	2006Q1	2013Q1
Genzyme[†]	2006Q1	2011Q1
Gilead Sciences[‡]	2006Q1	2012Q4
Incyte	2014Q1	2019Q4
Kite Pharma[†]	2014Q3	2017Q3
MedImmune[†]	2006Q1	2007Q1
Merck KGaA	2006Q1	2019Q4
Regeneron Pharmaceuticals[‡]	2016Q1	2018Q2
Servier	2010Q1	2019Q4
Shire[†]	2010Q1	2015Q3
Takeda[‡]	2006Q1	2011Q2
Teva Pharmaceutical Industries	2006Q1	2019Q4

Notes: To qualify as a Large Biotech, in a given year the firm must have over $15 billion in market capitalization, $2 billion in annual revenue, 20% of revenue invested in research and development, be independent from any Big Pharma firm, and have significant commercial operations in the US, Europe, China, and Japan.

[†]Acquired by an incumbent at the end of the span shown and mapped to its acquirer thereafter (Table 15): Baxalta by Shire (2016) and then Takeda (2019), Celgene by Bristol Myers Squibb (2019), Genentech by Roche (2009), Genzyme by Sanofi (2011), Kite Pharma by Gilead Sciences (2017), MedImmune by AstraZeneca (2007), Shire by Takeda (2019).

[‡]Crossed the Big Pharma thresholds after the span shown and appears in Table 15 from that quarter.

C Merger Disclosure Coding

In this appendix, we provide the codebook behind Sections 3.7 and 5.4, as well as the paired-price spreads of Section 7.2.

C.1 Retrieval and Extraction

The candidate set consists of all Evaluate M&A targets in our deal universe that could plausibly have filed a merger disclosure, namely US-listed targets acquired in a change-of-control transaction, which gives 171 deals. We match these targets to EDGAR by CIK and search their filings around the announcement date, which yields a Schedule 14D-9 or DEFM14A for 135 of them. We extract the "Background of the Merger" (or "Background of the Offer") section automatically and validate it. Of these 135 filings, 95 yield a chronology complete enough to code, and 7 of those are excluded from the reported distribution because the document describes a different transaction or the process was a related-party or otherwise unusable one, leaving the 88 processes of Figure 9.

C.2 Coding Rules

The unit of coding is the party. We read the chronologies party by party, which produces one row per named or aliased participant, for a total of 351 rows over the 95 coded processes. Every flag on a row must be supported by verbatim sentences copied from the filing, and this requirement allows the coding to be audited. A party counts as engaged, and therefore enters K^e, if it either (a) received confidential information under a confidentiality agreement executed for this transaction, or (b) made a price-bearing proposal, with or without an agreement. The acquirer always counts. We use three rules to decide the hard cases. First, a confidentiality agreement signed for a partnering or licensing discussion does not qualify, while one that was amended to cover an acquisition does. Second, an unsolicited approach counts only if it named a price. Third, parties in a parallel licensing or asset canvass are excluded from K^e unless they crossed over into the acquisition process. Where a filing describes informed parties without enumerating them (for instance, "several other parties conducted due diligence"), we record the count as a floor and flag it. Fifteen of the 88 counts are floors, which biases the measured K^e downward. Alongside K^e, we also record the number of parties that the banker contacted, the number that bid at all, and the number that submitted final bids.

C.3 Audit

We reproduce all counted parties in an audit sheet with one block per deal. Each block records the count and its basis (an enumerated list or a stated count), the funnel numbers, the source URL, an adjudication note that explains any judgment call, and the verbatim evidence for the flags on each party. We checked the machine-assisted reading by hand against this sheet, and ambiguous processes carry their resolution in the note. One caveat applies to all uses of these counts. The filings describe change-of-control sales of public companies, and the auction sample contains 91 deals of this form, so the funnel evidence bears directly on that part of the estimation sample. For the licenses that make up the majority of the sample, we rely on the partnering and licensing canvasses described in the same filings (Section 5.4).

C.4 Paired Prices

We build the paired prices used in Appendix K.5 from the same chronologies. For a given process, we record the winner's first offer, its final price at signing, and the best rival offer stated in the same unit (per share, or total consideration), and we keep a flag for whether the rival's price was a firm proposal as opposed to an indication of interest that was never formalized. We resolve unit mismatches and stale offers by hand, deal by deal, with notes. This procedure results in 22 processes with both prices in one unit, 15 of which have a firm rival proposal, and 37 processes in which we observe the winner's own offer path from first offer to signing.

D Proofs

In this appendix, we provide the proofs of Propositions 4.2, 4.3 and 4.4.

Throughout the proofs, the parties past the screen bid in the symmetric equilibrium of equation (5) (restated here):

$$b(v) = v - \frac{1}{F(v)^{n_{it}-1}} \int_{R_{it}}^{v} F(x)^{n_{it}-1}\, dx.$$

with each party's valuation drawn independently from the mixture $F_{\text{mix}} = (H_{jt}/K_t)\, F_H + (L_{jt}/K_t)\, F_L$ in the baseline and from F_L under the restriction. This is the treatment estimated in Section 6.2 and simulated in Appendix K.4. The proof of Proposition 4.2 establishes three steps that the other two proofs reuse.

Proposition 4.2. *Entry value is nondecreasing in the acquirer pool. Adding an incumbent to $\mathcal{K}_t$ weakly raises Π_{ij}, and the increase is weakly larger when the added incumbent is a high type.*

Proof. The proof proceeds in three steps. Step 1 expresses the option value of a process in terms of valuations in place of bids. Step 2 shows that the option value is nondecreasing in the pool at a fixed reserve. Step 3 extends the ordering through the Bellman equation by backward induction.

Step 1: The option value in valuations. Fix a process in which n parties past the screen draw valuations independently from F and the reserve is R. Let $v^{(1:n)}$ and $v^{(2:n)}$ denote the highest and second-highest of the n valuations. For simplicity, we define the convention that $v^{(2:1)} = -\infty$, the second-highest valuation in a process with a single party. We claim

$$S = \mathbb{E}\big[(v^{(2:n)} - R)^{+}\big] \tag{26}$$

In the symmetric equilibrium a party with $v \geq R$ bids $b(v)$ from equation (5) and wins when all rivals have a lower valuation or abstain. The allocation is therefore the same as in a second-price auction with reserve R, and the lowest type that bids, $v = R$, earns zero surplus in both formats. By the revenue equivalence theorem (Myerson, 1981; Riley and Samuelson, 1981), the seller's expected revenue is the same in the two formats. In the second-price format the price is $\max\{v^{(2:n)}, R\}$ when $v^{(1:n)} \geq R$, so

$$\mathbb{E}\big[b^{(1)}\, \mathbf{1}\{v^{(1:n)} \geq R\}\big] = \mathbb{E}\big[\max\{v^{(2:n)}, R\}\, \mathbf{1}\{v^{(1:n)} \geq R\}\big].$$

The entrant retains R when no party bids, so

$$\mathbb{E}\big[\max\{b^{(1)}, R\}\big] = R\,\Pr\big(v^{(1:n)} < R\big) + \mathbb{E}\big[\max\{v^{(2:n)}, R\}\,\mathbf{1}\{v^{(1:n)} \geq R\}\big] = R + \mathbb{E}\big[(v^{(2:n)} - R)^+\big],$$

where the last equality uses that $(v^{(2:n)} - R)^+ > 0$ only when $v^{(1:n)} \geq v^{(2:n)} > R$. Subtracting R gives (26). When n is random, as it is when the engaged count and the screen are drawn, the identity holds conditional on n and therefore in expectation.

Step 2: The option value is nondecreasing in the pool at a fixed reserve. Let two pools be given, with engaged counts drawn as in equation (3) and screen passes drawn independently at rate σ. Construct both pools on one probability space by assigning to each incumbent k three independent uniform draws (u_k^e, u_k^s, u_k^v). Incumbent k engages when $u_k^e \leq p$, passes the screen when $u_k^s \leq \sigma$, and draws valuation $F_H^{-1}(u_k^v)$ if a high type and $F_L^{-1}(u_k^v)$ if a low type. We then need two comparisons:

(a) If $\mathcal{K}' \supseteq \mathcal{K}$ with the same types for the common members, the set of parties past the screen under $\mathcal{K}'$ contains the set under $\mathcal{K}$ realization by realization, and the second-highest element of a superset weakly exceeds the second-highest element of the subset.

(b) If $\mathcal{K}'$ and $\mathcal{K}$ have the same members and one member is a high type in $\mathcal{K}'$ and a low type in $\mathcal{K}$, the sets of parties past the screen coincide realization by realization. First-order stochastic dominance gives $F_H^{-1}(u) \geq F_L^{-1}(u)$ for all u, so the switched member's valuation is weakly larger and all other valuations are unchanged.

In both cases $v^{(2)}$ is weakly larger under $\mathcal{K}'$ realization by realization. Since $x \mapsto (x-R)^+$ is nondecreasing, (26) gives $S_{it}(R;\mathcal{K}') \geq S_{it}(R;\mathcal{K})$ for all R.

Step 3: The ordering passes through the Bellman equation. Write the option value as a function of the reserve and the pool, $S_{it}(R;\mathcal{K})$, and define

$$\Phi(R;\mathcal{K}) = R + \big(S_{it}(R;\mathcal{K}) - \kappa\big)^+, \tag{27}$$

so that equation (8) reads $W_{it} = \Phi(R_{it};\mathcal{K}_t)$. By Step 2, Φ is nondecreasing in the pool at all R. Φ is also nondecreasing in R. By (26), $S_{it}(R)$ is nonincreasing in R with slope $-\Pr(v^{(2)} > R) \in [-1, 0]$ almost everywhere, so for $R' \geq R$, $S_{it}(R) - S_{it}(R') \leq R' - R$. Since $x \mapsto (x-\kappa)^+$ is nondecreasing with slope

at most one,

$$\Phi(R') - \Phi(R) = (R' - R) + \left[(S_{it}(R') - \kappa)^+ - (S_{it}(R) - \kappa)^+\right] \geq (R' - R) - \left(S_{it}(R) - S_{it}(R')\right) \geq 0.$$

The result now follows by backward induction on t. At $t = T$, $W_{iT} = \pi_j^{\text{app}}(\theta_{ij})\, V_{ij}^{\text{own}}$ does not depend on the pool. Suppose $W_{i,t+1}$ is nondecreasing in the pool in all states. Then $R_{it} = -c_{it} + \delta\, \mathbb{E}_t[W_{i,t+1}]$ is nondecreasing in the pool, because expectation preserves the ordering. Writing $R'_{it} \geq R_{it}$ for the reserves under $\mathcal{K} \cup \{k\}$ and $\mathcal{K}$,

$$W'_{it} = \Phi(R'_{it}; \mathcal{K} \cup \{k\}) \geq \Phi(R_{it}; \mathcal{K} \cup \{k\}) \geq \Phi(R_{it}; \mathcal{K}) = W_{it},$$

using first that Φ is nondecreasing in R and then that it is nondecreasing in the pool. Hence W_{i0}, and with it $\Pi_{ij} = W_{i0} - F_j$, is nondecreasing in the pool.

For the second statement, compare $\mathcal{K} \cup \{h\}$ with $\mathcal{K} \cup \{l\}$, where h is a high type and l a low type. Step 2(b) gives $S_{it}(R; \mathcal{K} \cup \{h\}) \geq S_{it}(R; \mathcal{K} \cup \{l\})$ for all R and all dates, and the same induction yields $\Pi_{ij}(\mathcal{K} \cup \{h\}) \geq \Pi_{ij}(\mathcal{K} \cup \{l\}) \geq \Pi_{ij}(\mathcal{K})$. □

Proposition 4.3. *Consider a restriction that removes $\mathcal{O}_{jt}$ from the pool in market j. Under Assumption 3, the expected number of engaged parties falls by pH_{jt}, and Π_{ij} weakly falls in every market where $H_{jt} > 0$ and is unchanged elsewhere.*

Proof. Under Assumption 3, the engaged count under the restriction is $m_{it}^L \sim \text{Bin}(L_{jt}, p)$, with mean $pL_{jt} = pK_t - pH_{jt}$. Removing the H_{jt} members of $\mathcal{O}_{jt}$ one at a time and applying Proposition 4.2 at each step gives Π_{ij} weakly lower under the restriction. In a market with $H_{jt} = 0$ the restriction removes no one, so Π_{ij} is unchanged. □

Proposition 4.4. *Let $\Pi_{ij}^{auction}$ denote entry value when $\kappa = 0$, in which case all assets are marketed at all dates. Then $\Pi_{ij}^{auction} \geq \Pi_{ij}$. Holding R_{it} fixed, a restriction that removes $\mathcal{O}_{jt}$ weakly lowers S_{it} at all dates, and the set of states in which a process runs therefore weakly shrinks. Under $\kappa = 0$ a process runs in all states, which means that $\Pi_{ij}^{auction}$ responds to the restriction only through the sale stage, and a counterfactual that holds the initiation factor of (11) at its baseline understates the fall in the acquisition payoff.*

Proof. For the first statement, write Φ_κ for the map in (27) at cost κ, and let W_{it}^0 and R_{it}^0 denote the value and reserve at $\kappa = 0$. At $t = T$ the two models coincide. Suppose $W_{i,t+1}^0 \geq W_{i,t+1}$ in all states. Then

$R^0_{it} \geq R_{it}$, and

$$W^0_{it} = \Phi_0(R^0_{it}) \geq \Phi_0(R_{it}) \geq \Phi_\kappa(R_{it}) = W_{it},$$

where the first inequality uses that Φ_0 is nondecreasing in R (Step 3 of the proof of Proposition 4.2) and the second uses $(S-0)^+ \geq (S-\kappa)^+$. Hence $W^0_{i0} \geq W_{i0}$ and $\Pi^{\text{auction}}_{ij} \geq \Pi_{ij}$.

For the second statement, fix R_{it}. The restriction replaces the pool $\mathcal{K}_t$ by $\mathcal{K}_t \setminus \mathcal{O}_{jt}$, which by Step 2(a) of the proof of Proposition 4.2, applied H_{jt} times, weakly lowers S_{it} in all states. By Proposition 4.1, a process runs when $S_{it} \geq \kappa$, so the set of states in which one runs weakly shrinks, and the first factor of (11) weakly falls. At $\kappa = 0$ the condition $S_{it} \geq 0$ holds in all states before and after the restriction, so that factor equals one in both and $\Pi^{\text{auction}}_{ij}$ responds only through the second factor and the price. A counterfactual that holds the first factor at its baseline therefore attributes to the restriction a weakly smaller fall in π^A_{ij} than the model implies, and with A_{ij} moving in the same direction, a weakly smaller fall in the acquisition term of (13). □

E The Free-Entry Engagement Regime

In this appendix, we derive the replacement ratio for the alternative to Assumption 3 that we discuss after Proposition 4.3, in which the engagement rate is an equilibrium object as opposed to the model primitive that we carry in the estimation.

By the envelope theorem, the ex-ante surplus of a low type bidder that incurs the diligence cost is given by

$$\Pi_{it}^{L} \;=\; \mathbb{E}\left[\int_{R_{it}}^{v} F_H(x)^{m_{it}^{H}}\, F_L(x)^{m_{it}^{L}-1}\, dx\right], \qquad v \sim F_L, \tag{28}$$

where the expectation is taken over the value draw and over the engagement counts in (3), and the integrand is the probability that all rivals sit below x. Free entry sets $\Pi_{it}^{L} = c$, and this condition determines the rate at which the remaining low types engage.

Removing $\mathcal{O}_{jt}$ sets $m_{it}^{H} = 0$ and raises the ex-ante low type surplus at all values of x. If we assume that the engagement probability adjusts in equilibrium in order to maintain the same level of surplus, then the equality requires more low types. Equating the integrand before and after the removal gives us a replacement ratio

$$\tilde{m}_{it}^{L} - m_{it}^{L} \;=\; m_{it}^{H}\,\lambda(x), \qquad \lambda(x) \;\equiv\; \frac{\ln F_H(x)}{\ln F_L(x)} \;>\; 1, \tag{29}$$

where $\lambda > 1$ because $F_H \succ_{\text{FOSD}} F_L$ implies $F_H(x) < F_L(x)$. In other words, a banned high type must be replaced by more than one low type. In terms of the engagement probability,

$$\tilde{p} \;\approx\; p\left(1 + \frac{H_{jt}}{L_{jt}}\,\bar{\lambda}\right), \tag{30}$$

where $\tilde{p}$ is the post-ban rate, $\bar{\lambda}$ is the average of $\lambda(x)$ over the relevant range of valuations, and the approximation holds when the rate is small enough that the engagement counts enter the surplus through their means. Under free entry, the ban raises the engagement rate of the remaining firms by the banned share scaled by the replacement ratio. As a result, the engaged count, and with it the participation channel, falls by less than the pH_{jt} of Proposition 4.3.

F Truncation of the Winning-Bid Distribution

In this appendix, we derive the correction referred to in Section 4.2.

We observe a transaction price only when a sale occurred. As a result, the winning bids in our data are draws from the distribution of the maximum bid, conditional on that maximum having cleared the reserve. The derivation below provides the correction term that this selection requires.

Fix a sale process at date t with m_{it} parties past the diligence screen (the n_{it} of Section 4.2) drawing valuations independently from F, and let R_{it} be the reserve price. Let $\tilde{G}$ denote the marginal bid distribution of a representative such party, which includes the convention that a party drawing a valuation below R_{it} abstains and is assigned a notional bid below the support, such that it counts toward $\tilde{G}(b)$ at all b. Then

$$\tilde{G}(b) = \begin{cases} F(R_{it}), & b < R_{it}, \\ F\big(v(b)\big), & b \geq R_{it}, \end{cases} \tag{31}$$

where $v(\cdot)$ inverts the bid function:

$$v(b) = b + \frac{\tilde{G}(b)}{(m_{it} - 1)\,\tilde{g}(b)}. \tag{32}$$

Note that $\tilde{G}(b)$ is continuous at the reserve value, where it takes the value $F(R_{it})$ instead of zero. Given that any draw below R_{it} contributes to $\tilde{G}(b)$ at all values of b, we then have that $\Pr\left(b_{it}^{(1)} \leq b\right) = \tilde{G}(b)^{m_{it}}$ holds without conditioning on the number of active bidders.

Across all processes, whether or not they produce a sale, the maximum bid satisfies $\Pr\left(b_{it}^{(1)} \leq b\right) = \tilde{G}(b)^{m_{it}}$. A sale occurs when at least one party clears the reserve, and this event has probability $1 - F(R_{it})^{m_{it}}$ by (7) applied to the parties past the screen. The distribution that we observe is therefore

$$G^{\text{win}}(b) \;=\; \Pr\left(b_{it}^{(1)} \leq b \mid \text{sale}\right) \;=\; \frac{\tilde{G}(b)^{m_{it}} - F(R_{it})^{m_{it}}}{1 - F(R_{it})^{m_{it}}}, \tag{33}$$

and inverting (33) gives the distribution that (32) requires:

$$\tilde{G}(b) \;=\; \Big[\, F(R_{it})^{m_{it}} \;+\; \big(1 - F(R_{it})^{m_{it}}\big)\, G^{\text{win}}(b) \,\Big]^{1/m_{it}}. \tag{34}$$

The uncorrected case is nested in equation (34) as a special case. When we set $F(R_{it})^{m_{it}} = 0$, we obtain $\tilde{G}(b) = \big[G^{\text{win}}(b)\big]^{1/m_{it}}$, the maximum order statistic inversion applied directly to the winning bids, and this is the estimator that ignores the reserve value. The correction is the additive term inside the bracket, and its size is given by the probability that a process yields no sale.

In Table 17, we report the size of the discrepancy at the median of the observed winning-bid distribution, setting $F(R_{it}) = 0.6$ for illustration. The correction is negligible when the engaged pool is large, but it is substantial at the pool sizes that the merger filings indicate.

Table 17: Corrected and uncorrected $\tilde{G}$ at $G^{\text{win}} = 0.5$

	$m_{it} = 2$	$m_{it} = 5$	$m_{it} = 20$
No-sale probability $F(R_{it})^{m_{it}}$	0.360	0.078	0.00004
Uncorrected $\big[G^{\text{win}}\big]^{1/m_{it}}$	0.707	0.871	0.966
Corrected, equation (34)	0.825	0.884	0.966

Current implementation. Applying (34) requires $F(R_{it})^{m_{it}}$, the probability that a process yields no sale. We are unable to estimate F and R_{it} separately in order to compute it, since F is the object being recovered. The current implementation instead treats the no-sale rate as a model primitive. The merger filings that supply the engagement probabilities in Section 5.4 also record sale processes that ran and produced no transaction, which could microfound this rate. However, doing so would ask a great deal of the filing data and would require assumptions that we are not prepared to defend.

The correction in equation (34) accounts for processes that ran and failed to clear the reserve. It does not address which compounds were marketed at all, which is the initiation margin of Section 4.3, and it does not address which completed transactions enter our data. We treat these two selections separately.

G Revenue at Risk: Timing, Placebo and Estimator Checks

In this appendix, we report three checks on the revenue-at-risk results of Section 5.2. The first two checks ask whether the acquisition response has the timing that anticipation implies, and the third asks whether the choice of count model is driving the results.

Timing. The timing of the acquisition response is consistent with anticipation, although it does not establish it. In a revenue-weighted event study that places each drug's pre-cliff sales in a window relative to the current quarter, the acquisition response is positive through the last eight quarters before exclusivity ends, 0.248 with a standard error of 0.236 at five to eight quarters out and 0.287 with a standard error of 0.290 at one to four, and 0.265 with a standard error of 0.207 when the two windows are pooled. No individual window is significant and we cannot reject that the response before the cliff equals the response after, $p = 0.47$. We read the profile as showing a response concentrated in the two years before the cliff, and we do not date it more finely.

Table 18: Acquisitions and entry by window relative to the cliff

	(1) Acquisitions	(2) Small-firm entry	(3) Incumbent entry
Pre-cliff sales of drugs losing exclusivity, by window (USD bn)			
25 to 32 quarters before	-0.811***	0.027	0.201
	(0.200)	(0.128)	(0.261)
17 to 24 quarters before	0.172	-0.159	0.204
	(0.410)	(0.270)	(0.175)
9 to 16 quarters before	-0.765**	0.151	-0.162
	(0.362)	(0.232)	(0.178)
1 to 8 quarters before	0.265	0.281	-0.223
	(0.207)	(0.191)	(0.170)
0 to 7 quarters after	0.133	0.007	-0.148
	(0.278)	(0.258)	(0.185)
8 to 15 quarters after	-0.262	0.170	-0.799***
	(0.382)	(0.220)	(0.247)
Mean of dependent variable	0.170	0.541	0.296
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	Yes	Yes	Yes

[1] As in the twelve-window table, with windows pooled to eight quarters. The 1 to 8 quarters window is the exposure measure of the main table.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Revenue that has already lost exclusivity. A placebo test also supports the anticipation interpretation. Revenue that has already lost exclusivity does not predict acquisitions (a coefficient of 0.240 with a standard

error of 0.361), while revenue that is still at risk in the same specification has a coefficient of 0.456 with a standard error of 0.174. In other words, incumbents buy ahead of the cliff and not after it. This pattern is consistent with franchise defense, and it is not consistent with a firm-wide cash motive.

Table 19: Revenue still at risk against revenue that has already lost exclusivity

	(1) Acquisitions	(2) Small-firm entry	(3) Incumbent entry
Revenue at risk, 8 quarters (USD bn)	0.456***	0.221***	-0.132
	(0.174)	(0.073)	(0.169)
Revenue that lost exclusivity in the past 8 quarters (USD bn)	0.240	-0.050	0.076
	(0.361)	(0.212)	(0.201)
Controls			
Market sales (USD bn)	-0.074	-0.041	0.175
	(0.188)	(0.081)	(0.132)
Likelihood of approval	0.005	-0.115	0.354**
	(0.176)	(0.101)	(0.145)
Mean of dependent variable	0.170	0.541	0.296
Observations	1,400	1,400	1,400
Year FE	Yes	Yes	Yes
Market FE	Yes	Yes	Yes
p-value, before = after	0.471	0.129	0.121

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales (USD bn) and the likelihood of approval.
[2] The second regressor is the sales (measured over the four quarters before the end of exclusivity) of drugs in the market whose exclusivity ended within the past 8 quarters. Column (1): incumbent acquisitions of development-stage drugs. Column (2): small-firm first Phase II trials naming fewer than four markets. Column (3): incumbent first Phase II trials.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

Horizon. In Table 20, we re-estimate the main specification with revenue at risk redefined over twelve and sixteen quarters ahead in place of the headline eight, for both outcomes. The acquisition response is concentrated at the short horizon, at 0.394 for eight quarters against 0.079 for twelve. This pattern is what we would expect from a dated cliff, as an incumbent buys when the expiry is near, while sales that expire three or four years out enter the longer-horizon measures and contribute little to the response. Entry, on the other hand, responds over a longer window, with a coefficient of 0.160 at twelve quarters that fades by sixteen quarters. This pattern is consistent with entrants positioning themselves earlier in the approach to a cliff than the point at which acquirers transact.

Estimator. In Tables 21 and 22, we compare least squares, Poisson and a zero-inflated negative binomial on the main specification, and the three estimators agree. The zeros in these data are what a count model with this mean would produce, as opposed to excess mass. For instance, the fitted Poisson implies 1,201

Table 20: Revenue at risk at eight, twelve and sixteen quarters

	Incumbent acquisitions			Small-firm entry		
	(1)	(2)	(3)	(4)	(5)	(6)
Revenue at risk, 8 quarters (USD bn)	0.394***			0.179**		
	(0.136)			(0.074)		
Revenue at risk, 12 quarters (USD bn)		0.079			0.160***	
		(0.254)			(0.060)	
Revenue at risk, 16 quarters (USD bn)			-0.153			0.078
			(0.165)			(0.086)
Controls						
Market sales (USD bn)	-0.073	-0.072	-0.012	-0.045	-0.065	-0.054
	(0.182)	(0.196)	(0.189)	(0.116)	(0.124)	(0.116)
Likelihood of approval	0.011	0.030	0.043	-0.116	-0.116	-0.111
	(0.176)	(0.175)	(0.182)	(0.104)	(0.102)	(0.102)
Mean of dependent variable	0.170	0.170	0.170	0.688	0.688	0.688
Observations	1,400	1,400	1,400	1,400	1,400	1,400

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Poisson regressions; coefficients are the proportional change in the rate per USD billion at risk. Standard errors clustered by market in parentheses. Controls are market sales and the likelihood of approval; all columns carry year and market fixed effects, the main specification of the acquisitions and entry tables.
[2] Each column redefines revenue at risk as the sales of approved drugs in the market whose exclusivity ends within the stated number of quarters, in USD billions per quarter. The headline tables use eight quarters.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

zero market-quarters for acquisitions against 1,211 observed, and 786 against 775 for entry. The inflation equation is accordingly close to degenerate, and for entry it returns the Poisson coefficient to three decimals under all of the inflation specifications that we tried. We report the zero-inflated model for completeness, although the data do not call for it.

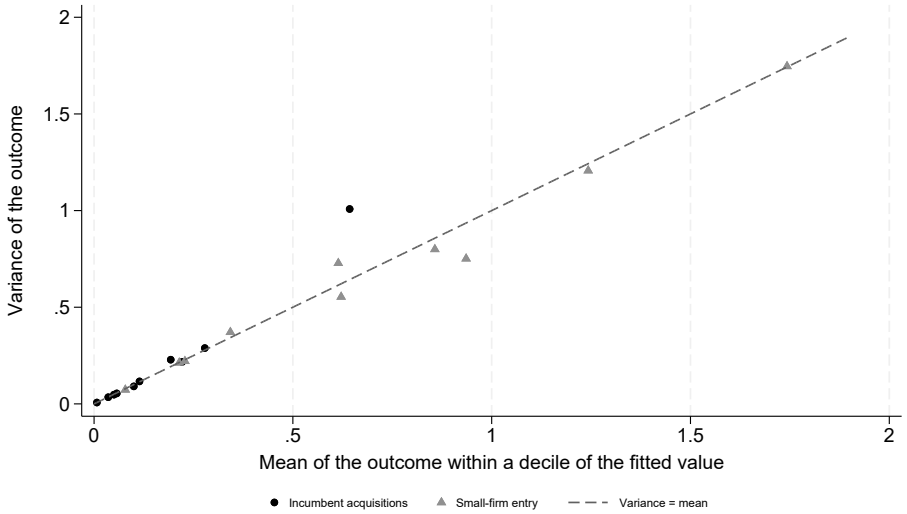
The firm level. In Table 23, we report the firm-level regressions referenced in Section 5.2. An incumbent's own revenue at risk does not predict its own acquisitions at any of the four horizons, nor does it when the regressor enters in levels in place of a share of the firm's oncology sales, nor for control-transferring alliances. Estimating the level specification at the longer horizons, which the table does not display, gives the least precise estimates of the set, at 0.040 with a robust standard error of 0.026 at twenty quarters and 0.027 with a standard error of 0.019 at thirty-two quarters. We therefore interpret the firm-level null as a failure to detect a response, as opposed to evidence that no response exists. Given that the panel holds about 20 incumbents, a firm-level design has little power at this sample size, and we read these estimates alongside the market-level estimates as opposed to a separate result.

Table 21: Estimator comparison: incumbent acquisitions

	Incumbent acquisitions of development-stage drugs		
	(1)	(2)	(3)
Revenue at risk, 8 quarters (USD bn)	0.091**	0.394***	0.362***
	(0.039)	(0.136)	(0.138)
Controls			
Market sales (USD bn)	-0.016	-0.073	-0.066
	(0.048)	(0.182)	(0.166)
Likelihood of approval	-0.026	0.011	-0.029
	(0.021)	(0.176)	(0.177)
Estimator	OLS	Poisson	ZINB
Inflation equation	–	–	Market sales
Overdispersion α			0.000
Mean of dependent variable	0.170	0.170	0.170
Observations	1,400	1,400	1,400

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Every column has year and market fixed effects and controls for market sales (USD bn) and the likelihood of approval. Standard errors clustered by market in parentheses. The dependent variable counts acquisitions by a Big Pharma or Large Biotech incumbent of drugs in Phase II, Phase III or NDA review in the market that quarter.
[2] Column (1) is ordinary least squares in levels, so its coefficient is events per USD billion at risk. Columns (2) and (3) are multiplicative, so their coefficients are proportional changes in the count. The zero-inflated negative binomial in column (3) reports the count equation; its inflation equation is a logit on market sales, and the coefficient is conditional on the market-quarter not being in the always-zero state rather than being the unconditional mean the other columns estimate.
[3] The counts are not zero inflated in the sense the third model is built for: the Poisson fit in column (2) implies 1,201 market-quarters with no acquisition against 1,211 observed, and 786 with no entry against 775 observed. The zero-inflated model also fails to converge under several inflation specifications, including revenue at risk in the inflation equation; the variants attempted are reported in the log.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

Figure 12: The variance of each count tracks its mean



Notes: Market-quarters are sorted into deciles of the fitted value from the Poisson regression in column (2) of Table 3 (and its entry counterpart), and the figure plots the variance of the outcome within each decile against its mean. The dashed line is variance equal to mean. Ordinary least squares in levels assumes instead that the variance is constant, and it places 199 of 1,400 fitted acquisition counts and 49 of 1,400 fitted entry counts below zero, and a counterfactual that lowers revenue at risk would make such negative predictions more frequent. Least-squares and Poisson fitted values are close in sample, so in-sample fit does not separate the two models. The count model is chosen for its estimand and its zero bound.

Table 22: Estimator comparison: small-firm first Phase II trials

	Small-firm first Phase II trials		
	(1)	(2)	(3)
Revenue at risk, 8 quarters (USD bn)	0.380***	0.179**	0.179**
	(0.109)	(0.074)	(0.074)
Controls			
Market sales (USD bn)	0.070	-0.045	-0.045
	(0.160)	(0.116)	(0.116)
Likelihood of approval	-0.041	-0.116	-0.116
	(0.040)	(0.104)	(0.104)
Estimator	OLS	Poisson	ZINB
Inflation equation	–	–	Market sales
Overdispersion α			0.000
Mean of dependent variable	0.688	0.688	0.688
Observations	1,400	1,400	1,400

[1] Parent indication by quarter, 2006q1 to 2019q4, 25 markets. Every column has year and market fixed effects and controls for market sales (USD bn) and the likelihood of approval. Standard errors clustered by market in parentheses. The dependent variable counts first Phase II trials by firms that are neither Big Pharma nor Large Biotech, counting each market the trial names.

[2] Column (1) is ordinary least squares in levels, so its coefficient is events per USD billion at risk. Columns (2) and (3) are multiplicative, so their coefficients are proportional changes in the count. The zero-inflated negative binomial in column (3) reports the count equation; its inflation equation is a logit on market sales, and the coefficient is conditional on the market-quarter not being in the always-zero state rather than being the unconditional mean the other columns estimate.

[3] The counts are not zero inflated in the sense the third model is built for: the Poisson fit in column (2) implies 1,201 market-quarters with no acquisition against 1,211 observed, and 786 with no entry against 775 observed. The zero-inflated model also fails to converge under several inflation specifications, including revenue at risk in the inflation equation; the variants attempted are reported in the log.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

Table 23: Incumbent acquisitions and the firm's own revenue at risk

	M&A					Alliances
	(1) $h = 8$	(2) $h = 12$	(3) $h = 20$	(4) $h = 32$	(5) $h = 8$	(6) $h = 8$
Share of own oncology sales at risk	0.014					0.000
	(0.056)					(0.052)
Share of own oncology sales at risk		-0.025				
		(0.042)				
Share of own oncology sales at risk			0.017			
			(0.036)			
Share of own oncology sales at risk				0.005		
				(0.028)		
Own revenue at risk (USD bn)					0.016	
					(0.028)	
Mean of dependent variable	0.040	0.040	0.040	0.040	0.040	0.039
Observations	1,419	1,419	1,419	1,419	1,419	1,419

[1] Incumbent by quarter, firm and year fixed effects. The dependent variable counts the incumbent's acquisitions of development-stage drugs (columns 1 to 5, M&A; column 6, control-transferring alliances). Heteroskedasticity-robust standard errors in parentheses. We report robust rather than cluster-by-firm standard errors because the panel holds about 20 incumbents, and at that number of clusters the cluster-robust estimator is downward biased: on the level regressor in column (5) it returns 0.008 against a robust 0.028 on the same coefficient. The inference in the share specifications does not turn on the choice, as each is far from significance under either estimator.

[2] The regressor is the share of the incumbent's own oncology sales from approved drugs whose exclusivity ends within h quarters; column (5) uses the level in USD billions.

$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01.$

H The Competitor-Approval Event Study in Levels

In Table 24, we report the stacked event study of Section 5.5 with the outcomes measured in levels, namely dollars of attributed sales, patient-months, and the share of the cancer's treated patients in percentage points. The stacks, window, fixed effects and clustering are the same as in Table 7. The levels estimates agree with the log estimates in sign and in implied magnitude once the coefficients are compared with the treated pre-period mean. However, they are less precise for sales and for the share, because a levels regression gives more weight to the largest incumbents. We report the log and share forms in the main text because the self-commercialization payoff uses the share of patients, which is proportional by construction.

Table 24: The competitor-approval event study in levels

	Sales in c (USD m)		Patient-months in c		Share of c	
	(1)	(2)	(3)	(4)	(5)	(6)
Post x treated	-21.984	-8.339	-304.670***	-77.305*	-0.004	-0.003
	(15.123)	(5.236)	(110.624)	(46.052)	(0.005)	(0.002)
Control cancers	clean	all	clean	all	clean	all
Treated pre-period mean	101.64	101.64	2,206.59	2,206.59	0.06	0.06
Observations	31,974	155,156	31,970	155,143	31,970	155,143
Clusters	81	84	80	83	80	83

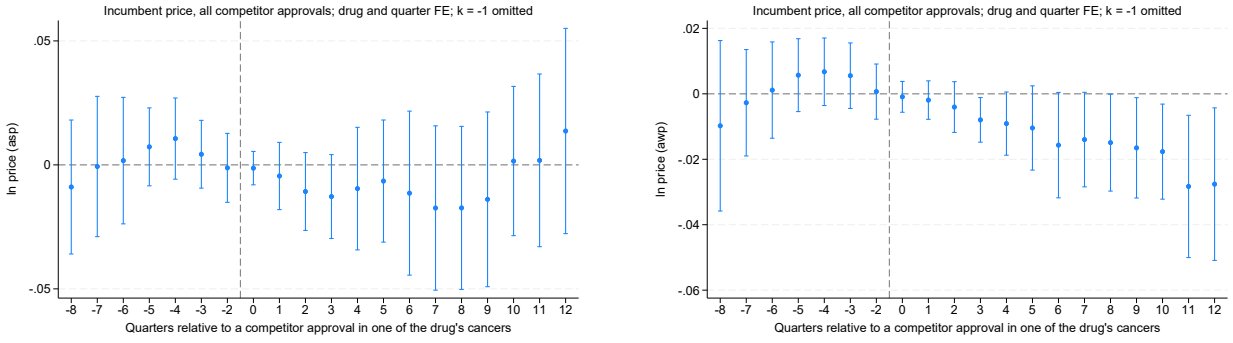
Notes: The competitor-approval event study of Section 5.5 with the outcomes in levels rather than logs: US sales attributed to the cancer in millions of 2015q1 dollars, patient-months among the cancer's patients, and the drug's share of the cancer's treated patients. Same stacks, window, fixed effects and clustering as the log version. The treated pre-period mean converts each coefficient to an implied percent. * p<0.10, ** p<0.05, *** p<0.01.

I Incumbent Prices and the Competitor Count

In this appendix, we report the price evidence behind the choice in Section 5.5 to summarize competition among approved drugs with a competitor count. We use two price series. For infused and injected drugs, which carry most oncology revenue, we use the CMS Part B average sales price (ASP), the manufacturer's average price to all US purchasers net of most concessions, quarterly with a two-quarter lag, one HCPCS code per molecule, net of the statutory add-on, covering 56 drugs from 2009q1 to 2019q4. For branded oral drugs we use RED BOOK average wholesale price (AWP), a list price chained across NDCs, covering 55 drugs. Under Medicare Part B, providers are reimbursed ASP plus six percent, and prior work shows that privately negotiated physician services are set as a percent markup from ASP (Clemens and Gottlieb, 2017).

In Figure 13, we plot the distributed-lag coefficients of log price on the number of competitor approvals k quarters earlier in any cancer in which the drug is approved. The unit of observation is the drug and not the cancer, as there is no indication-specific pricing in the US. Averaged over the twelve quarters after an approval, the ASP response is -0.007 and is statistically indistinguishable from zero. Oral list prices continue to rise throughout, at a rate that is about one point a year slower than they otherwise would.

Figure 13: Incumbent prices around a competitor's approval



(a) Infused drugs: average sales price (ASP)

(b) Oral drugs: list price (AWP)

Notes: Distributed-lag coefficients on the number of competitor approvals k quarters earlier in any cancer the drug is approved in, $k = -8$ to 12 with bins beyond, $k = -1$ omitted; drug and quarter fixed effects; standard errors clustered by drug. Left: CMS Part B average sales price per dosage unit, net of the statutory add-on, 56 infused and injected drugs, 2009q1 to 2019q4. Right: RED BOOK average wholesale price per unit for branded oral drugs, 55 drugs, 2009q1 to 2019q4. Bars are 95 percent intervals.

In Table 25, we report the cross-sectional counterpart, which regresses incumbent revenue and volume on the number of approved competitors in the cancer. The competitor count is imprecisely estimated once indication effects absorb the cross-section, and this is one reason that we take the entrant's share from the event study as opposed to these regressions (Appendix J.4).

Table 25: Incumbent revenue and volume on the number of approved competitors

	ln sales in c		ln patient-months in c			share of cancer c		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Competitors approved in c	-0.010	-0.010	-0.009	-0.009	-0.008	-0.001	-0.001	-0.000
	(0.019)	(0.020)	(0.014)	(0.013)	(0.020)	(0.001)	(0.001)	(0.001)
Pair and quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age-in-cancer FE	No	Yes	No	Yes	No	No	Yes	No
Drug $imes$ quarter FE	No	No	No	No	Yes	No	No	Yes
Observations	3,664	3,664	3,701	3,701	2,233	3,748	3,748	2,259
Drugs	103	103	101	101	37	101	101	38

Notes: Competitors approved in c = running count of other drugs with an FDA approval in the same cancer, excluding drugs sold by a company that also sells the focal drug. Sample: drug x cancer pairs where the focal drug is approved in c and selling, 2009q1-2019q4. Sales in c = US sales x Marketscan share of the drug's patient-months in cancer c; patient-months and share of cancer c are Marketscan counts. Age = quarters since the focal drug was approved in c and selling. Drug x quarter fixed effects use drugs approved in two or more cancers. SEs clustered by drug. * p<0.10, ** p<0.05, *** p<0.01.

J Measurement Detail for the Estimated Model

In this appendix, we describe the construction of the objects that we take to the data in Section 6. These are match quality and the sample rules behind the choice set, the bid index, the hedonic price, and the alternatives that we carry for the entrant's share. The robustness and diagnostic exercises referenced in the results are in Appendix K.

J.1 Match Quality

Match quality is defined in equation (25) of Section 6.5. There are three tiers, which we list from coarse to fine. The first is the EphMRA ATC level-4 class, with 72 classes defined by modality and target family, such as "protein kinase inhibitor, ALK" or "monoclonal antibody, CD20". The second is the target family from Evaluate's mechanism-of-action fields, with 404 values. The third is the specific action, with 584 values over 2,410 matched compounds. At a given tier, we take the leave-one-out share in the full development panel (all firms) over compounds with a Phase II or later record in j. Most specific actions are held by a handful of compounds, and for this reason the shrinkage matters. A compound whose exact mechanism is rare borrows first from mechanisms aimed at the same target and then from its broad class, as opposed to standing on its own few observations. The ATC-class tier alone, which is the coarsest measure with any mechanism content, is the alternative that we report alongside the headline measure in Section 7.3.

Both pieces of this construction are standard in the literature. The leave-one-out share is a conditional co-occurrence probability on a class-by-indication incidence matrix, which is the quantity that the relatedness literature refers to as proximity and uses to predict what a unit does next (Hidalgo et al., 2007). The back-off in (25) is a hierarchical shrinkage of a noisy fine cell toward the coarser cell that contains it, as in the county and commuting-zone estimates of Chetty and Hendren (2018). Our contribution in this respect is to apply these tools to the class-by-indication network, as opposed to either component on its own.

The recursion also serves to keep a first-in-class compound in the estimation. A compound whose mechanism is genuinely new has no class-mates to average over, and a measure defined as a share of same-class compounds would be undefined or zero for it in all markets, and this would be read as scientific infeasibility everywhere as opposed to an absence of evidence. With the fallback described in Section 6.5, the measure instead returns the indication's own development rate, which is the correct prior for a compound about which the class structure says nothing. Of the 2,356 compounds in the reference panel, 1,834 resolve

at the specific-action tier, 242 at the target family, 164 at the ATC class, and 116 at the base rate. In the estimation sample the corresponding counts are 866, 130 and 98 of 1,195 cases, with 37 at the base rate and a further 64 compounds that fall outside the reference panel and take $\bar{\pi}_j$ as well. For these 101 cases, q_{ij} varies across indications only through $\bar{\pi}_j$, which the alternative constants absorb, and as a result these cases inform the coefficient through the decision of whether to enter at all, as opposed to the choice among markets. Wu and Ono (2025), estimating a conditional logit over cancer indications, instead drop compounds whose mechanism has only one observed indication. Our recursion keeps these compounds in the sample, at the cost of a measure that contains no information about where they go.

J.2 The Choice Set and the Treatment of Basket Trials

We treat a first Phase II trial that names k indications as k entries, one per named indication, with each counted in full. There is no solid-tumor approval to enter, and as a result a compound trialing in three cancers has entered three markets, and in the estimation a basket trial is k cases that face the full choice set. The registry placeholders are not markets and leave the estimation. The parent category "hematologic cancer", which a trial names when it names no specific blood cancer, has nothing approved for it and no revenue, which means that the payoff terms built from approved drugs do not exist there. The same holds for "other solid tumors". For the latter, the ClinicalTrials.gov record of the dating trial often names the cancers that the basket enrolled, and we assign the entry to those cancers, which places 27 of 57 placeholder entries. We drop the 30 entries that the registry is unable to place, as well as the eight compounds whose trials name more than four markets, as these are tumor-agnostic baskets whose market choice the record is unable to identify. The choice set holds the 27 remaining indications and the outside option.

J.3 The Bid Index and the Hedonic Price

The bid index is a regression of the surplus bid on the drug's phase, the deal's form (a whole-company indicator), rights and royalty terms, orphan status, an indicator that the compound's analyst likelihood of approval ever sat above its indication-year average, the market's sales and pipeline, and year and indication effects. We exclude the acquirer's own position in the market, because the counterfactual acts on this margin.

The hedonic price A_{jt} is a regression of total potential deal value (the upfront payment plus development and sales milestones at their nominal amounts for a license, and the company deal value for a whole-

company acquisition) on all priced deals in the Section 3.3 sample, against the pool mix, market sales, the deal's form, phase, rights and royalty terms, orphan status and drug count, and a year factor in three-year bins. We then predict the price at the pool mix and market sales of a given indication-year, with the deal characteristics held at their sample means. We use the bins in place of free year effects, which would put sampling noise into the price given that there are about fifteen priced deals a year. We do not risk-adjust the milestones at the deal level, because the data do not break them down by trigger, and in Section 7.2 we report the risk-adjusted price that splits the difference.

How the Two Components Load on the Pool Mix

In Tables 26 and 27, we show how the two components load on the pool mix. The hazard rises with the high-type share, and it is 81 percent higher per unit of H/K with year effects (with a standard error of 0.28 on the log rate) and roughly doubles under indication effects. The incumbents' revenue at risk, which raises the number of acquisitions in an indication in Section 5.2, does not raise the rate per compound, at -0.15 per \$1 billion with a standard error of 0.22. Revenue at risk is instead correlated with the stock of small-firm compounds in the indication, which means that the count response of Section 5.2 operates through the number of compounds at risk as opposed to the chance that any one of them is bought. The price does not load on the pool mix at all once the whole-company deals are in the sample, at \$104 million per unit of H/K on total deal value with year bins (with a standard error of \$269 million) and \$609 million with indication effects (with a standard error of \$721 million), and the upfront columns are equally uninformative. Indication and year constants still explain 88 percent of the fitted hazard, which means that most of the level is cross-sectional. Both of these facts are what the design anticipates. These regressions measure the levels of the payoff, while the price response to the pool in the counterfactual comes from the auction, as opposed to the hedonic regression.

J.4 Alternatives for the Entrant's Share

The entrant's share s in equation (23) comes from the stacked event study of Section 5.5, and we carry two alternatives through the results. The first is the clean-control estimate of 10.6 percent, which uses only the events whose control cancers had no approval of their own within four quarters. The second is the equal-share benchmark, which sets $s = 1/(n_{jt} + 1)$ for a market with n_{jt} approved products. We do not use

Table 26: The acquisition hazard and the high-type share of the pool

	(1) Year FE	(2) + indication FE
High-type share of the pool H/K	0.813^{***}	1.353^{***}
	(0.280)	(0.517)
Market revenue (USD bn per quarter)	0.054	0.030
	(0.052)	(0.085)
Incumbents' revenue at risk (USD bn)	-0.149	-0.056
	(0.220)	(0.178)
Indication-years	592	592
Log likelihood	-546.9	-526.6

Poisson regression of incumbent acquisitions of small-firm Phase I to NDA compounds on the high-type share of the pool, market revenue and the incumbents' revenue at risk (sales on products within eight quarters of patent expiry), exposure the mean stock of such compounds in the indication-year. Coefficients are proportional changes in the hazard. Standard errors clustered by indication.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

Table 27: The hedonic price and the high-type share of the pool: total deal value and upfront payment

	Total deal value		Upfront payment	
	(1) Year bins	(2) + indication FE	(3) Year bins	(4) + indication FE
High-type share of the pool H/K	103.62	609.04	-11.02	-52.52
	(269.11)	(720.72)	(52.59)	(138.37)
Market sales (USD bn per quarter)	32.22	38.29	9.44	11.37
	(25.36)	(32.39)	(6.18)	(7.10)
Priced deals	337	337	246	246
R-squared	0.114	0.190	0.261	0.375

Total potential deal value (upfront plus milestones, nominal; the company deal value for whole-company deals) and the upfront payment, USD m 2015q1, on every priced deal 2006 to 2019: exclusive licenses plus the whole-company deals whose price can be tied to one compound. Controls: a whole-company indicator, phase at the deal, worldwide rights, royalty in the deal, orphan status, drugs in the deal, and a year factor in three-year bins. The acquirer's own position in the market is excluded by design. Standard errors clustered by indication.
$^{*}\ p < 0.10,\ ^{**}\ p < 0.05,\ ^{***}\ p < 0.01$.

a cross-sectional alternative, in which an incumbent's quarterly US sales in a cancer are regressed on the number of approved competitors there, for two reasons. First, the competitor count is imprecisely estimated once indication effects absorb the cross-section. Second, it answers a different question from the one that the entry decision poses, namely how much one more approved product takes from the market (Section 5.5).

K Entry and Auction Robustness

In this appendix, we report the robustness and diagnostic exercises behind the estimation results of Section 7. These are the entry model estimated on entrants alone and on aggregate shares, the variation that the entry coefficients require, the simulation of the counterfactual ratios, the alternative bid variables together with the paired prices from the filings, and the slack in the measured reserve.

K.1 Entry Estimated on Entrants Alone

In Table 28, we drop the outside option from the choice set. The coefficient then governs where a compound enters conditional on entering, as opposed to both margins at once. Column 1 uses the headline match-quality measure and gives 0.037 with a standard error of 0.010 on this sample, against 0.032 in the body. Conditioning on entry therefore neither creates nor removes the response. In Column 2, we replace the specific-action tier with the coarse EphMRA ATC level-4 class and find 0.013 with a standard error of 0.007, the same sign at a third of the size. This attenuation is what we would expect from a blunt feasibility measure, as a single class lumps together mechanisms whose feasible indications differ. In Column 3, we price the acquisition at the upfront payment and find 0.22 with a standard error of 0.06. This estimate is larger because the upfront-based payoff is about a sixth of the total-value payoff in levels.

K.2 Aggregate-Share Estimators

In Table 29, we report the aggregate-share estimators on 392 indication-years from 2006 to 2019. The coefficient on the acquisition payoff is small and imprecise in all columns, at -0.0002 per \$1 million of expected payoff by Poisson with year effects and -0.004 with indication effects (with standard errors of 0.002 and 0.005), and at 0.001 and 0.000 in the two log-odds columns (with standard errors of 0.002 and 0.005). Crowding accounts for the cross-indication pattern, at 0.78 log points per log unit without indication effects and 0.48 with them, and promise loads only without indication effects. At the indication-year level, the payoff varies mostly with the constants, and as a result the aggregate estimators have little variation left to work with.

Table 28: Entry at the compound level, entrants only

	(1) Entrants, q	(2) Entrants, q^{ATC}	(3) Entrants, upfront
$q\times$ acquisition payoff	0.0366***		
	(0.0100)		
$q\times$ own payoff	0.0049		0.0030
	(0.0048)		(0.0057)
$\ln q$	0.3311***		0.3436***
	(0.0611)		(0.0559)
Crowding (log)	-0.0771	0.0134	-0.0507
	(0.1454)	(0.1378)	(0.1506)
Promise (log)	-0.0033	-0.0003	-0.0039
	(0.1103)	(0.1063)	(0.1120)
$q^{ATC}\times$ acquisition payoff		0.0126*	
		(0.0071)	
$q^{ATC}\times$ own payoff		0.0094	
		(0.0076)	
$\ln q^{ATC}$		0.7384***	
		(0.1233)	
$q\times$ acq. payoff, upfront			0.2177***
			(0.0584)
Cases	633	633	633
Choice rows	17,091	17,091	17,091
Clusters (years)	24	24	24
Log likelihood	-1835.0	-1874.5	-1837.4

Notes: Market choice among entrants, the headline entry model with the outside option dropped from the choice set, so the coefficient governs where a compound enters conditional on entering. Column 1 is the headline match-quality measure, column 2 the coarse EphMRA ATC level-4 class alone, and column 3 prices the acquisition at the upfront payment instead of total deal value. Standard errors clustered by decision year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 29: Entry on indication-year shares

	Entrants: Poisson count		Log odds vs no entry, to 2015	
	(1) Year FE	(2) + indication FE	(3) Year FE	(4) + indication FE
Acquisition payoff Π^A (USD m)	-0.0002	-0.0035	0.0017	0.0004
	(0.0023)	(0.0049)	(0.0022)	(0.0046)
Own payoff $\pi^{app}V^{own}$ (USD m)	0.0009	0.0001	0.0006	0.0009
	(0.0007)	(0.0020)	(0.0010)	(0.0018)
Crowding (log)	0.7860***	0.3834	0.5166***	0.1457
	(0.0659)	(0.2382)	(0.0475)	(0.2977)
Promise (log)	0.1473*	0.0428	0.0827	-0.0168
	(0.0845)	(0.0831)	(0.0707)	(0.0848)
Indication-years	378	378	270	270
Log likelihood	-503.4	-489.9	-257.6	-240.9
R-squared			0.593	0.640

Columns 1 and 2: Poisson regression of first Phase II trials by small firms on the indication-year payoff terms, which is the multinomial logit over indications on aggregate counts; year effects absorb the denominator. Columns 3 and 4: log odds of entering the indication relative to no entry, Laplace-smoothed, decision years 2006 to 2015 so that the outside option is not right-censored. The acquisition payoff is the discounted probability of acquisition over the program's horizon times the hedonic total deal value (USD m); the own payoff is the Phase II approval probability times V^{own} (USD m). Crowding is log(1 + small-firm Phase II to NDA programs in the indication the prior year); promise is log(1 + non-industry trials started in the prior year). Standard errors clustered by indication (29).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

K.3 What the Entry Coefficients Require

Two assumptions are required for the variation of Section 6.5 to be informative about λ_1. The first is that a compound's mechanism-specific feasibility affects where it enters only through the payoff terms that it scales and through $\ln q_{ij}$. In other words, once those terms are in the model, feasibility has no direct effect on the taste for an indication. The second is that the composition of the acquirer pool affects entry only through the acquisition payoff, conditional on the indication's scientific promise and the alternative constants.

The second is the more demanding of the two, because a market where many incumbents hold overlapping positions may be a market that independently looks promising to an entrant. Three features of the specification speak to this concern. First, the promise control measures non-industry trial activity in the indication directly. Second, the aggregate-share estimators, which we report with and without indication effects, bracket λ_1 from the pool-mix side. Finally, the compound-level model identifies the coefficient from match quality, which the pool mix does not drive. An instrument for the payoff itself would settle the question, and we leave this to future work.

It is also worth noting one restriction of the functional form. The outside option is in the same logit as the 27 markets, which means that the counterfactual's aggregate-entry effect is the entry coefficient passed through the logit's substitution structure, as opposed to a separately estimated response. A nested logit would relax this restriction. We estimated one with hematologic and solid-tumor nests, but the dissimilarity parameters were not identified once the alternative constants were included. We therefore keep the plain logit, with the implication for Section 7.4 that a compound which is easily priced out of its best market is also easily priced out of development altogether.

K.4 Simulating the Counterfactual Ratios

The two ratios of equation (22) are simulated indication-year by indication-year. Each draw takes an engaged count from equation (17), from the full pool K_t in the baseline and from the permitted pool L_{jt} under the ban. Each engaged party is then a high type with probability H_{jt}/K_t in the baseline, and all engaged parties are low types under the ban, with valuations drawn from the recovered F_H and F_L.

Within a process, the engaged parties pass the diligence screen independently, and competition is the equilibrium bid among the screen-passers facing the bid-stage share F. A party that is alone past the screen is not informed of this, and it shades against a phantom rival as opposed to dropping its bid to the reserve

(Section 4.2). The simulated price is the winning bid returned to dollars, which is the reserve plus the fitted surplus plus the equilibrium bid on the residual scale. A sale requires a party to pass both walk-away margins, which happens with probability $1 - 0.65^m$ in a process with m engaged parties. This probability is the denominator of η^{π}_{jt} in the baseline and its numerator under the ban.

In Table 30, we report the composition effect summarized in Section 7.2, in which we hold the engaged count fixed and draw all parties from the low-type distribution in place of the sample's mix.

Table 30: The composition effect at a fixed engaged count, bid-stage share $F = 0.35$

	Expected winning bid (USD m)			Expected maximum valuation (USD m)		
Engaged parties m	Baseline	Restricted	Change (%)	Baseline	Restricted	Change (%)
2	631.4	543.6	-13.9	2463.9	2078.8	-15.6
3	891.2	727.3	-18.4	3013.7	2546.1	-15.5
5	1373.6	1086.0	-20.9	4041.0	3419.0	-15.4

Notes: Baseline draws m engaged parties from the inversion sample's type mix (81 percent high type); restricted draws all m from the low-type distribution. Prices are expected winning bids from the symmetric first-price equilibrium at the level reserve + fitted surplus + residual-scale bid, in 2015q1 $m. The expected maximum valuation is shown for comparison and is not a price.

In Table 31, we report the ban's auction-stage effect across the walk-away grid of Section 7.2. The price ratio runs from 0.57 when all of the walk-away occurs at the diligence screen to 0.77 when all of it occurs at the bid stage. The price effect of the restriction is therefore between 23 and 43 percent, with the working split giving 32 percent. The probability-of-sale ratio, on the other hand, does not vary across the grid. It is 0.82 in all cells, because a sale requires one engaged party to pass through both margins, and the filings determine the product of the two pass rates at any split. This ratio instead responds to the permitted pool shrinking from K_t to L_{jt}, which lowers the distribution of the engaged count, and this channel is separate from the split. The combined ratio therefore runs from 0.49 to 0.65, with 0.58 at the working value, and we use the working cell in the entry counterfactual of Section 7.4.

K.5 The Bid Variable and the Paired Prices

In this appendix, we report two checks on the recovered valuation level that the auction-stage results summarize. The first check uses the other price measures. The upfront payment alone is what the seller receives with certainty, and it is priced for 246 deals. On the same 246 deals, we build a risk-adjusted price, defined as the upfront plus half of the milestone schedule. Given that most milestones never pay, the nominal total overstates the price, while the upfront alone discards the contingent payments entirely, and the middle

Table 31: The ban's auction-stage effect across the walk-away split

F (bid stage)	Price base (USD m)	Price ratio	Pr(sale) ratio	Combined
0.00	284.5	0.573	0.821	0.492
0.25	416.8	0.626	0.820	0.535
0.35*	551.7	0.681	0.820	0.579
0.50	881.2	0.746	0.820	0.630
0.65	1391.2	0.766	0.820	0.645

Notes: Each row inverts the surplus bids at the stated bid-stage share F and simulates the restriction at the same split, which means that the inversion and the counterfactual use one value of F throughout. Columns are means over 588 indication-years at the calibrated engagement rate $p = 0.16$: the expected winning bid in the baseline in 2015q1 $m, its ratio under the ban, the ratio of the probability that at least one engaged party bids, and the product of the two ratios. The price base rises with F because a larger bid-stage share places more of the value distribution below the reserve and the inversion recovers higher valuations. Section 7.2 reports the bid-fit check that bears on which split the bids support. * Working value.

price is therefore closer than either endpoint to what a deal actually transfers. In Table 32, we run the same reserve rule, grid and engagement rate on all three bids. On the upfront, the bid index explains 53 percent of the variance, the value-to-bid ratio is 4.7 at the mean and 4.0 at the median (against 3.5 and 2.5 at the risk-adjusted price and 3.9 and 2.7 on total value), and the reserve discount calibrates to 0.40 in place of one. No price measure brings the ratio near the takeover benchmark, but the counterfactual ratios are close across the three bids, with a combined ratio of 0.58 on total value, 0.61 at the risk-adjusted price and 0.62 on the upfront. The second check uses the paired prices coded from the merger filings, which cover 22 processes with the winner's price and a rival's best offer in the same unit (Appendix C). Where the winner was on top, its price exceeded the runner-up's by a median of 10 percent, with an interquartile range of 2 to 20 percent, and across the 37 processes in which the winner revised its offer, the rise from first offer to signing was 12 percent at the median. In seven processes the runner-up's price exceeded the winner's, which the filings attribute to certainty of closing, timing, or a proposal that was stale or never firm. It is worth noting that neither spread is a valuation. The price on each side bounds that side's value from below, and as a result the gap between them bounds the winner's rent only in the sense of Haile and Tamer (2003). Both spreads are of the same order as the slack in Gorbenko and Malenko (2014).

The comparison with the license prices is not like for like, and the difference cuts in the model's favor. The benchmark deals are whole-company takeovers of listed targets whose market price places a floor under all bids. A license has no such floor, its reserve is the seller's continuation value, which we measure at a small fraction of the winning bid, and the acquirer's valuation includes the commercialization the seller cannot fund. A valuation that is several times the upfront payment is therefore not in itself evidence against

the model. The ratio instead measures the division of the surplus between the two parties. A value-to-bid ratio of 4.7 says that the winning incumbent keeps about four fifths of its own valuation as rent and the seller receives one fifth. This is a statement about competition in the licensing market, and it may be true, since a licensing canvass often ends with one party pricing. However, a symmetric inversion is unable to separate thin competition from unobserved asset heterogeneity in the residual on winning bids alone, and for this reason we do not report the level in the paper.

Table 32: The bid variable: total deal value, the risk-adjusted price and the upfront payment

	Total deal value	Risk-adjusted ($\delta = 0.5$)	Upfront payment
Priced deals (licenses and whole-company)	337	246	246
Inversion sample, after trimming	126	100	96
Mean bid in the inversion sample (USD m)	760.7	462.1	142.0
Mean recovered value over mean bid	3.88	3.47	4.65
Median value-to-bid ratio	2.72	2.46	4.03
Fit gap (USD m)	125.2	53.6	23.7
Rank correlation, bid and value	1.000	1.000	1.000
Price ratio under the ban	0.681	0.719	0.734
Pr(sale) ratio under the ban	0.820	0.820	0.820
Combined ratio	0.579	0.608	0.619

Notes: The auction stage estimated three times, with total potential deal value (upfront plus development and sales milestones, nominal; the company deal value for whole-company deals), with the risk-adjusted price (upfront plus half the milestone schedule, on the 246 deals reporting both components), and with the upfront payment alone as the bid, all in 2015q1 $m, on the priced-deal sample of Section 3.3. Same reserve rule (ϕ calibrated so 95 percent of pool deals clear the reserve; 1.00 on total value and the risk-adjusted price, 0.40 on the upfront), same grid over the walk-away split and engagement rate $p = 0.16$; rows report the working cell, a bid-stage share $F = 0.35$. The value-to-bid ratios are the mean recovered valuation over the mean winning bid and the median of the deal-level ratio, in the trimmed inversion sample. The fit gap and rank correlation are defined as in Table 10.

K.6 How Far the Reserve Sits Below the Price

In Section 6.2, we state that the reserve is slack by design, and in this appendix we provide the magnitudes. The comparison depends on which price the reserve is held against, because the three measures are not equally certain money. Across deals by pool members, the reserve averages $34 million. The upfront payment, which is the only component paid regardless of what happens next, averages $163 million against it, a factor of 4.8 at the means and 6.1 at the median deal. The risk-adjusted price averages $551 million, a factor of 16.2 and a median of 19.1. Total potential deal value averages $900 million, a factor of 25.9 and a median of 32.2, but much of this is milestone money that never pays, and the upfront comparison is therefore the conservative one. Even on that comparison, no plausible offer struggles to clear the seller's accept threshold.

There are three features of the construction that make the remaining gap unsurprising. First, the reserve

values the compound at the entrant's share of its market, 3.1 percent, while the acquirer brings a franchise, a sales force and a higher chance of approval, and this wedge is the reason that the biotech sells as opposed to developing alone. Second, the reserve is evaluated in the compound's lead market only, which means that a multi-indication compound's other markets are missing from the seller's outside option while the price covers the whole asset. Third, the priced sample is selected toward large disclosed deals by pool members, which raises the price side of all of the ratios above what a randomly chosen transfer would show.

We calibrate the commercialization discount ϕ of equation (18) against the same comparison, and it is constrained from one side only. The requirement that the winning bid clears the reserve in 95 percent of pool deals rules out values that are too high. When bids sit far above continuation values, as they do on total deal value and the risk-adjusted price, all values up to one satisfy the requirement, and ϕ therefore lands at one as an upper bound as opposed to an estimate. On the upfront payment alone, the requirement binds and sets ϕ at 0.40. Little rides on where ϕ sits within this range, as forcing $\phi = 0.40$ everywhere leaves all of the estimated counterfactual ratios unchanged to the second decimal.